



COURSE SLIDES · WSQ

Google Associate Cloud Engineer Training

Course Code TGS-2023041024

Synchronous e-Learning
4 days · 30 training hrs · 2 assessment hrs

VERSION 8.0

Conducted by Tertiary Infotech Academy Pte Ltd
UEN 201200696W



Digital attendance with TRAQOM

1

AM

Scan the session QR and verify the submitted record.

2

PM

Repeat after lunch; do not assume the morning record carries forward.

3

Assessment

Scan the separate assessment attendance QR before the papers open.

4

Evidence

Keep the confirmation screen available until the trainer verifies the roster.

DECISION RULE

Attendance is valid only when the learner identity and the correct session window match.

About the Trainer

1 Name	2 Credentials	3 Cloud experience	4 Contact
--	---	--	---

DECISION RULE

The trainer confirms current credentials, course role and escalation channel before delivery.

About Dr Alfred Ang

1

Role

Course developer and trainer

2

Academic

PhD-qualified technology educator

3

Practice

Cloud, AI and software engineering

4

Delivery

Tertiary Infotech Academy Pte Ltd

DECISION RULE

Trainer evidence and certificates are maintained in the protected course-application records.

Learner introductions as a cloud inventory

1

Current role

What systems do you operate?

2

Cloud exposure

Console, CLI, networking, data or containers?

3

Target outcome

Certification, deployment or operations?

4

Constraint

Device, account, billing or corporate policy?

DECISION RULE

The trainer uses this inventory to pace demonstrations without changing assessed outcomes.

Ground rules for a shared cloud classroom

1

Scope

Use only your assigned training project.

2

Cost

Stop resources after each activity and check Billing.

3

Security

Never paste keys, passwords or tokens into chat or screenshots.

4

Evidence

Capture project ID, region and timestamps with results.

DECISION RULE

When unsure, stop the command and ask before creating a billable or externally reachable resource.

Download course material from the LMS

1

Sign in

<https://lms-tms.tertiaryinfotech.com/>

2

Open course

Select this course by title and TGS code.

3

Choose resources

Open Learner Slides, Learner Guide and Activities.

4

Verify version

Confirm v8.0 and TGS-2023041024 before use.

DECISION RULE

The learner copy is the slide PDF in the Learner Guide folder; activity procedures are not embedded in the PPT.

Approved course envelope



DECISION RULE

Course design follows ICT-DIT-3020-1.1 and the approved 32-hour envelope.

Skills Framework mapping

1 TSC title Cloud Computing	2 TSC code ICT-DIT-3020-1.1	3 Knowledge K1–K5	4 Abilities A1–A6
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DECISION RULE

Each topic, activity and assessment item prints its mapped K/A codes.

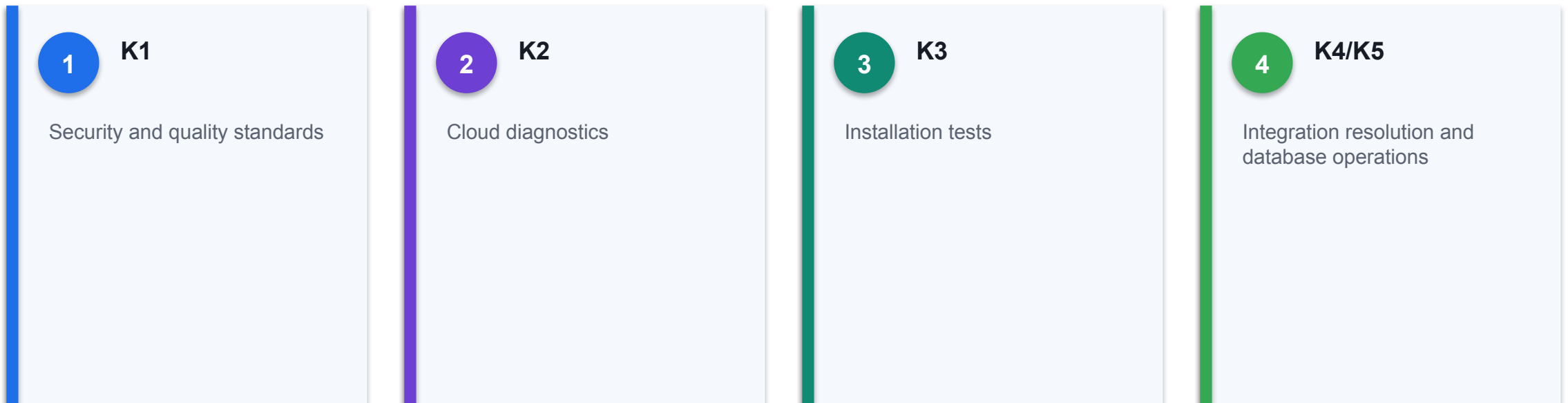
Ability evidence map



DECISION RULE

Practical Performance tasks map A1–A6 to Activities 1, 5, 8, 9 and 10.

Knowledge evidence map



DECISION RULE

Written Assessment has five open-ended questions: one mapped question for each K code.

Learning outcomes

1

LO1

Set up and test Google Cloud solutions in accordance with security standards.

2

LO2

Detect and resolve Google Cloud integration issues using technical solutions and database administration techniques.

3

LO3

Conduct implementation tests against defined metrics and escalate unresolved Google Cloud issues with evidence.

4

Evidence

Every outcome ends in an inspectable artifact.

DECISION RULE

Slides teach mechanisms; the Learner Guide and Activity PDFs contain the complete procedures.

Course roadmap · Topics 1–5



DECISION RULE

Day 1 establishes the environment, identity and network; Day 2 moves into compute and data.

Course roadmap · Topics 6–10

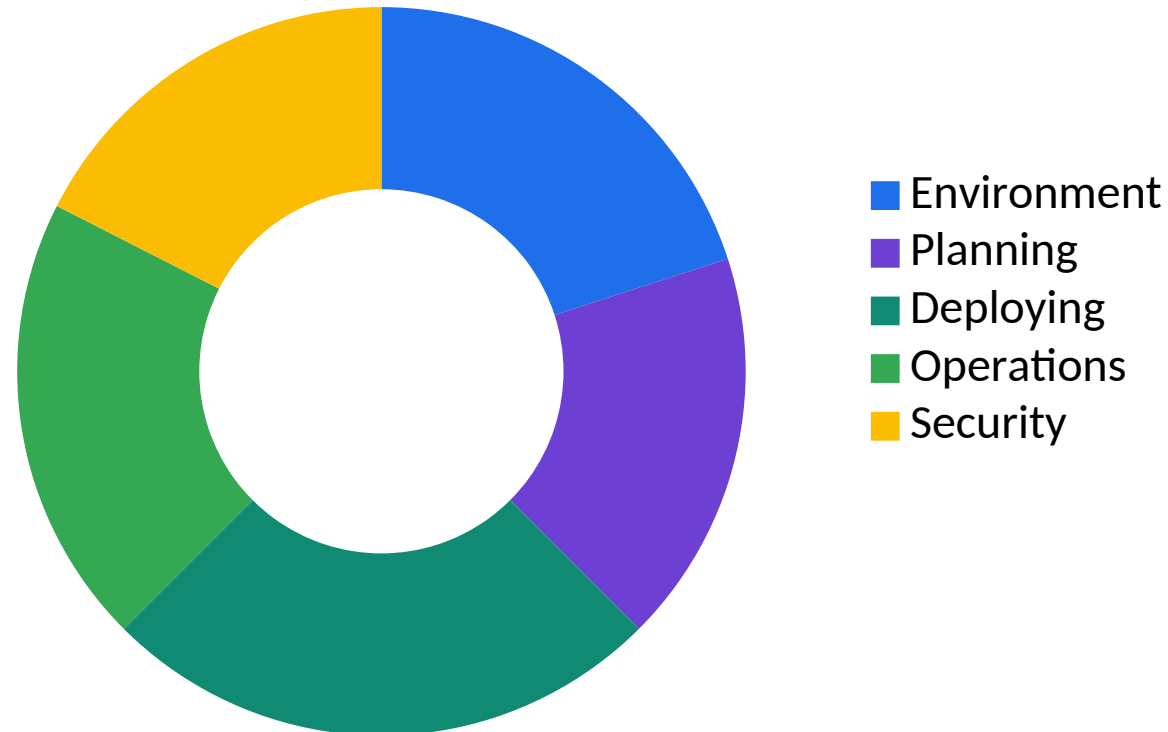


DECISION RULE

Hybrid connectivity, managed services, Kubernetes, monitoring and scaling complete the operational path.

Associate Cloud Engineer domain weighting

Approx. exam weight



WHAT THE DATA SHOWS

Deploying and implementing carries the largest approximate share, but the exam deliberately spans setup, planning, operations and security. Practice must therefore cross service boundaries instead of memorising one product.

Account and billing readiness gate

1

Identity

Use a Google Account you control and enable MFA.

2

Trial

Review current eligibility and terms before accepting free credits.

3

Billing

A payment method may be required; link only the training project.

4

Guardrail

Create budget alerts before billable resources.

DECISION RULE

A budget is an alert, not a hard spending cap. Delete resources and verify Billing after each activity.

Briefing for assessment

1

Individual

Complete both instruments yourself.

2

Open book

Use approved slides, LG and Activity material only.

3

Evidence

Use project ID, commands and observed outputs.

4

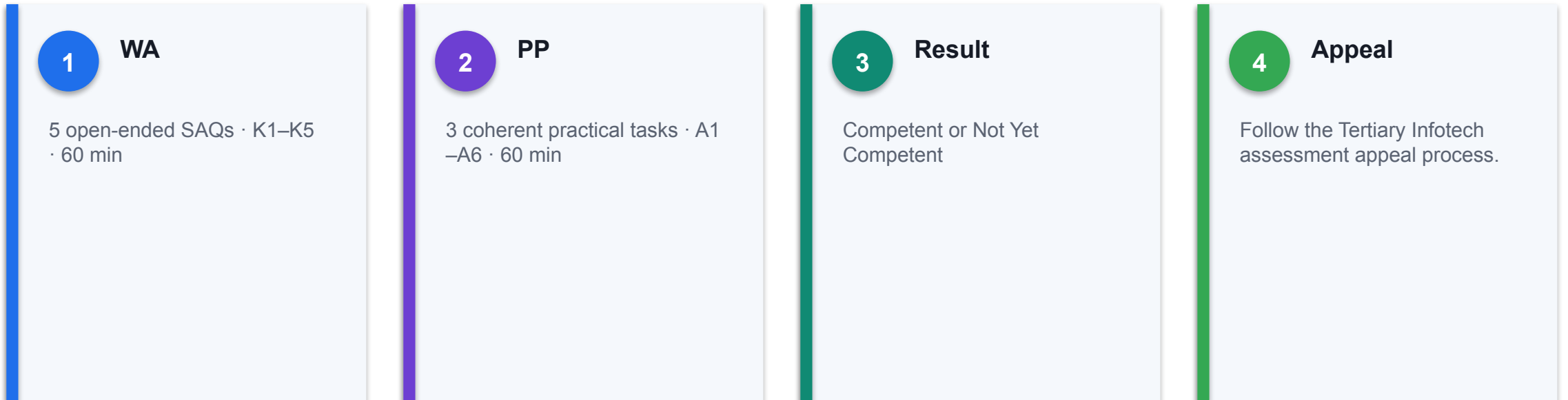
Submission

Upload completed papers to <https://lms-tms.tertiaryinfotech.com/>

DECISION RULE

No answer keys are learner-facing; oral clarification may be used only under the approved assessment process.

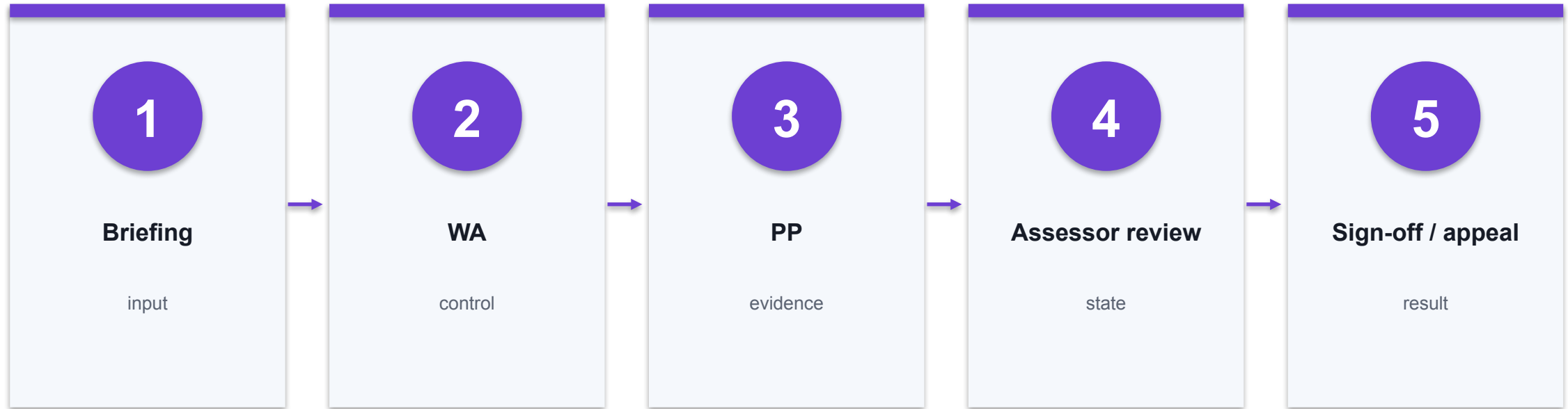
Final assessment instruments



DECISION RULE

The PP reuses Activity evidence patterns; it does not introduce unseen services or procedures.

Assessment flow and sign-off



Question papers go to learners; answer keys remain trainer-only.

Practice exam for Associate Cloud Engineer

1

Platform

Tertiary Exams

2

Exam

Google Associate Cloud Engineer

3

Mode

Practice and timed exam modes

4

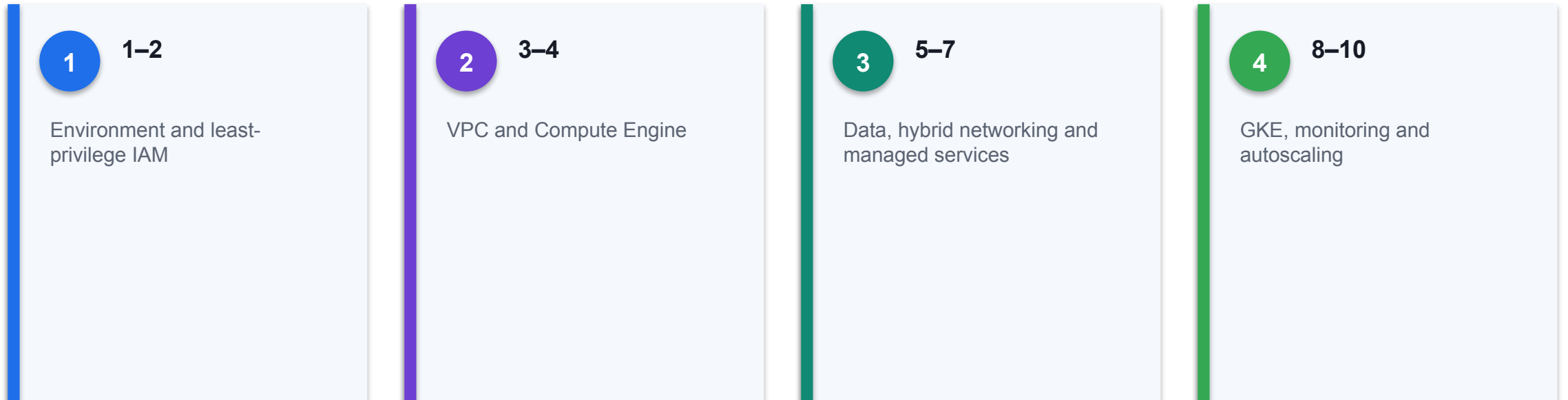
Link

<https://exams.tertiaryinfotech.com/practice-exams/google/google-ace>

DECISION RULE

Practice questions are original and blueprint-aligned; they are not exam dumps.

Activity roadmap



DECISION RULE

Ten folders contain mock data or scripts plus one Activity Guide PDF each.

TOPIC 01 · A2 · K3

01

Google Cloud and Cloud Computing

Set up a cloud solution environment

Projects as operating and trust boundaries · architecture

1

Project Id

Boundary component 1. State and ownership must be explicit.

2

Enabled Apis

Boundary component 2. State and ownership must be explicit.

3

Billing Link

Boundary component 3. State and ownership must be explicit.

4

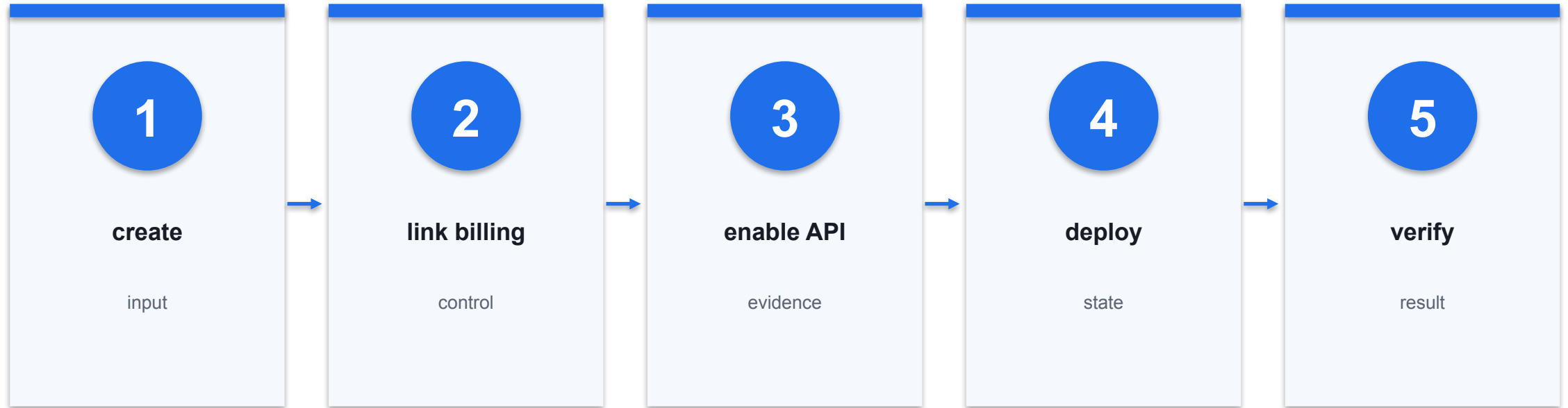
Iam Policy

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Project state is ACTIVE; intended APIs are enabled; resources resolve to the same project ID.

Projects as operating and trust boundaries · runtime trace



Project state is ACTIVE; intended APIs are enabled; resources resolve to the same project ID.

Projects as operating and trust boundaries · concrete contract

```
gcloud projects describe $PROJECT_ID 01
gcloud services list --enabled --project=$PROJECT_ID 02
```

1

Scope: Google Cloud and Cloud Computing

2

Mapped: A2 · K3

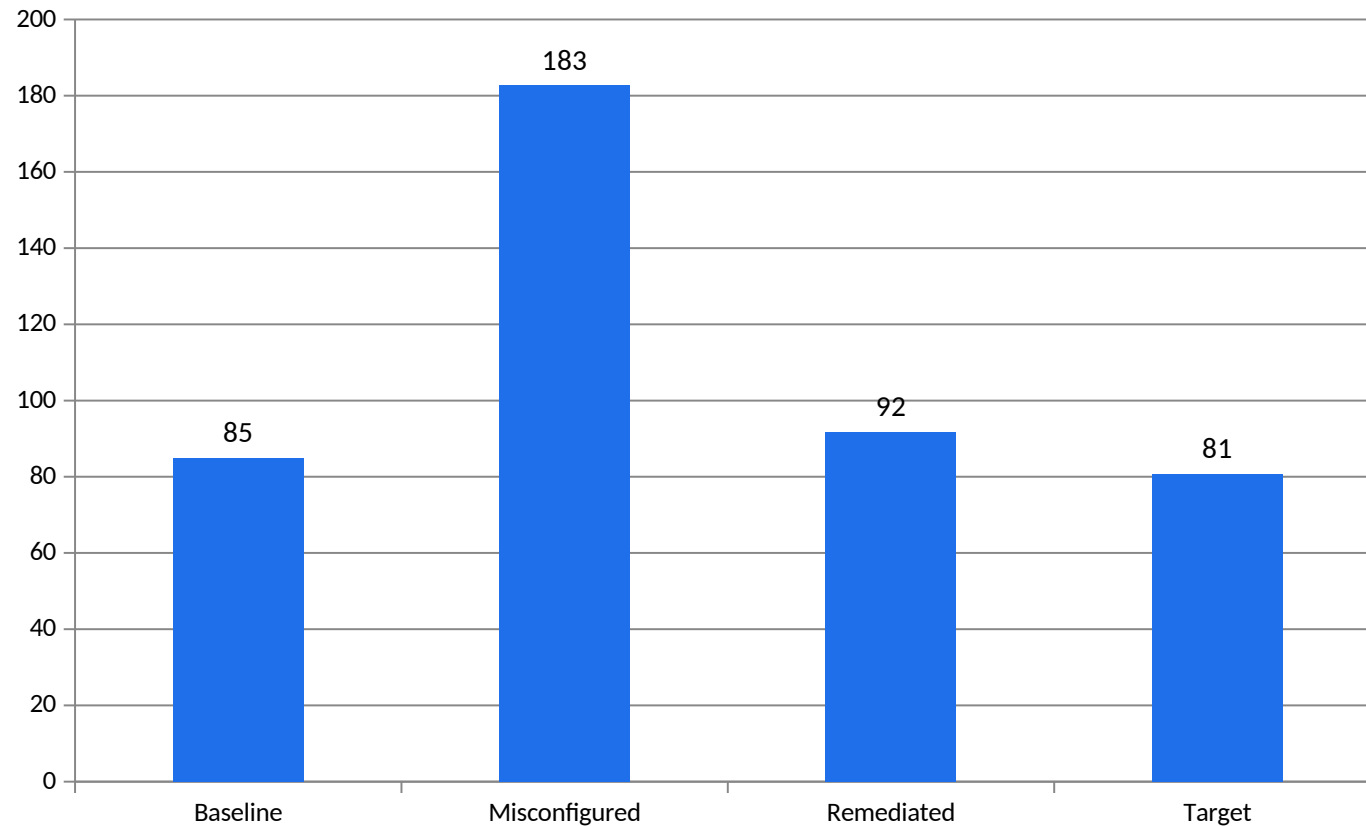
3

Capture stdout and exit status

4

Source: official resource-manager documentation

Topic 1 evidence profile



WHAT THE DATA SHOWS

A controlled fault should move the signal away from baseline. The remediation is accepted only when it returns close to target and the exact configuration change is recorded.

Indicative teaching data. Learners replace it with measurements from the aligned activity.

Regions, zones and resource scope · architecture

1

Global Resources

Boundary component 1. State and ownership must be explicit.

2

Region

Boundary component 2. State and ownership must be explicit.

3

Zone

Boundary component 3. State and ownership must be explicit.

4

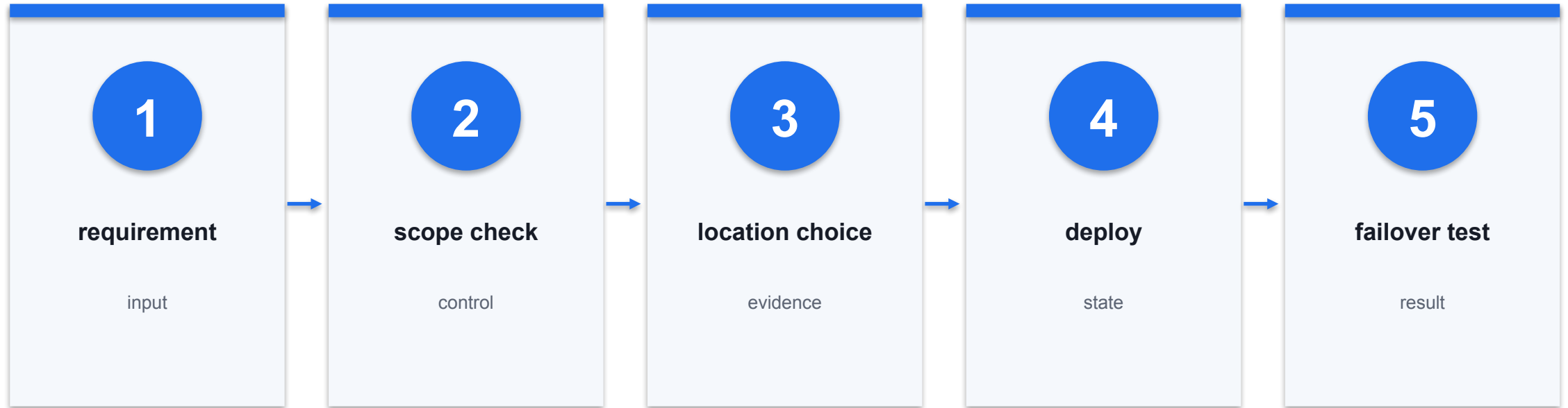
Multi-Region

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Location choice matches latency, residency and availability requirements.

Regions, zones and resource scope · runtime trace



Location choice matches latency, residency and availability requirements.

Regions, zones and resource scope · concrete contract

```
gcloud compute zones list --filter='region:(asia-southeast1)' 01
gcloud compute regions describe asia-southeast1 02
```

1

Scope: Google Cloud and Cloud Computing

2

Mapped: A2 · K3

3

Capture stdout and exit status

4

Source: official locations documentation

Regions, zones and resource scope · failure and control

FAILURE MODES



zonal single point



unsupported service location



cross-region egress

VERIFICATION EVIDENCE



Expected state is explicit



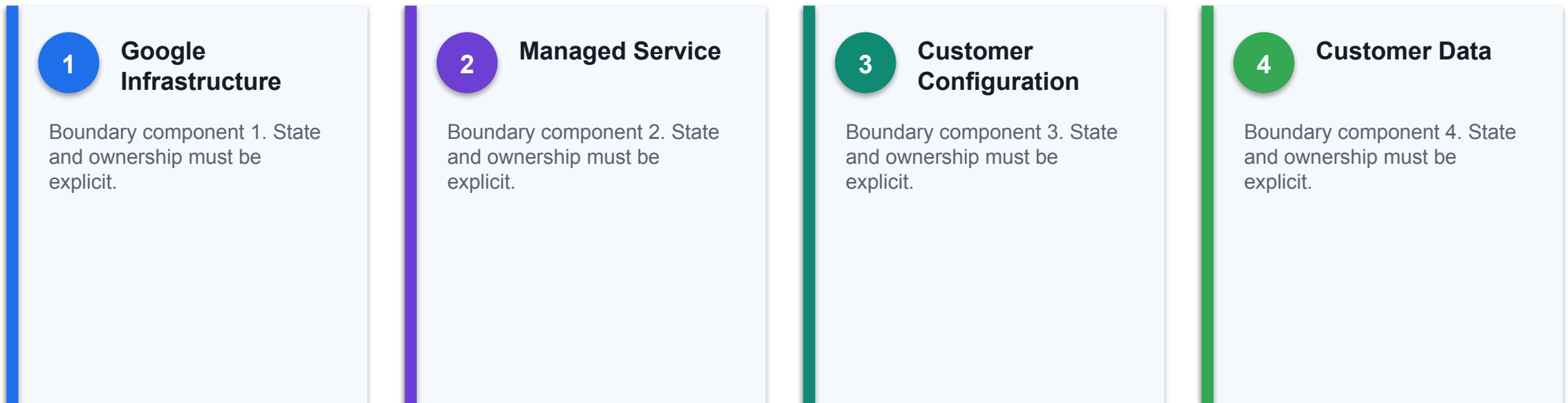
Observed output is captured



Timestamp and scope are recorded

Location choice matches latency, residency and availability requirements.

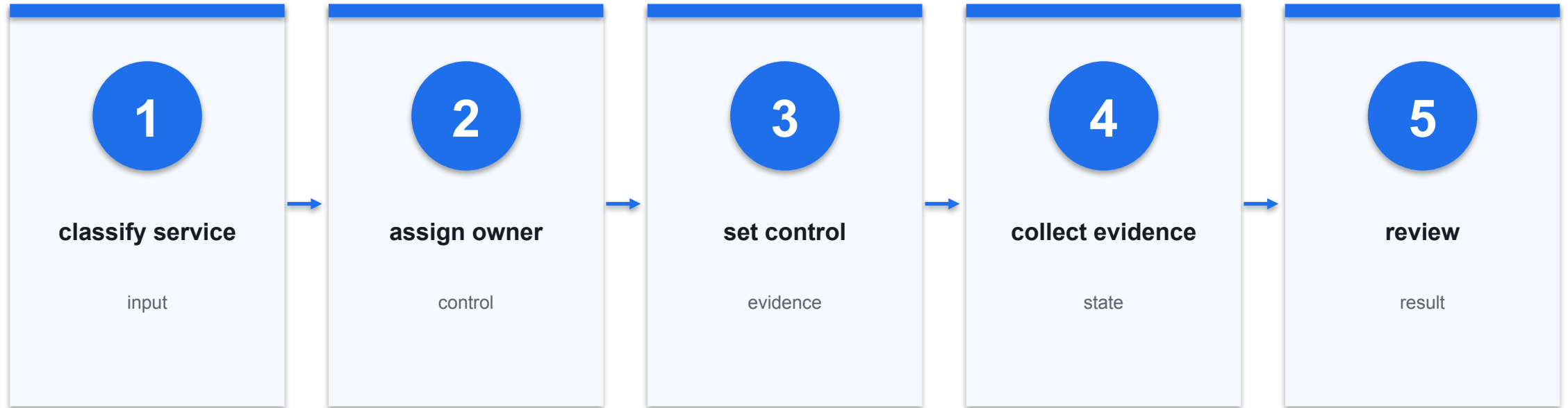
Shared-responsibility control map · architecture



DECISION RULE

Scope first: Every resource has an accountable owner and a testable configuration control.

Shared-responsibility control map · runtime trace



Every resource has an accountable owner and a testable configuration control.

Shared-responsibility control map · concrete contract

```
gcloud asset search-all-resources  
--scope=projects/$PROJECT_NUMBER --  
format='table(assetType,location) '
```

01

1

Scope: Google Cloud and Cloud Computing

2

Mapped: A2 · K3

3

Capture stdout and exit status

4

Source: official security documentation

Shared-responsibility control map · failure and control

FAILURE MODES



assuming managed means secured



unowned data



no evidence trail

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Every resource has an accountable owner and a testable configuration control.

Console, Cloud Shell and local CLI surfaces · architecture

1

Cloud Console

Boundary component 1. State and ownership must be explicit.

2

Cloud Shell

Boundary component 2. State and ownership must be explicit.

3

Gcloud Cli

Boundary component 3. State and ownership must be explicit.

4

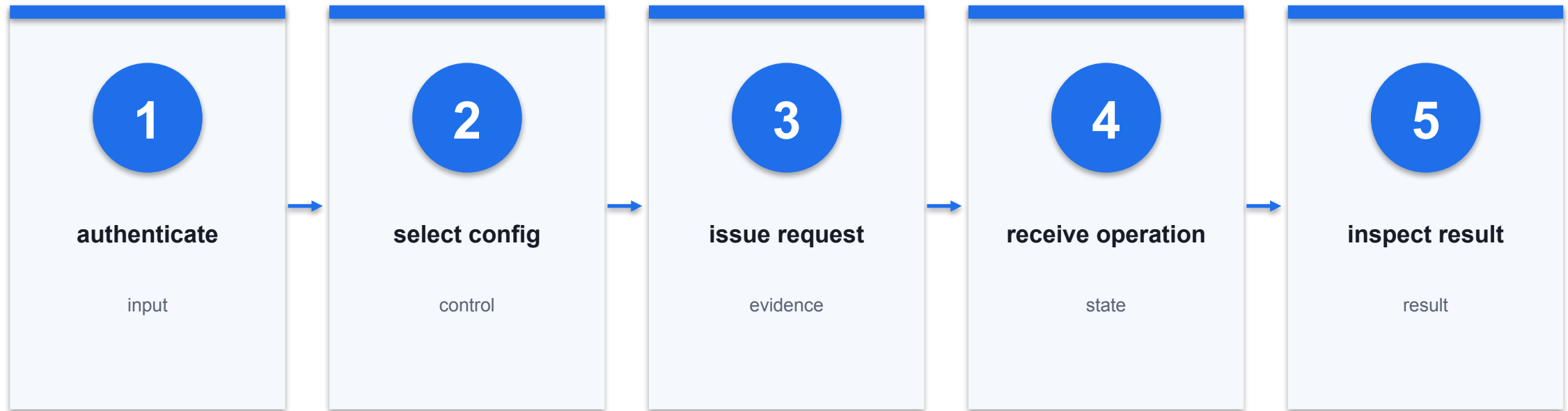
Rest Api

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: The intended account and project are active before any mutation.

Console, Cloud Shell and local CLI surfaces · runtime trace



The intended account and project are active before any mutation.

Console, Cloud Shell and local CLI surfaces · concrete contract

```
gcloud auth list                                01
gcloud config configurations list                02
gcloud config list                               03
```

1

Scope: Google Cloud and Cloud Computing

2

Mapped: A2 · K3

3

Capture stdout and exit status

4

Source: official interact documentation

Console, Cloud Shell and local CLI surfaces · failure and control

FAILURE MODES



wrong account



wrong configuration



expired credentials

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

The intended account and project are active before any mutation.

API enablement and service identities · architecture

1 Service Usage Api

Boundary component 1. State and ownership must be explicit.

2 Service Agent

Boundary component 2. State and ownership must be explicit.

3 Iam Grant

Boundary component 3. State and ownership must be explicit.

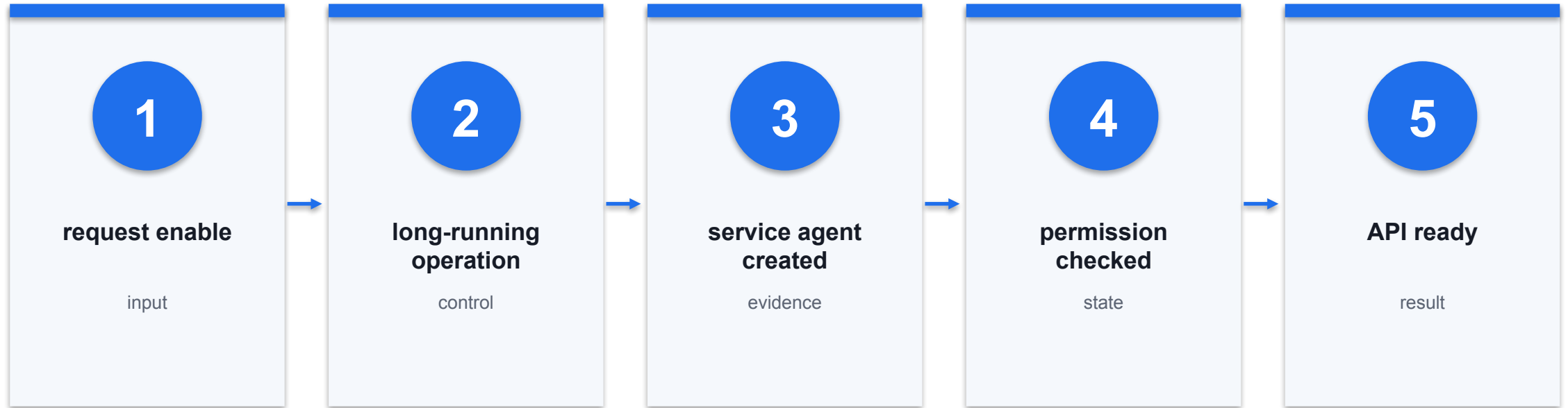
4 Service Endpoint

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Enabled service appears in the service list and its agent receives only documented roles.

API enablement and service identities · runtime trace



Enabled service appears in the service list and its agent receives only documented roles.

API enablement and service identities · concrete contract

```
gcloud services enable compute.googleapis.com 01  
monitoring.googleapis.com logging.googleapis.com
```

1

Scope: Google Cloud and Cloud Computing

2

Mapped: A2 · K3

3

Capture stdout and exit status

4

Source: official service-usage documentation

API enablement and service identities · failure and control

FAILURE MODES



SERVICE_DISABLED



permission denied



quota project mismatch

VERIFICATION EVIDENCE



Expected state is explicit



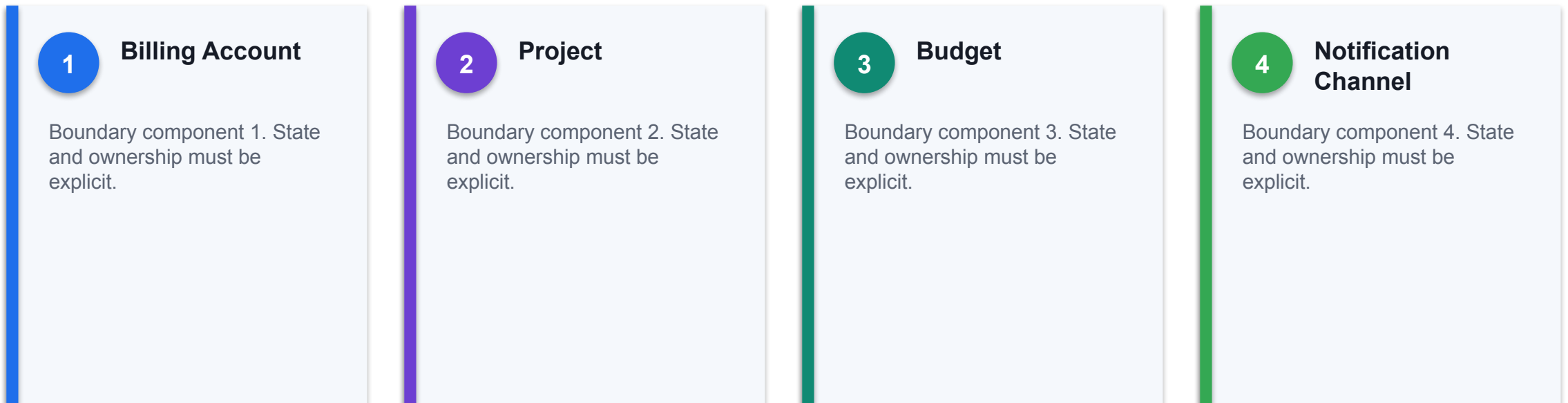
Observed output is captured



Timestamp and scope are recorded

Enabled service appears in the service list and its agent receives only documented roles.

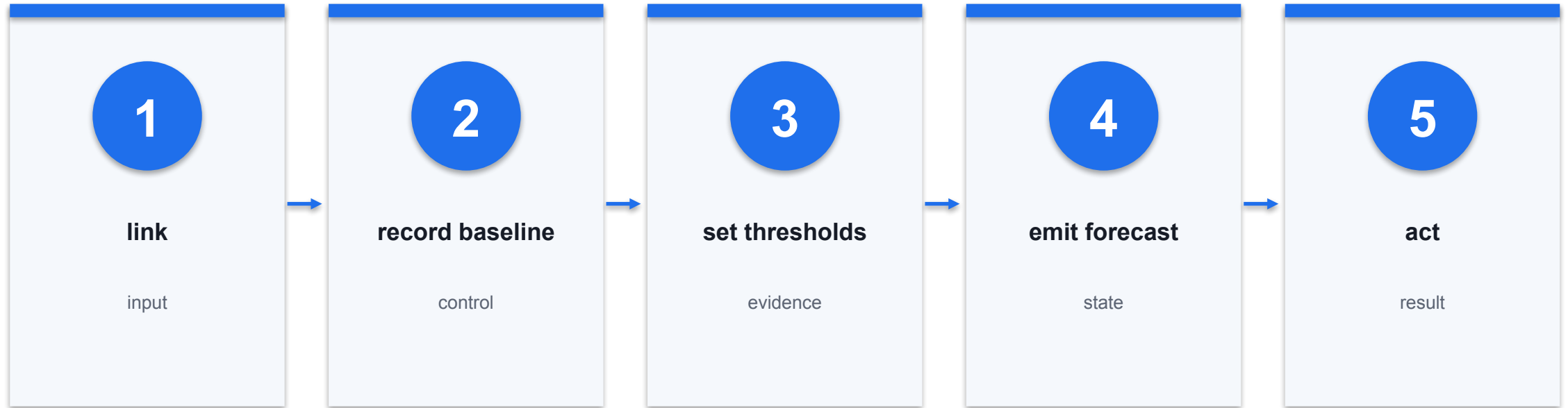
Billing attachment and budget signals · architecture



DECISION RULE

Scope first: Billing is enabled and budget thresholds are visible before billable resources are created.

Billing attachment and budget signals · runtime trace



Billing is enabled and budget thresholds are visible before billable resources are created.

Billing attachment and budget signals · concrete contract

```
gcloud billing projects describe $PROJECT_ID    01
gcloud billing budgets list --billing-         02
account=$BILLING_ACCOUNT
```

1

Scope: Google Cloud and Cloud Computing

2

Mapped: A2 · K3

3

Capture stdout and exit status

4

Source: official billing documentation

Billing attachment and budget signals · failure and control

FAILURE MODES



budget is not a cap



closed account



wrong currency assumption

VERIFICATION EVIDENCE



Expected state is explicit



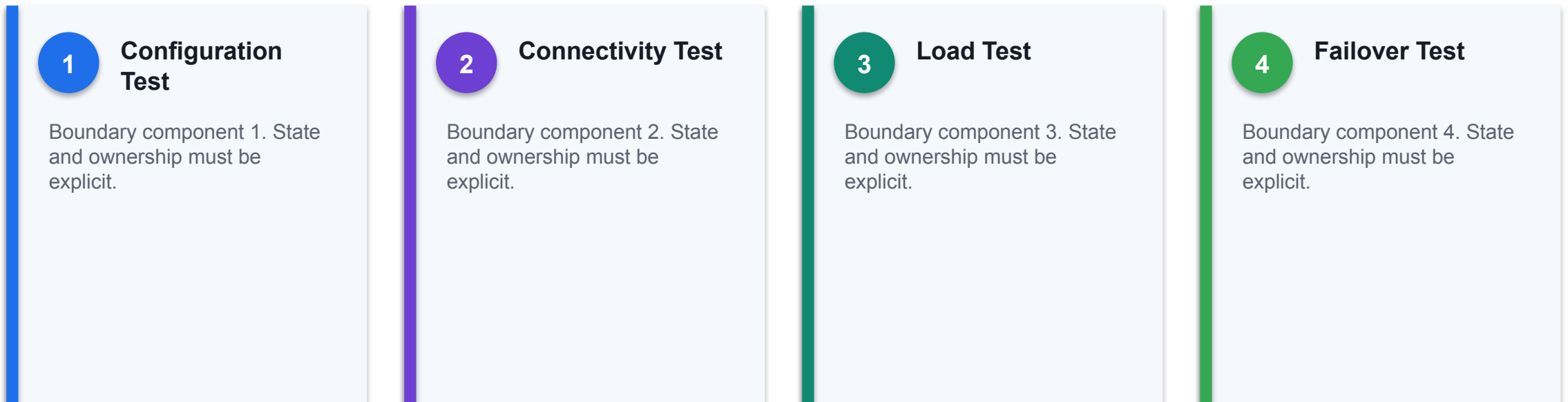
Observed output is captured



Timestamp and scope are recorded

Billing is enabled and budget thresholds are visible before billable resources are created.

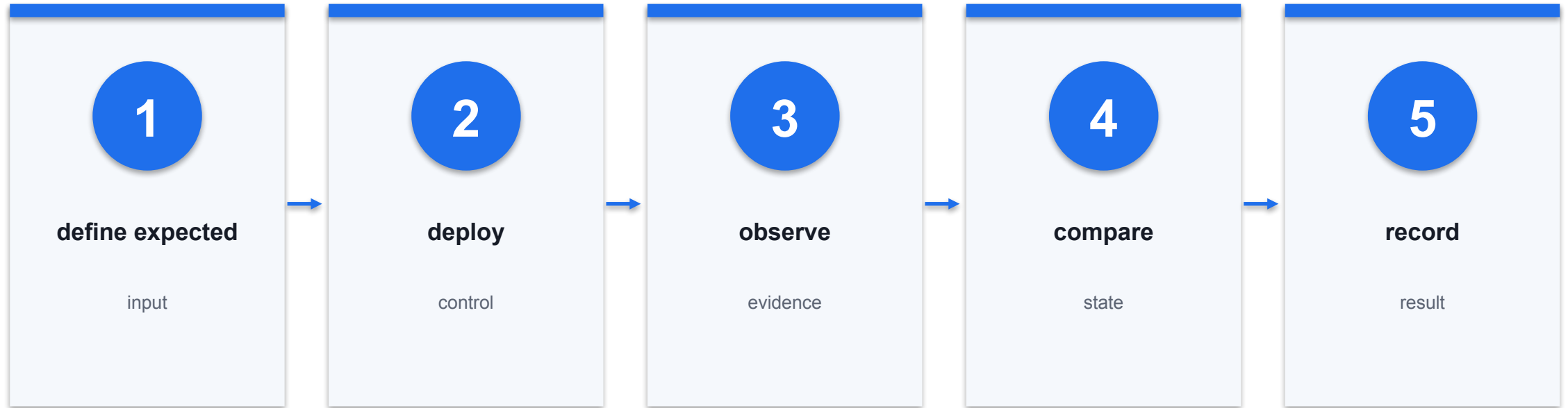
Installation test pyramid for cloud resources · architecture



DECISION RULE

Scope first: Each test has expected output, observed output, timestamp and owner.

Installation test pyramid for cloud resources · runtime trace



Each test has expected output, observed output, timestamp and owner.

Installation test pyramid for cloud resources · concrete contract

```
gcloud asset search-all-resources 01  
--scope=projects/$PROJECT_NUMBER --format=json >  
inventory.json
```

1

Scope: Google Cloud and Cloud Computing

2

Mapped: A2 · K3

3

Capture stdout and exit status

4

Source: official testing documentation

Installation test pyramid for cloud resources · failure and control

FAILURE MODES



test without baseline



single happy path



no rollback

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Each test has expected output, observed output, timestamp and owner.

ACE task-domain traceability · architecture

1

Environment 20%

Boundary component 1. State and ownership must be explicit.

2

Planning 17.5%

Boundary component 2. State and ownership must be explicit.

3

Deploying 25%

Boundary component 3. State and ownership must be explicit.

4

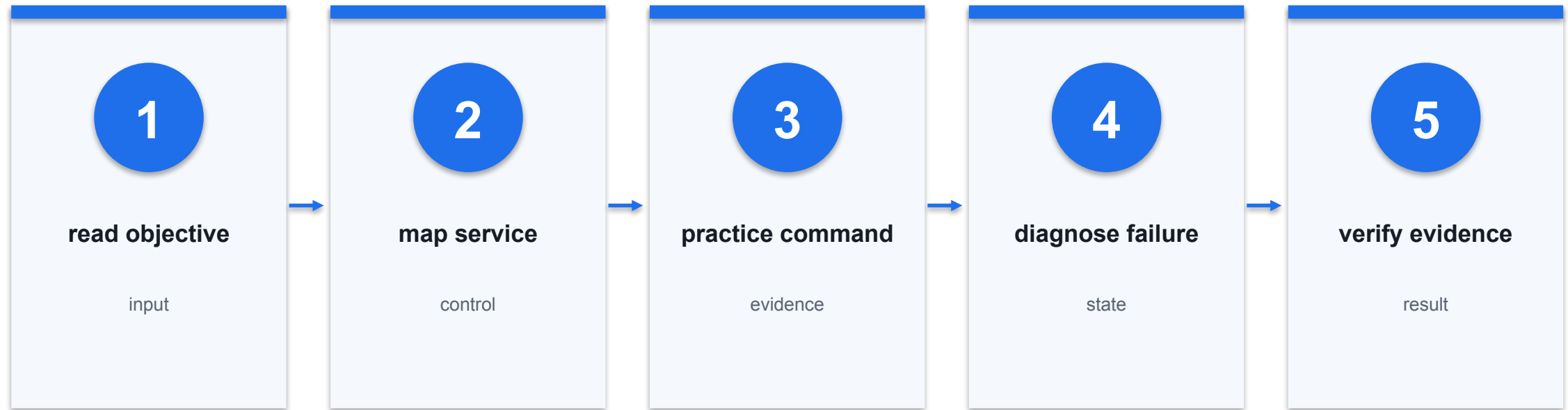
Operations 20%

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Practice covers both console interpretation and CLI evidence across all five exam domains.

ACE task-domain traceability · runtime trace



Practice covers both console interpretation and CLI evidence across all five exam domains.

ACE task-domain traceability · concrete contract

```
gcloud version                                01
gcloud info --                                02
format='value(config.paths.global_config_dir)'
```

1

Scope: Google Cloud and Cloud Computing

2

Mapped: A2 · K3

3

Capture stdout and exit status

4

Source: official ace-guide documentation

ACE task-domain traceability · failure and control

FAILURE MODES



memorising product names



ignoring operations



no hands-on evidence

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Practice covers both console interpretation and CLI evidence across all five exam domains.

Create a Safe Google Cloud Training Environment

SCENARIO

Create an account, project, billing guardrails and CLI configuration, then prove the environment is safe to use.

YOU WILL PRODUCE

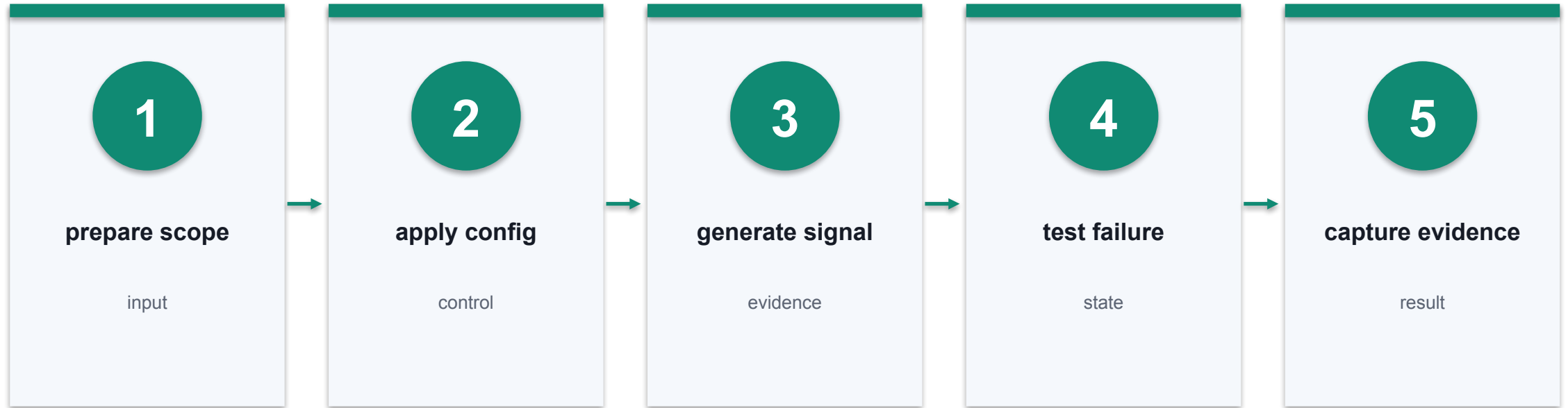
environment-check.txt, budget-policy.json and a project inventory

`check_environment.sh`

`budget-policy.json`

Full click-by-click instructions are in the Learner Guide and Activity PDF. This slide shows the work product, not the procedure.

Activity 1 system trace



Create an account, project, billing guardrails and CLI configuration, then prove the environment is safe to use.

Activity 1 worked artifact

```
#!/usr/bin/env bash 01
set -euo pipefail 02
: "${PROJECT_ID:?Set PROJECT_ID}" 03
gcloud auth list 04
gcloud config set project "$PROJECT_ID" 05
gcloud projects describe "$PROJECT_ID" 06
gcloud services list --enabled --project "$PROJECT_ID" 07
```

1

Use placeholders for project-specific values

2

Run only in the isolated training project

3

Capture both success and failure output

4

Keep secrets out of submitted evidence

Activity 1 acceptance gate

FAILURE MODES



wrong project or region



insufficient IAM or disabled API



observed output does not meet target

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Accept when environment-check.txt, budget-policy.json and a project inventory are complete, timestamped and reproducible.

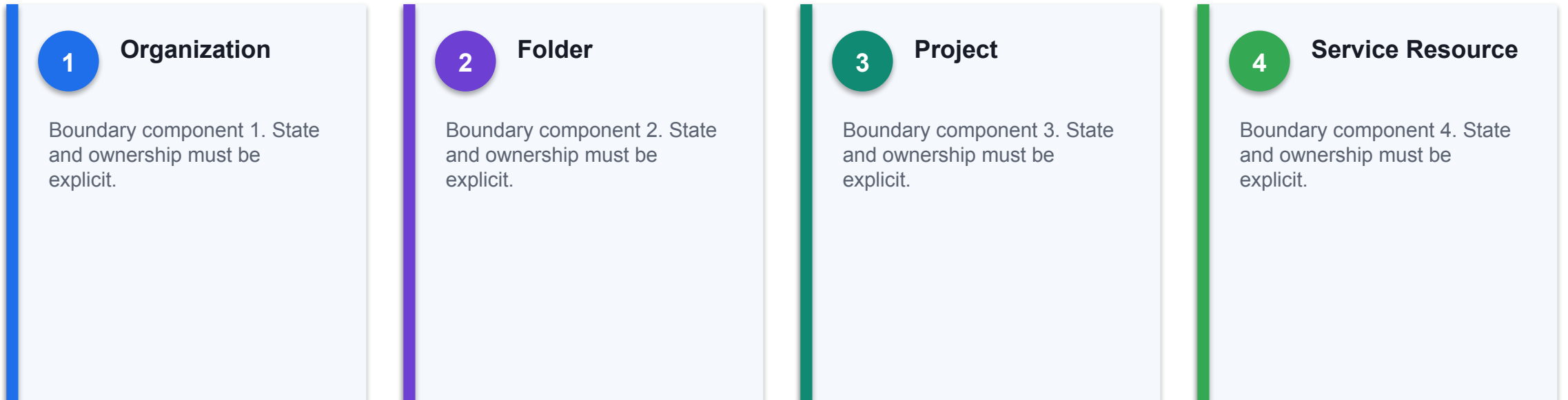
TOPIC 02 · A1 · K1

02

GCP Basics and IAM

Configure access and security

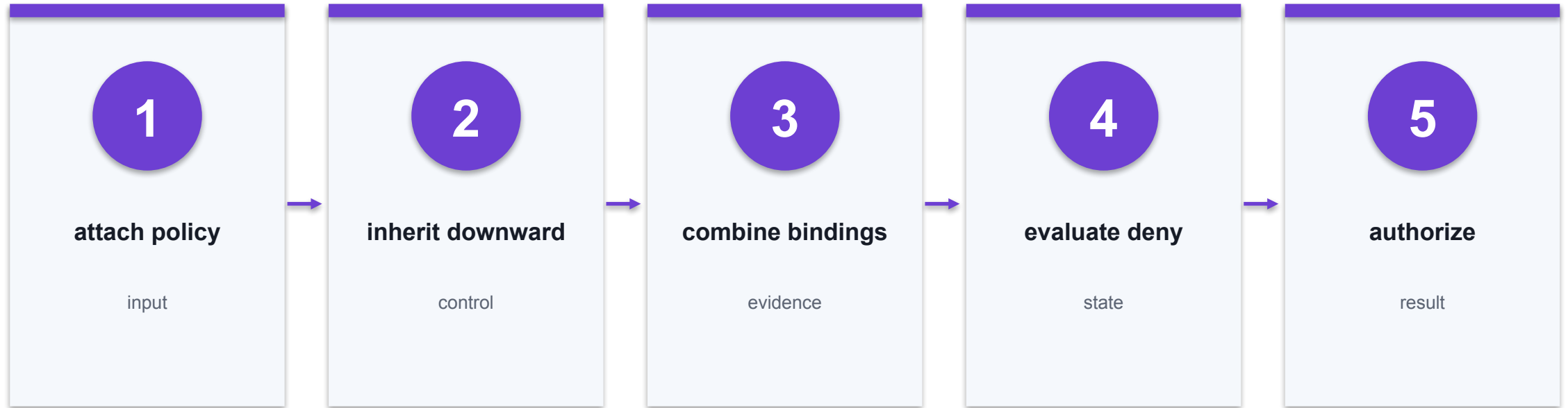
Resource hierarchy and inherited policy · architecture



DECISION RULE

Scope first: Effective access includes direct and inherited allow policies, then applies applicable deny constraints.

Resource hierarchy and inherited policy · runtime trace



Effective access includes direct and inherited allow policies, then applies applicable deny constraints.

Resource hierarchy and inherited policy · concrete contract

```
gcloud projects get-iam-policy $PROJECT_ID --  
format=json
```

01

1

Scope: GCP Basics and IAM

2

Mapped: A1 · K1

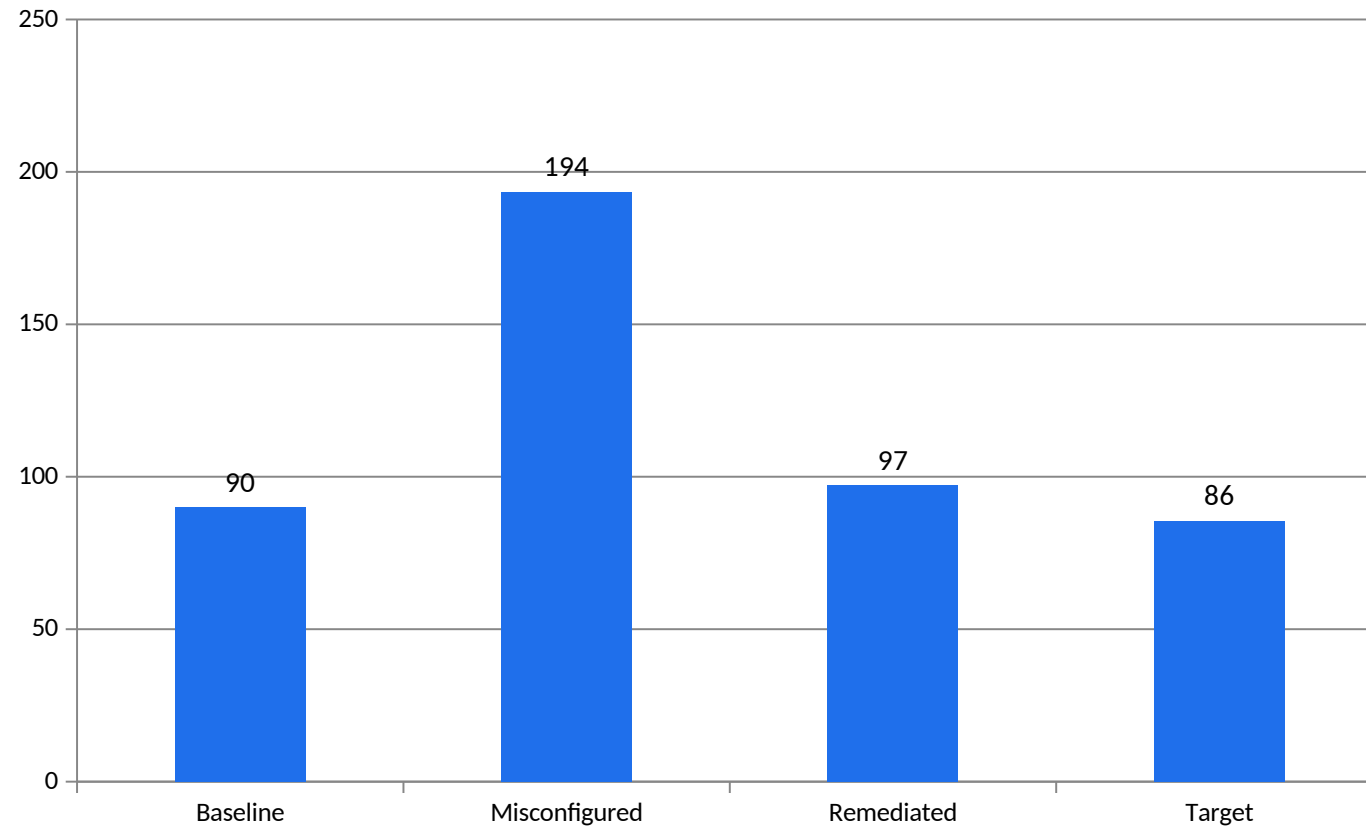
3

Capture stdout and exit status

4

Source: official iam documentation

Topic 2 evidence profile

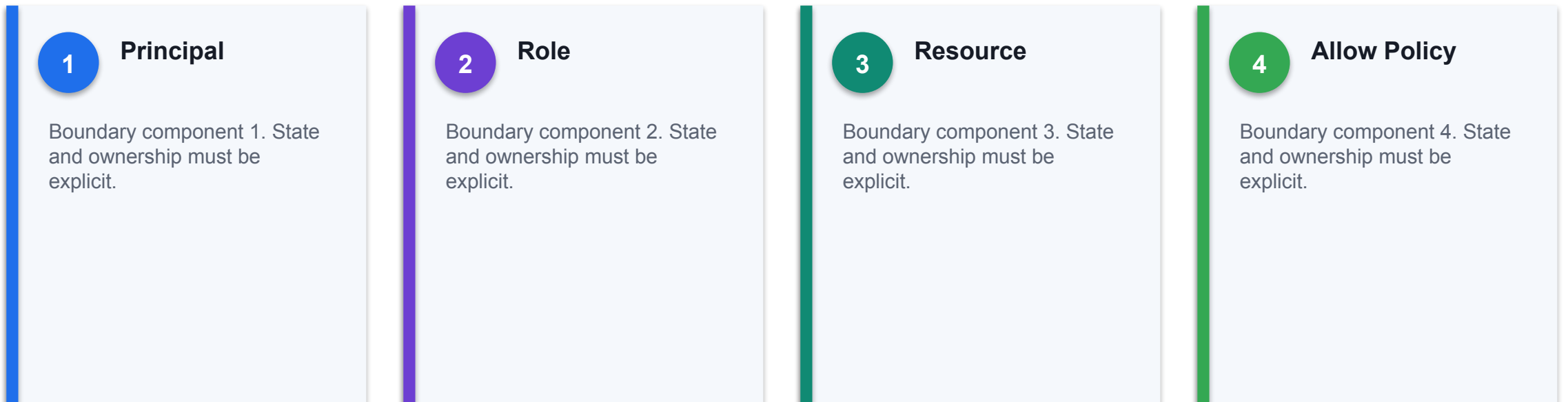


WHAT THE DATA SHOWS

A controlled fault should move the signal away from baseline. The remediation is accepted only when it returns close to target and the exact configuration change is recorded.

Indicative teaching data. Learners replace it with measurements from the aligned activity.

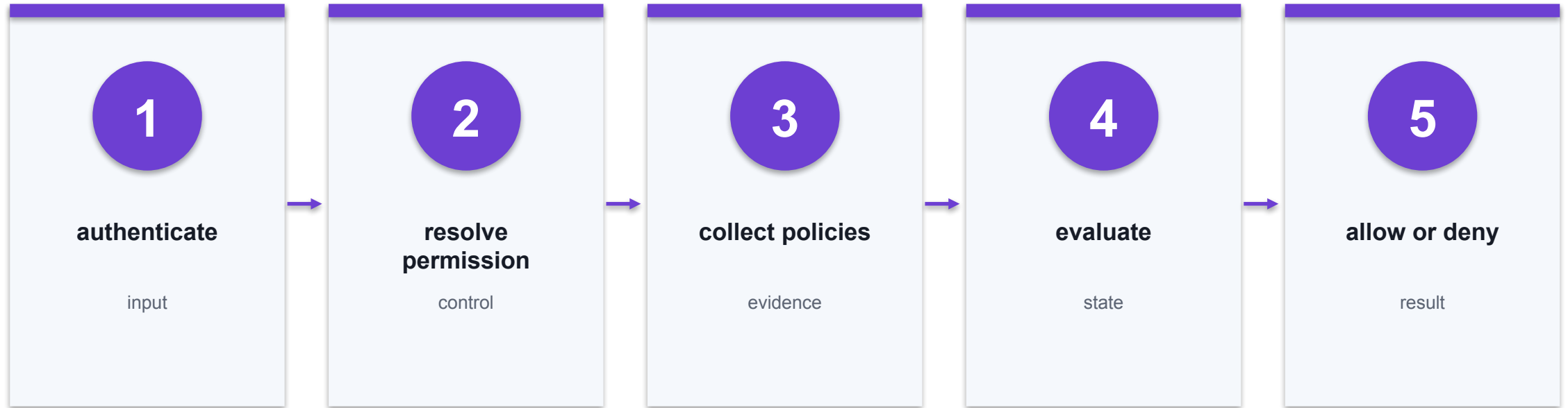
IAM authorization tuple · architecture



DECISION RULE

Scope first: Binding identifies one principal, one role and the narrowest practical resource scope.

IAM authorization tuple · runtime trace



Binding identifies one principal, one role and the narrowest practical resource scope.

IAM authorization tuple · concrete contract

```
gcloud projects add-iam-policy-binding $PROJECT_ID --member=user:learner@example.com --role=roles/viewer
```

1

Scope: GCP Basics and IAM

2

Mapped: A1 · K1

3

Capture stdout and exit status

4

Source: official iam documentation

IAM authorization tuple · failure and control

FAILURE MODES



basic Owner role



misspelled principal



binding on wrong project

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Binding identifies one principal, one role and the narrowest practical resource scope.

Predefined, custom and basic roles · architecture

1

Permissions

Boundary component 1. State and ownership must be explicit.

2

Predefined Role

Boundary component 2. State and ownership must be explicit.

3

Custom Role

Boundary component 3. State and ownership must be explicit.

4

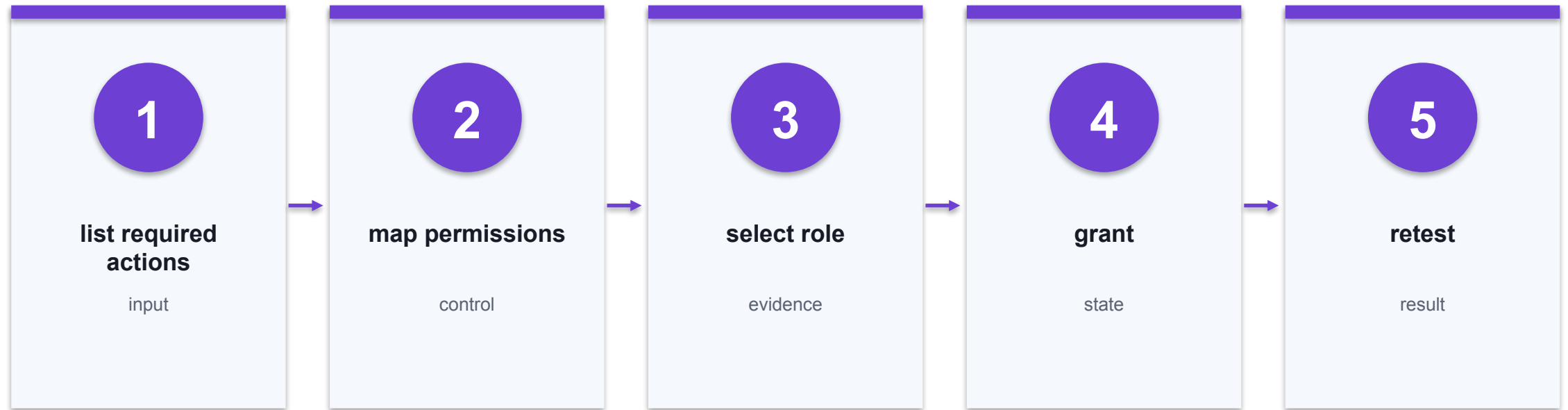
Basic Role

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Selected role contains required permissions without unrelated administrative power.

Predefined, custom and basic roles · runtime trace



Selected role contains required permissions without unrelated administrative power.

Predefined, custom and basic roles · concrete contract

```
gcloud iam roles describe roles/storage.objectViewer01  
gcloud iam roles list --project=$PROJECT_ID 02
```

1

Scope: GCP Basics and IAM

2

Mapped: A1 · K1

3

Capture stdout and exit status

4

Source: official iam documentation

Predefined, custom and basic roles · failure and control

FAILURE MODES



role drift



unsupported custom permission



basic role in production

VERIFICATION EVIDENCE



Expected state is explicit



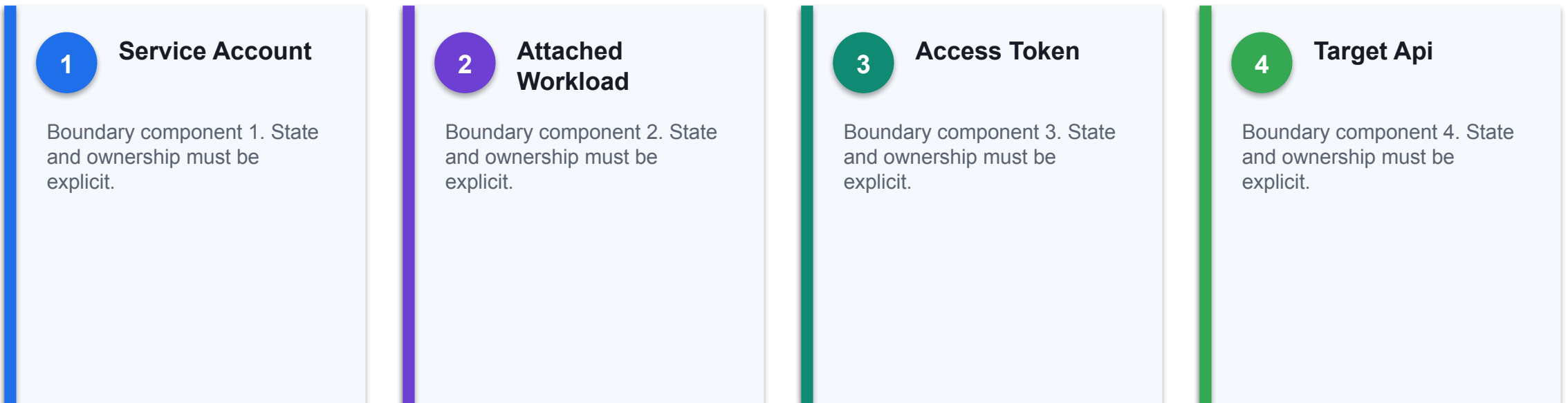
Observed output is captured



Timestamp and scope are recorded

Selected role contains required permissions without unrelated administrative power.

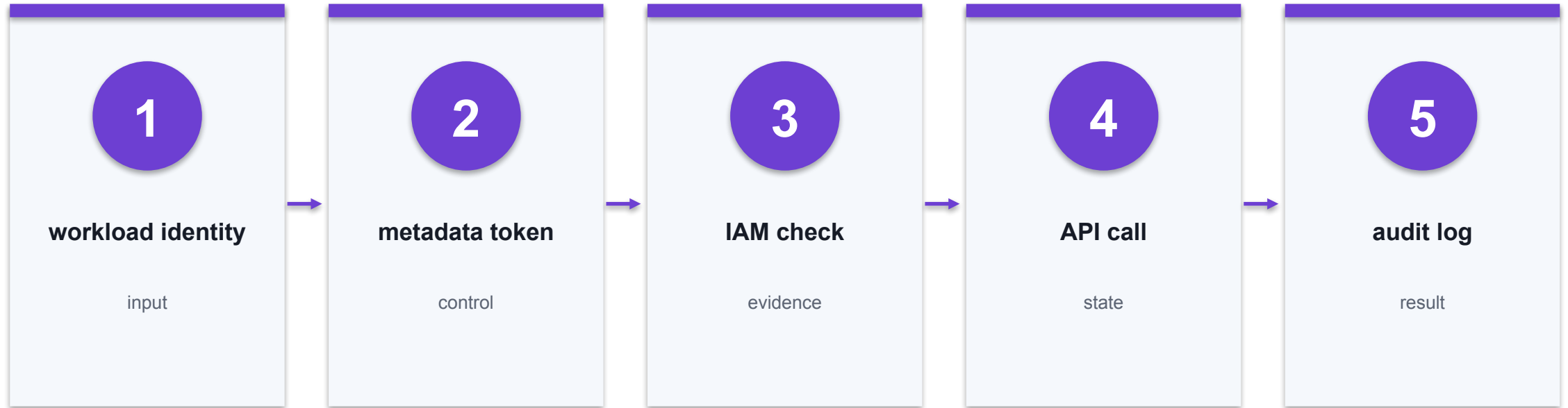
Service accounts as workload principals · architecture



DECISION RULE

Scope first: Workload uses an attached user-managed service account; no long-lived key file is required.

Service accounts as workload principals · runtime trace



Workload uses an attached user-managed service account; no long-lived key file is required.

Service accounts as workload principals · concrete contract

```
gcloud iam service-accounts create app-runtime -- 01  
display-name='App_runtime'  
gcloud projects add-iam-policy-binding $PROJECT_ID --02  
member=serviceAccount:app-  
runtime@$PROJECT_ID.iam.gserviceaccount.com --  
role=roles/storage.objectViewer
```

1

Scope: GCP Basics and IAM

2

Mapped: A1 · K1

3

Capture stdout and exit status

4

Source: official service-accounts documentation

Service accounts as workload principals · failure and control

FAILURE MODES



default service account



downloaded key



token audience mismatch

VERIFICATION EVIDENCE



Expected state is explicit



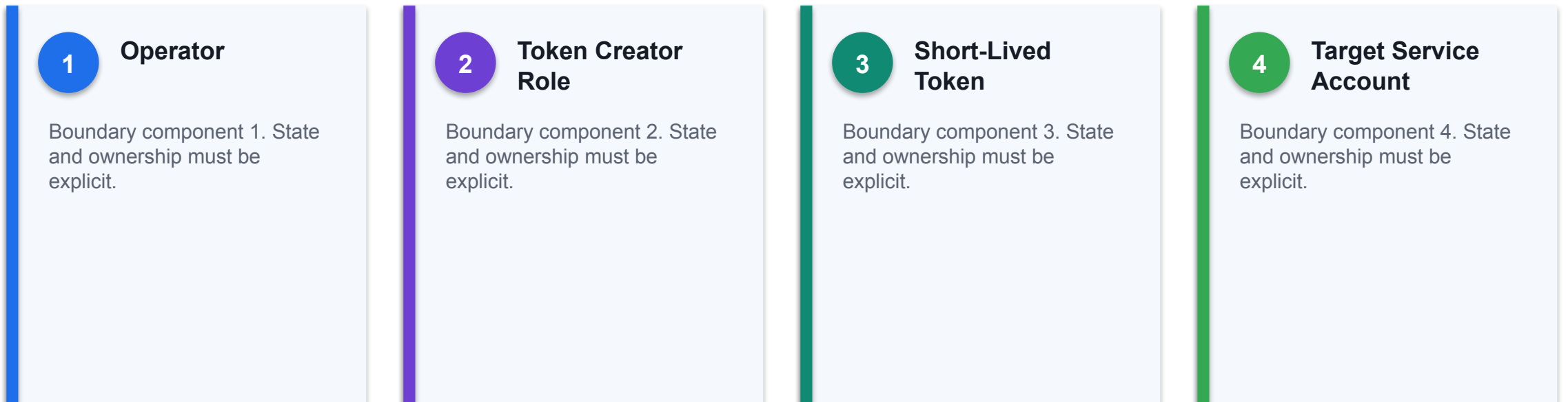
Observed output is captured



Timestamp and scope are recorded

Workload uses an attached user-managed service account; no long-lived key file is required.

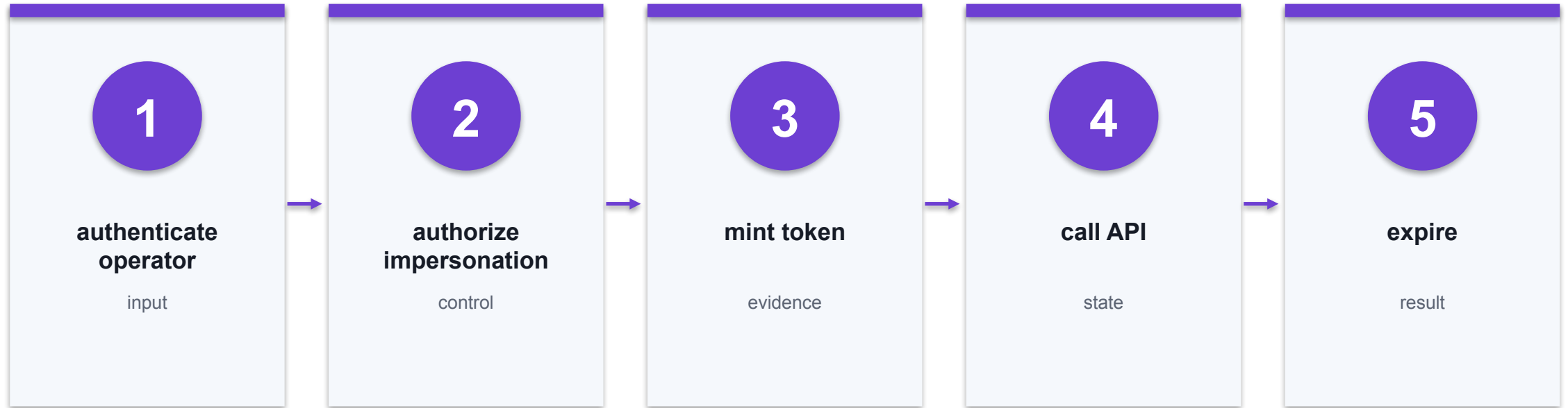
Service account impersonation · architecture



DECISION RULE

Scope first: Token is short-lived and the impersonation event is auditable.

Service account impersonation · runtime trace



Token is short-lived and the impersonation event is auditable.

Service account impersonation · concrete contract

```
gcloud auth print-access-token --impersonate-serviceQ1  
account=app-  
runtime@$PROJECT_ID.iam.gserviceaccount.com
```

1

Scope: GCP Basics and IAM

2

Mapped: A1 · K1

3

Capture stdout and exit status

4

Source: official service-accounts documentation

Service account impersonation · failure and control

FAILURE MODES



missing Token Creator



self-impersonation confusion



cached wrong identity

VERIFICATION EVIDENCE



Expected state is explicit



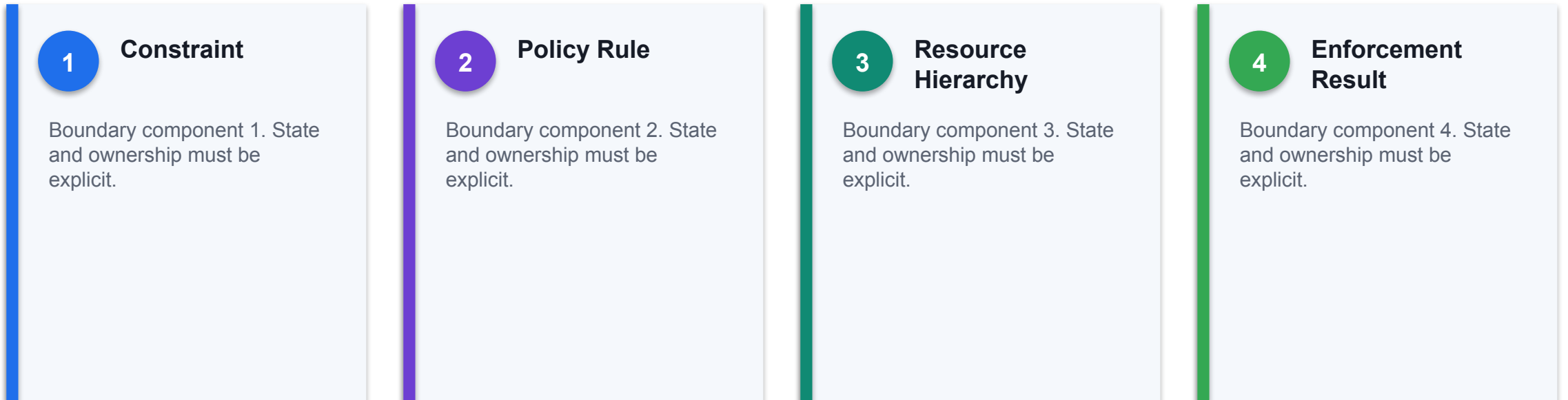
Observed output is captured



Timestamp and scope are recorded

Token is short-lived and the impersonation event is auditable.

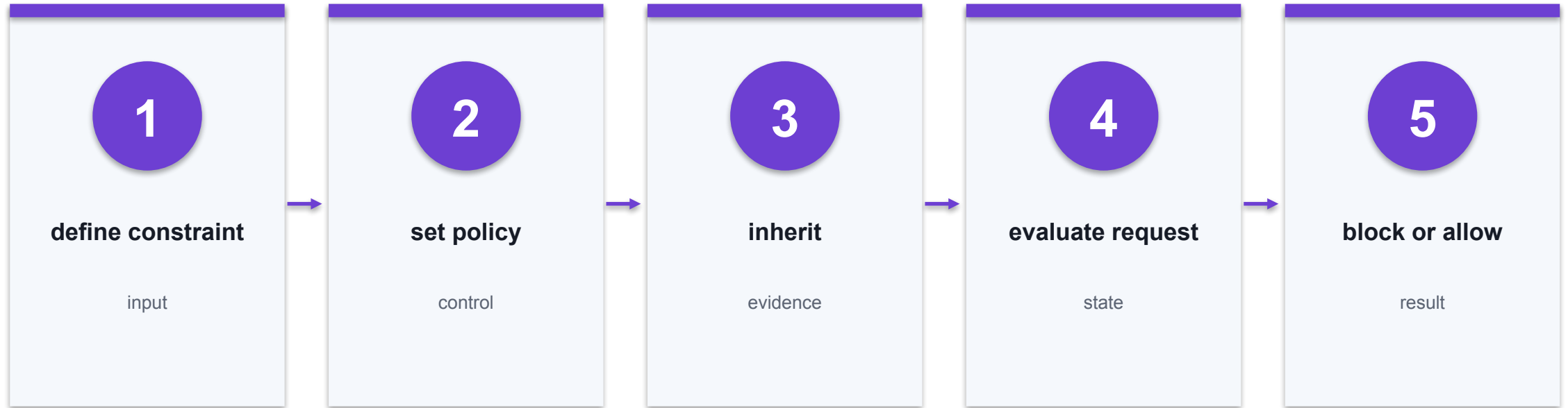
Organization policy and deny guardrails · architecture



DECISION RULE

Scope first: Policy state and inheritance explain why a request is allowed or rejected.

Organization policy and deny guardrails · runtime trace



Policy state and inheritance explain why a request is allowed or rejected.

Organization policy and deny guardrails · concrete contract

```
gcloud resource-manager org-policies describe 01  
constraints/compute.vmExternalIpAccess --  
project=$PROJECT_ID
```

1

Scope: GCP Basics and IAM

2

Mapped: A1 · K1

3

Capture stdout and exit status

4

Source: official org-policy documentation

Organization policy and deny guardrails · failure and control

FAILURE MODES



constraint unset



policy attached above project



exception not documented

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Policy state and inheritance explain why a request is allowed or rejected.

IAM Conditions and time-bound access · architecture

1

Role Binding

Boundary component 1. State and ownership must be explicit.

2

Cel Expression

Boundary component 2. State and ownership must be explicit.

3

Request Context

Boundary component 3. State and ownership must be explicit.

4

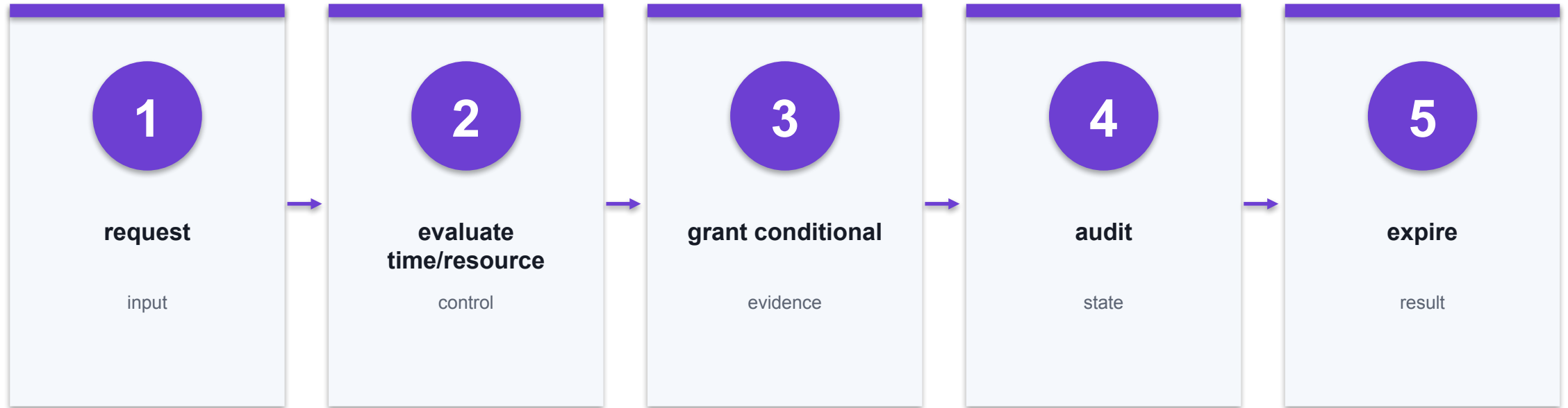
Resource

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Condition title, expression and scope match the intended temporary access window.

IAM Conditions and time-bound access · runtime trace



Condition title, expression and scope match the intended temporary access window.

IAM Conditions and time-bound access · concrete contract

```
gcloud projects get-iam-policy $PROJECT_ID -- 01  
format='table(bindings.role,bindings.members,bindings  
.condition) '
```

1

Scope: GCP Basics and IAM

2

Mapped: A1 · K1

3

Capture stdout and exit status

4

Source: official iam-conditions documentation

IAM Conditions and time-bound access · failure and control

FAILURE MODES



condition on unsupported role



timezone error



broad inherited grant remains

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Condition title, expression and scope match the intended temporary access window.

Audit evidence for access changes · architecture

1

Admin Activity Log

Boundary component 1. State and ownership must be explicit.

2

Principalemail

Boundary component 2. State and ownership must be explicit.

3

Methodname

Boundary component 3. State and ownership must be explicit.

4

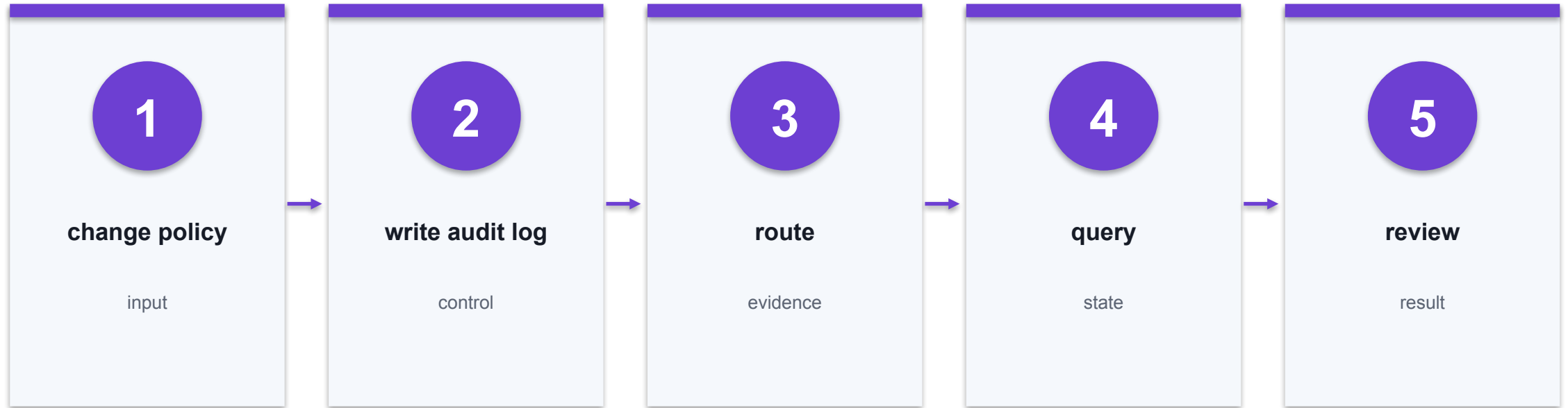
Resourcename

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Evidence identifies who changed which policy, when, and on which resource.

Audit evidence for access changes · runtime trace



Evidence identifies who changed which policy, when, and on which resource.

Audit evidence for access changes · concrete contract

```
gcloud logging read                                01  
'protoPayload.methodName:SetIamPolicy' --limit=20 --  
format=json
```

1

Scope: GCP Basics and IAM

2

Mapped: A1 · K1

3

Capture stdout and exit status

4

Source: official audit-logs documentation

Audit evidence for access changes · failure and control

FAILURE MODES



query too broad



wrong project



data-access logs not enabled where needed

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Evidence identifies who changed which policy, when, and on which resource.

Implement Least-Privilege IAM

SCENARIO

Create a workload service account, grant a narrow role and verify both allowed and denied actions.

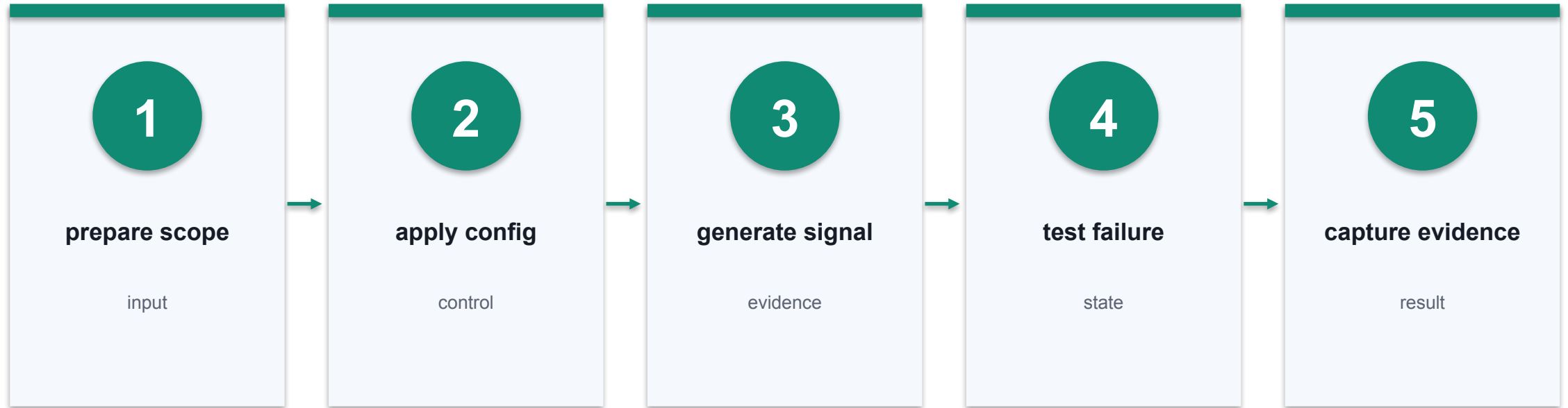
YOU WILL PRODUCE

iam-policy-before.json, iam-policy-after.json and access-test.txt

`iam_setup.sh`

Full click-by-click instructions are in the Learner Guide and Activity PDF. This slide shows the work product, not the procedure.

Activity 2 system trace



Create a workload service account, grant a narrow role and verify both allowed and denied actions.

Activity 2 worked artifact

```
#!/usr/bin/env bash 01
set -euo pipefail 02
SA=app-runtime@$PROJECT_ID.iam.gserviceaccount.com 03
gcloud iam service-accounts create app-runtime --display- 04
name='ACE app runtime'
gcloud projects add-iam-policy-binding "$PROJECT_ID" -- 05
member=serviceAccount:$SA --role=roles/storage.objectViewer
gcloud projects get-iam-policy "$PROJECT_ID" --format=json 06
iam-policy-after.json
```

1

Use placeholders for project-specific values

2

Run only in the isolated training project

3

Capture both success and failure output

4

Keep secrets out of submitted evidence

Activity 2 acceptance gate

FAILURE MODES



wrong project or region



insufficient IAM or disabled API



observed output does not meet target

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Accept when iam-policy-before.json, iam-policy-after.json and access-test.txt are complete, timestamped and reproducible.

TOPIC 03 · A3 · K4

03

Virtual Networks

Plan, deploy and manage networking

Global VPC and regional subnet model · architecture

1

Global Vpc

Boundary component 1. State and ownership must be explicit.

2

Regional Subnet

Boundary component 2. State and ownership must be explicit.

3

Route Table

Boundary component 3. State and ownership must be explicit.

4

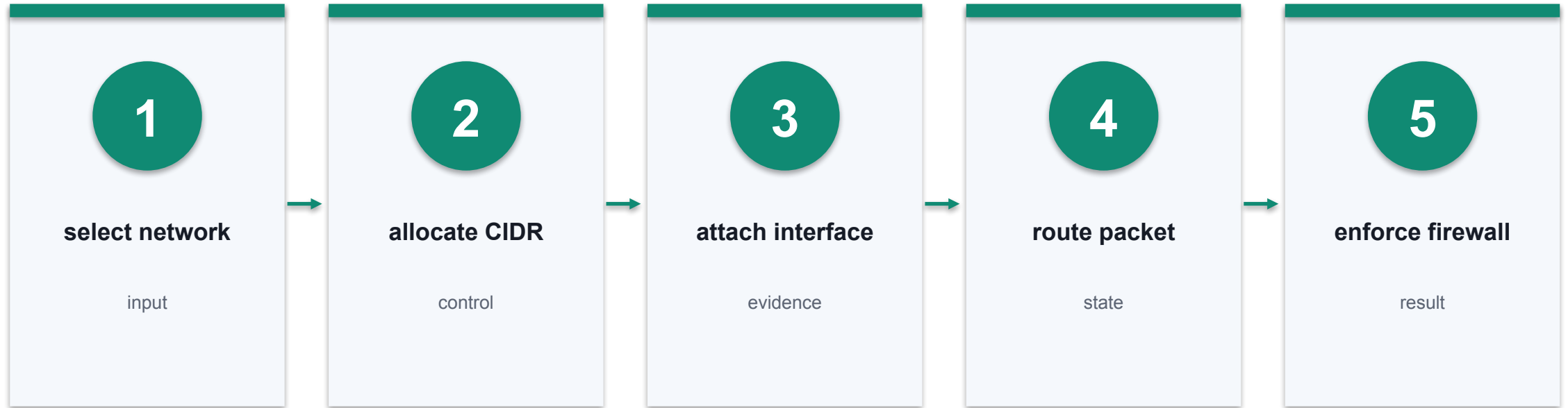
Firewall Policy

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Subnet range is unique, regional and visible in the intended global VPC.

Global VPC and regional subnet model · runtime trace



Subnet range is unique, regional and visible in the intended global VPC.

Global VPC and regional subnet model · concrete contract

```
gcloud compute networks create ace-vpc --subnet- 01
mode=custom
gcloud compute networks subnets create app-sg -- 02
network=ace-vpc --region=asia-southeast1 --
range=10.20.0.0/24
```

1

Scope: Virtual Networks

2

Mapped: A3 · K4

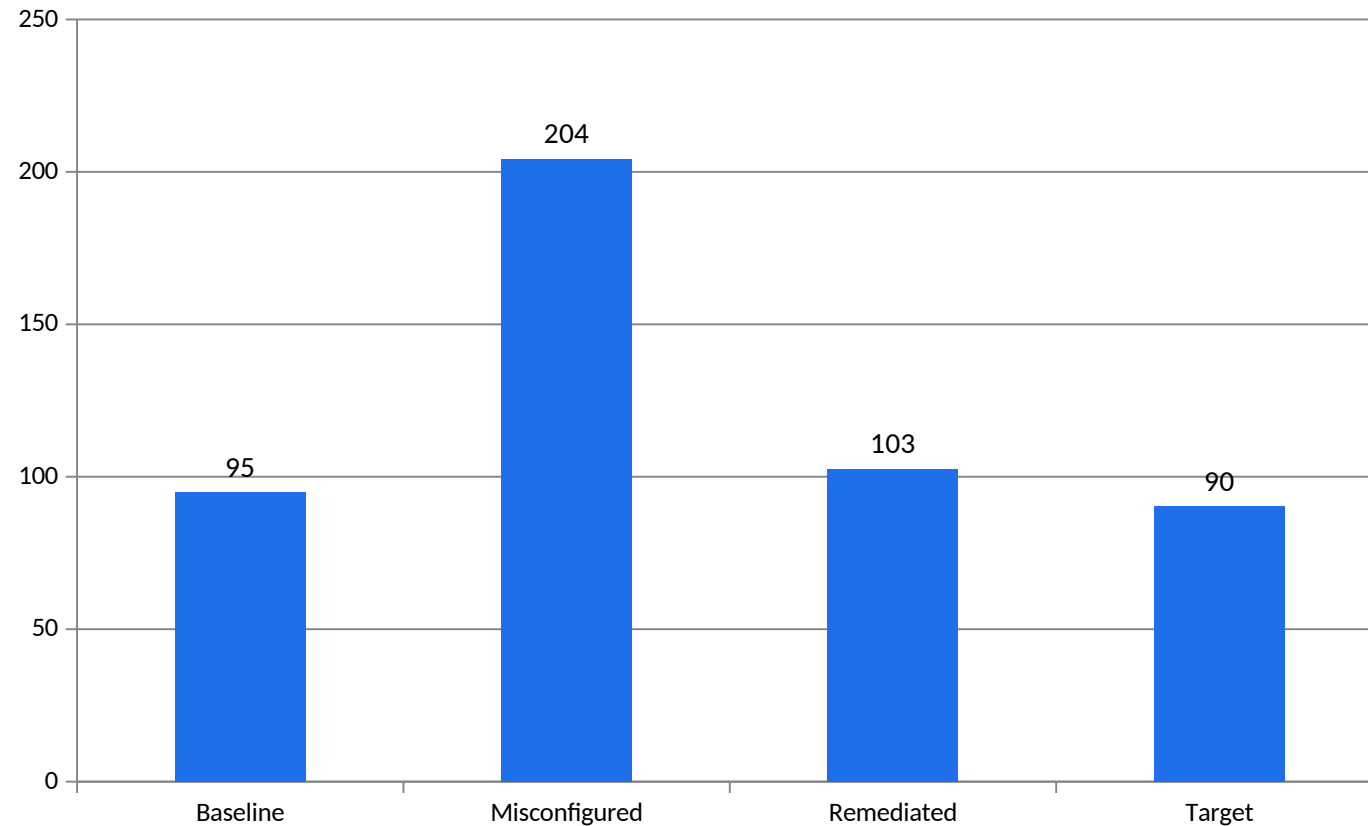
3

Capture stdout and exit status

4

Source: official vpc documentation

Topic 3 evidence profile



WHAT THE DATA SHOWS

A controlled fault should move the signal away from baseline. The remediation is accepted only when it returns close to target and the exact configuration change is recorded.

Indicative teaching data. Learners replace it with measurements from the aligned activity.

CIDR capacity and reserved addresses · architecture

1

Prefix Length

Boundary component 1. State and ownership must be explicit.

2

Host Bits

Boundary component 2. State and ownership must be explicit.

3

Primary Range

Boundary component 3. State and ownership must be explicit.

4

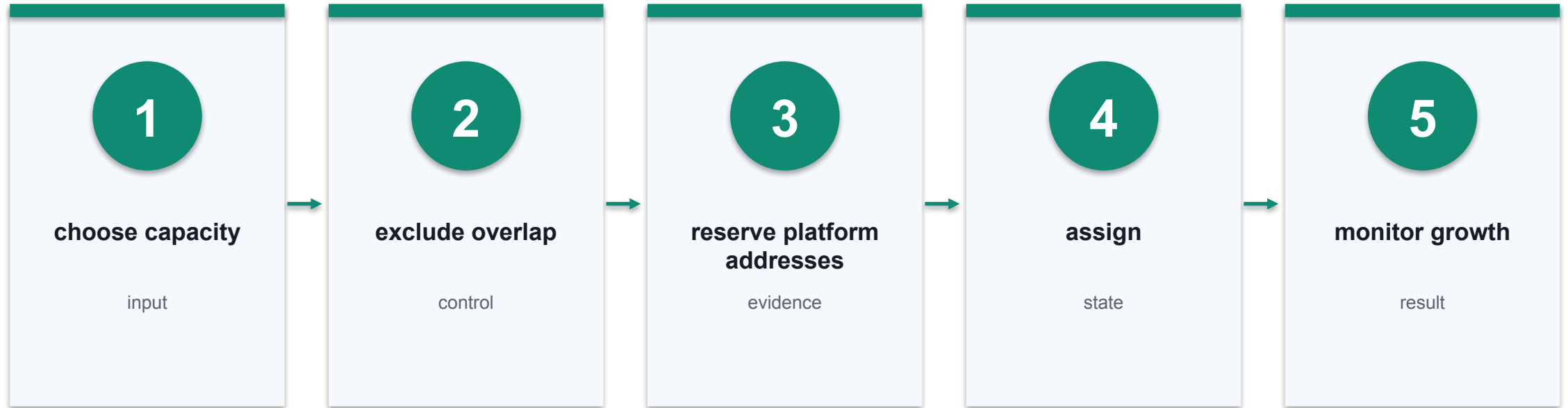
Secondary Range

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Address plan records total range, usable design capacity and expansion path.

CIDR capacity and reserved addresses · runtime trace



Address plan records total range, usable design capacity and expansion path.

CIDR capacity and reserved addresses · concrete contract

```
python3 -c 'import ipaddress;  
n=ipaddress.ip_network("10.20.0.0/24");  
print(n.num_addresses)'
```

01

1

Scope: Virtual Networks

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official vpc documentation

CIDR capacity and reserved addresses · failure and control

FAILURE MODES



undersized subnet



overlap with on-prem



secondary range omitted for GKE

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Address plan records total range, usable design capacity and expansion path.

Route selection and next hops · architecture

1

Destination Prefix

Boundary component 1. State and ownership must be explicit.

2

Priority

Boundary component 2. State and ownership must be explicit.

3

Next Hop

Boundary component 3. State and ownership must be explicit.

4

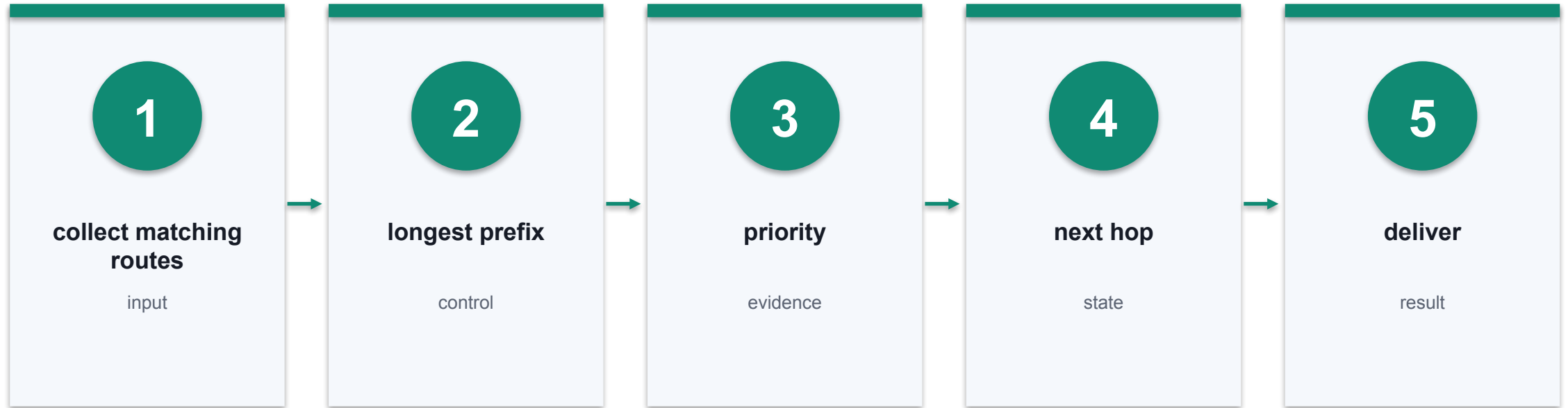
Instance Interface

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: The winning route has the most specific destination, then the applicable priority.

Route selection and next hops · runtime trace



The winning route has the most specific destination, then the applicable priority.

Route selection and next hops · concrete contract

```
gcloud compute routes list --filter='network:ace-vpc'  
--  
format='table(name,destRange,priority,nextHopGateway,  
nextHopInstance) '
```

1

Scope: Virtual Networks

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official routes documentation

Route selection and next hops · failure and control

FAILURE MODES



missing custom route



asymmetric return



route exists but firewall denies

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

The winning route has the most specific destination, then the applicable priority.

Firewall policy evaluation · architecture

1

Direction

Boundary component 1. State and ownership must be explicit.

2

Priority

Boundary component 2. State and ownership must be explicit.

3

Target

Boundary component 3. State and ownership must be explicit.

4

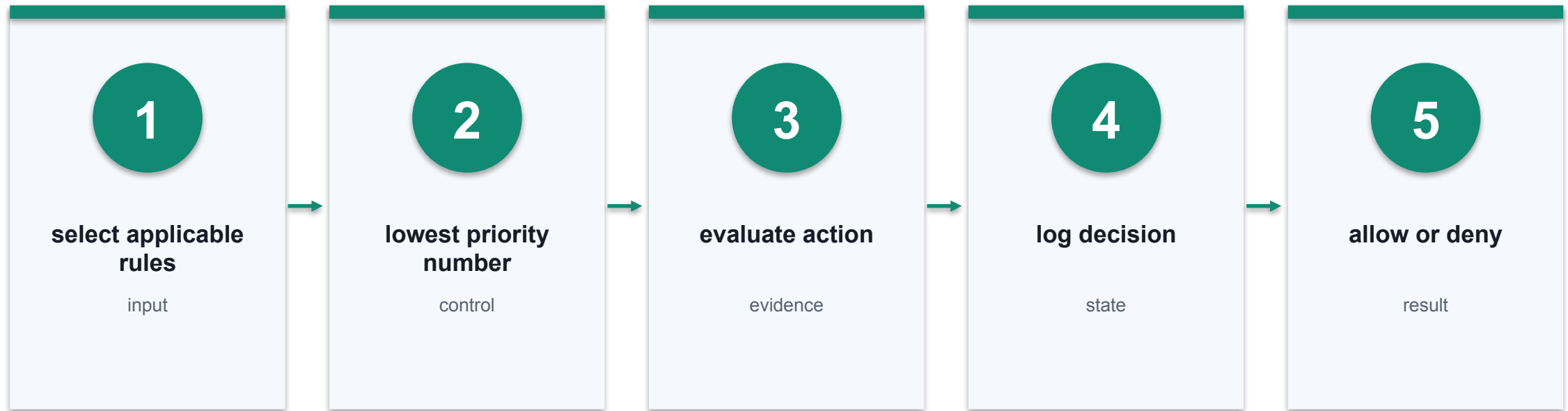
Source/ Destination

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Rule target and source range are narrower than 0.0.0.0/0 and firewall logging can prove the decision.

Firewall policy evaluation · runtime trace



Rule target and source range are narrower than 0.0.0.0/0 and firewall logging can prove the decision.

Firewall policy evaluation · concrete contract

```
gcloud compute firewall-rules create allow-web -- 01  
network=ace-vpc --direction=INGRESS --priority=1000  
--action=ALLOW --rules=tcp:80 --source-  
ranges=10.20.0.0/16 --target-tags=web
```

1

Scope: Virtual Networks

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official firewall documentation

Firewall policy evaluation · failure and control

FAILURE MODES



wrong network tag



higher-priority deny



implied ingress deny

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Rule target and source range are narrower than 0.0.0.0/0 and firewall logging can prove the decision.

Private Google Access and Cloud NAT · architecture

1

Internal-Only Vm

Boundary component 1. State and ownership must be explicit.

2

Subnet Setting

Boundary component 2. State and ownership must be explicit.

3

Google Apis

Boundary component 3. State and ownership must be explicit.

4

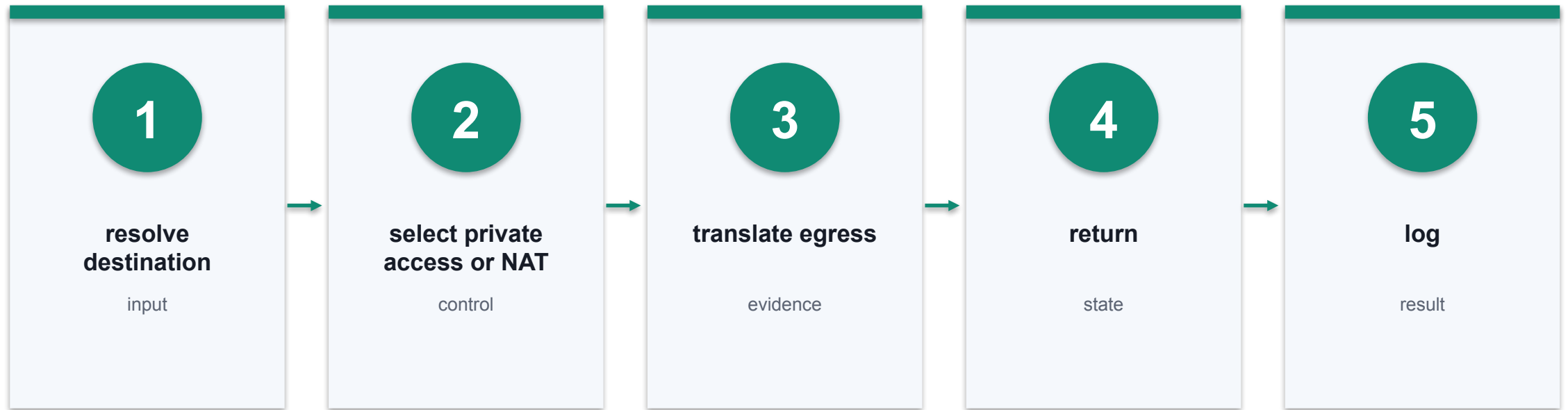
Cloud Nat

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: VM without external IP can reach intended Google APIs; general internet egress uses controlled NAT.

Private Google Access and Cloud NAT · runtime trace



VM without external IP can reach intended Google APIs; general internet egress uses controlled NAT.

Private Google Access and Cloud NAT · concrete contract

```
gcloud compute networks subnets update app-sg -- 01  
region=asia-southeast1 --enable-private-ip-google-  
access
```

1

Scope: Virtual Networks

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official nat documentation

Private Google Access and Cloud NAT · failure and control

FAILURE MODES



no default route



PGA disabled



NAT port exhaustion

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

VM without external IP can reach intended Google APIs; general internet egress uses controlled NAT.

VPC Flow Logs as packet evidence · architecture

1

5-Tuple

Boundary component 1. State and ownership must be explicit.

2

Reporter

Boundary component 2. State and ownership must be explicit.

3

Bytes

Boundary component 3. State and ownership must be explicit.

4

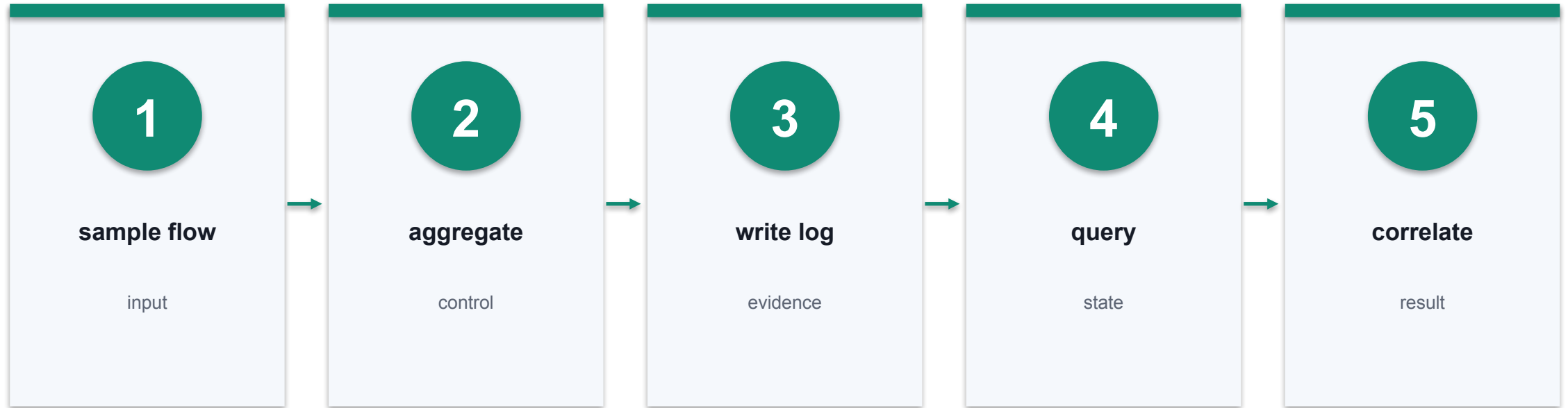
Action Context

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Flow record supports a connectivity hypothesis with source, destination, ports and bytes.

VPC Flow Logs as packet evidence · runtime trace



Flow record supports a connectivity hypothesis with source, destination, ports and bytes.

VPC Flow Logs as packet evidence · concrete contract

```
gcloud logging read 'logName:vpc_flows AND  
jsonPayload.connection.src_ip="10.20.0.10"' --  
limit=20
```

01

1

Scope: Virtual Networks

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official flow-logs documentation

VPC Flow Logs as packet evidence · failure and control

FAILURE MODES



sampling hides short flow



query wrong log bucket



firewall log confused with flow log

VERIFICATION EVIDENCE



Expected state is explicit



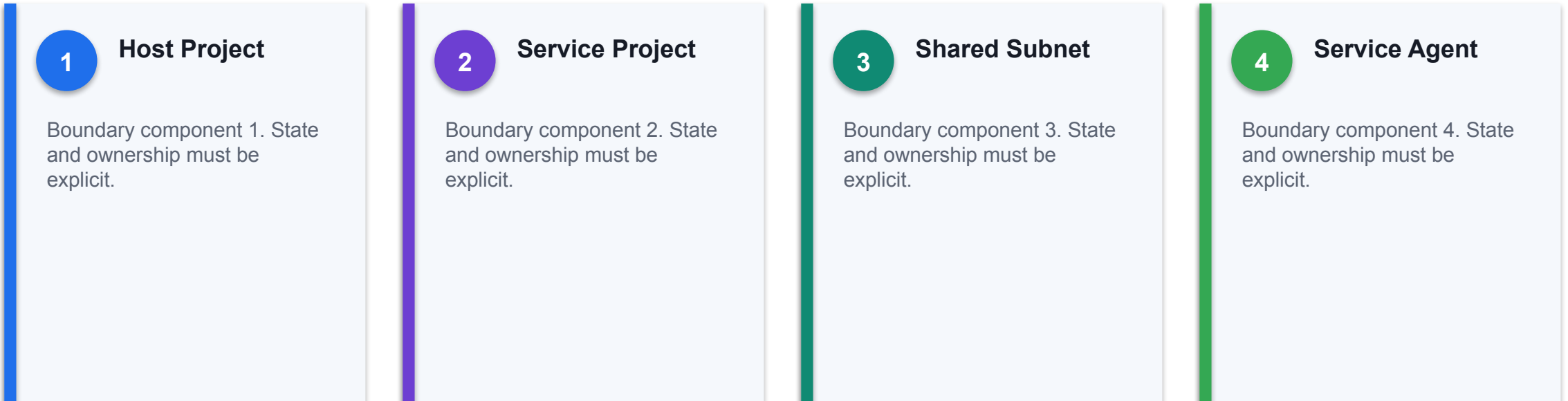
Observed output is captured



Timestamp and scope are recorded

Flow record supports a connectivity hypothesis with source, destination, ports and bytes.

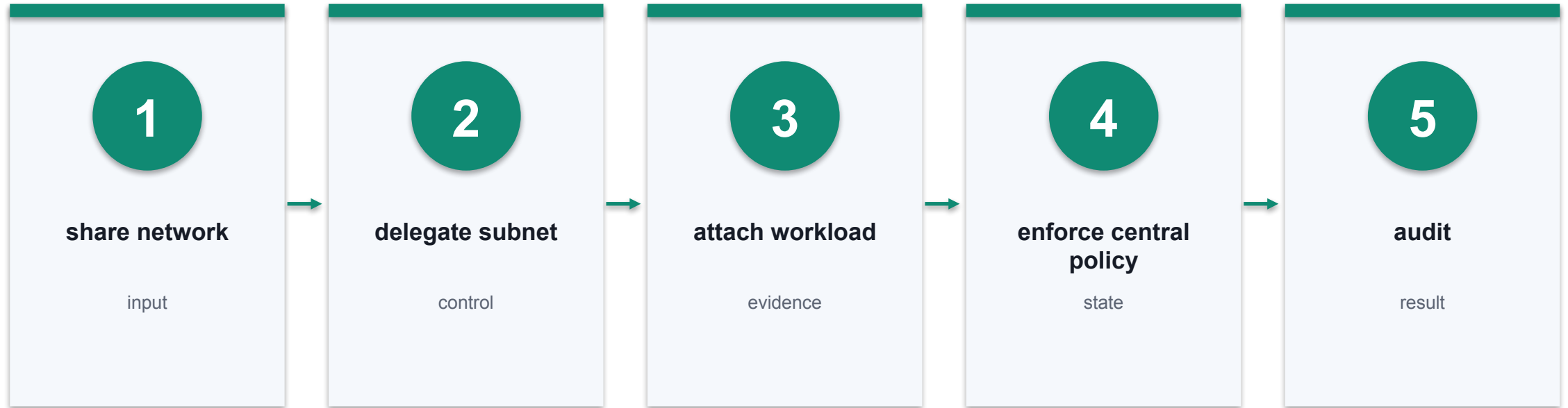
Shared VPC ownership boundaries · architecture



DECISION RULE

Scope first: Network ownership and workload ownership are separately documented and permissioned.

Shared VPC ownership boundaries · runtime trace



Network ownership and workload ownership are separately documented and permissioned.

Shared VPC ownership boundaries · concrete contract

```
gcloud compute shared-vpc get-host-project  
$PROJECT_ID
```

01

1

Scope: Virtual Networks

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official shared-vpc documentation

Shared VPC ownership boundaries · failure and control

FAILURE MODES



service project not associated



missing networkUser role



subnet in wrong host

VERIFICATION EVIDENCE



Expected state is explicit



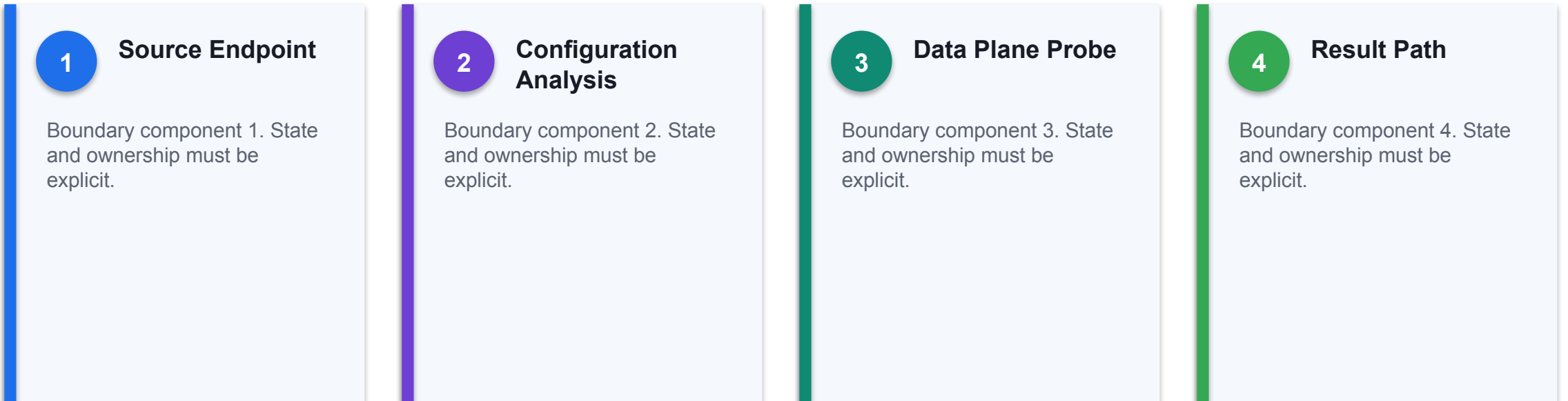
Observed output is captured



Timestamp and scope are recorded

Network ownership and workload ownership are separately documented and permissioned.

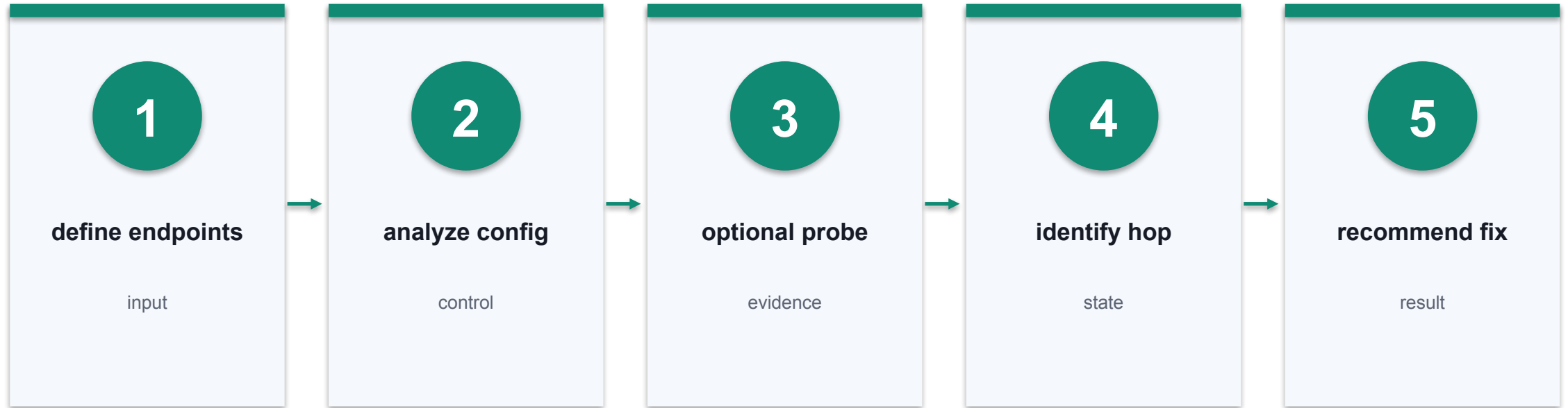
Connectivity Tests diagnostic graph · architecture



DECISION RULE

Scope first: Test result names the failing hop or policy instead of relying on ping alone.

Connectivity Tests diagnostic graph · runtime trace



Test result names the failing hop or policy instead of relying on ping alone.

Connectivity Tests diagnostic graph · concrete contract

```
gcloud network-management connectivity-tests list --01  
format='table(name,source, destination)'
```

1

Scope: Virtual Networks

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official connectivity-tests documentation

Connectivity Tests diagnostic graph · failure and control

FAILURE MODES



ICMP blocked but TCP works



stale test



source identity lacks visibility

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Test result names the failing hop or policy instead of relying on ping alone.

Build and Diagnose a Custom VPC

SCENARIO

Build a custom VPC and regional subnets, apply narrow firewall rules and diagnose a failed connection.

YOU WILL PRODUCE

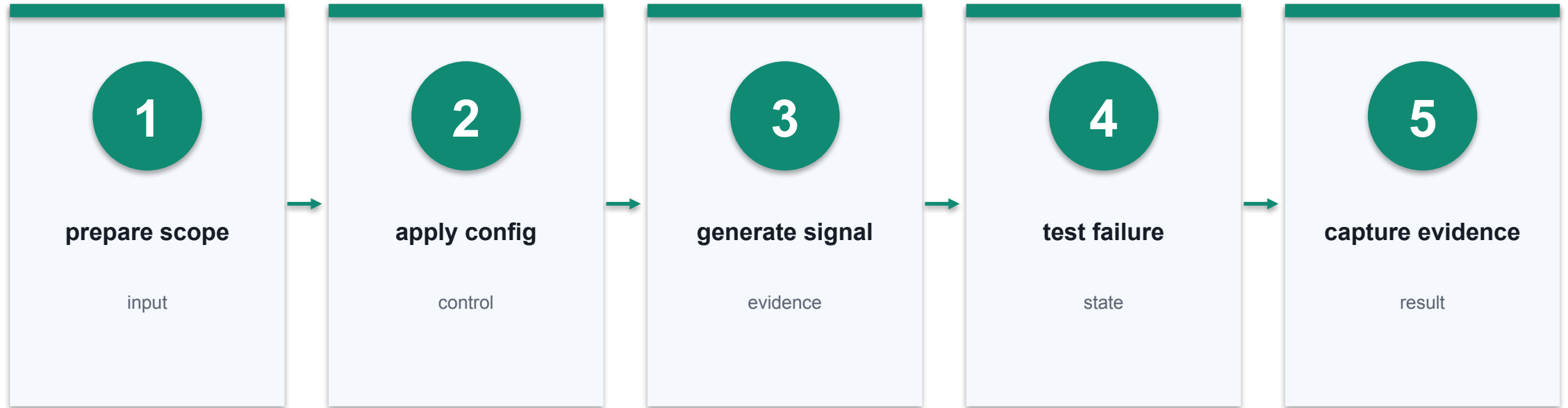
network inventory, connectivity-test result and firewall evidence

`network_setup.sh`

`network_plan.csv`

Full click-by-click instructions are in the Learner Guide and Activity PDF. This slide shows the work product, not the procedure.

Activity 3 system trace



Build a custom VPC and regional subnets, apply narrow firewall rules and diagnose a failed connection.

Activity 3 worked artifact

```
#!/usr/bin/env bash 01
set -euo pipefail 02

gcloud compute networks create ace-vpc --subnet- 03
mode=custom
gcloud compute networks subnets create app-sg -- 04
network=ace-vpc --region=asia-southeast1 --
gcloud compute firewall-rules create allow-web- 05
range=10.20.0.0/24 --enable-private-ip-google-access
internal --network=ace-vpc --allow=tcp:80 --source-
ranges=10.20.0.0/16 --target-tags=web --enable-
logging
```

1

Use placeholders for project-specific values

2

Run only in the isolated training project

3

Capture both success and failure output

4

Keep secrets out of submitted evidence

Activity 3 acceptance gate

FAILURE MODES



wrong project or region



insufficient IAM or disabled API



observed output does not meet target

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Accept when network inventory, connectivity-test result and firewall evidence are complete, timestamped and reproducible.

Lunch and resource check

Before you step away

Stop test traffic

Check active resources

Save evidence

Do not leave tokens on screen

Day 1 evidence recap

1

Topic 1

Google Cloud and Cloud Computing

2

Topic 2

GCP Basics and IAM

3

Topic 3

Virtual Networks

4

Evidence

Record expected versus observed output.

DECISION RULE

A configuration is not complete until its observable state matches the acceptance criterion.

TOPIC 04 · A3 · K4

04

Virtual Machines

Deploy and operate Compute Engine

Instance contract and dependencies · architecture

1

Machine Type

Boundary component 1. State and ownership must be explicit.

2

Boot Disk Image

Boundary component 2. State and ownership must be explicit.

3

Nic

Boundary component 3. State and ownership must be explicit.

4

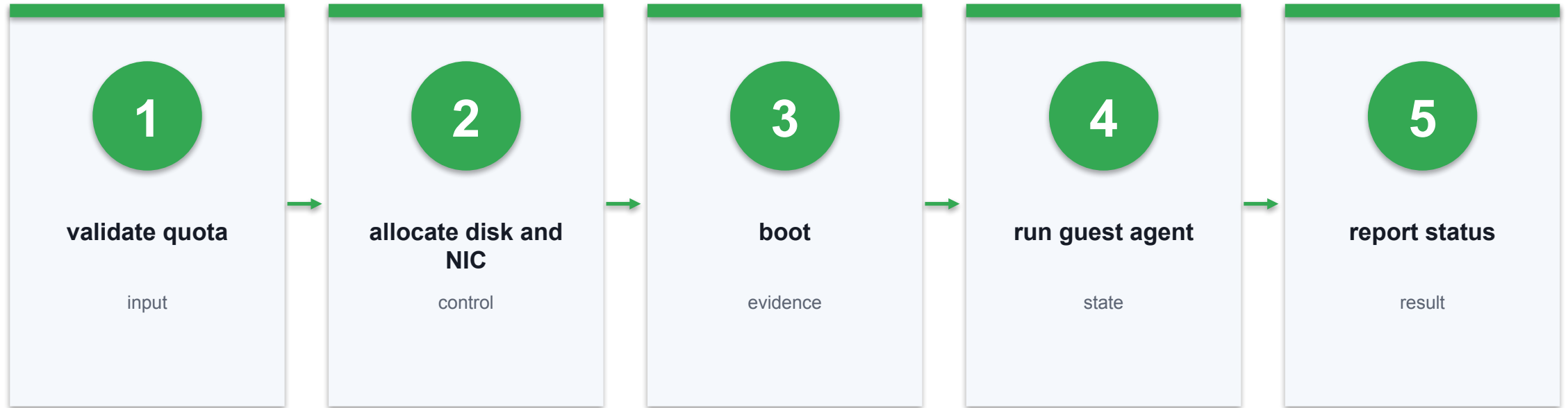
Service Account

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Instance RUNNING state is paired with the intended zone, network, disk and service account.

Instance contract and dependencies · runtime trace



Instance RUNNING state is paired with the intended zone, network, disk and service account.

Instance contract and dependencies · concrete contract

```
gcloud compute instances create web-1 --zone=asia-01  
southeast1-b --machine-type=e2-micro --subnet=app-sg  
--no-address --service-account=app-  
runtime@$PROJECT_ID.iam.gserviceaccount.com --  
scopes=cloud-platform
```

1

Scope: Virtual Machines

2

Mapped: A3 · K4

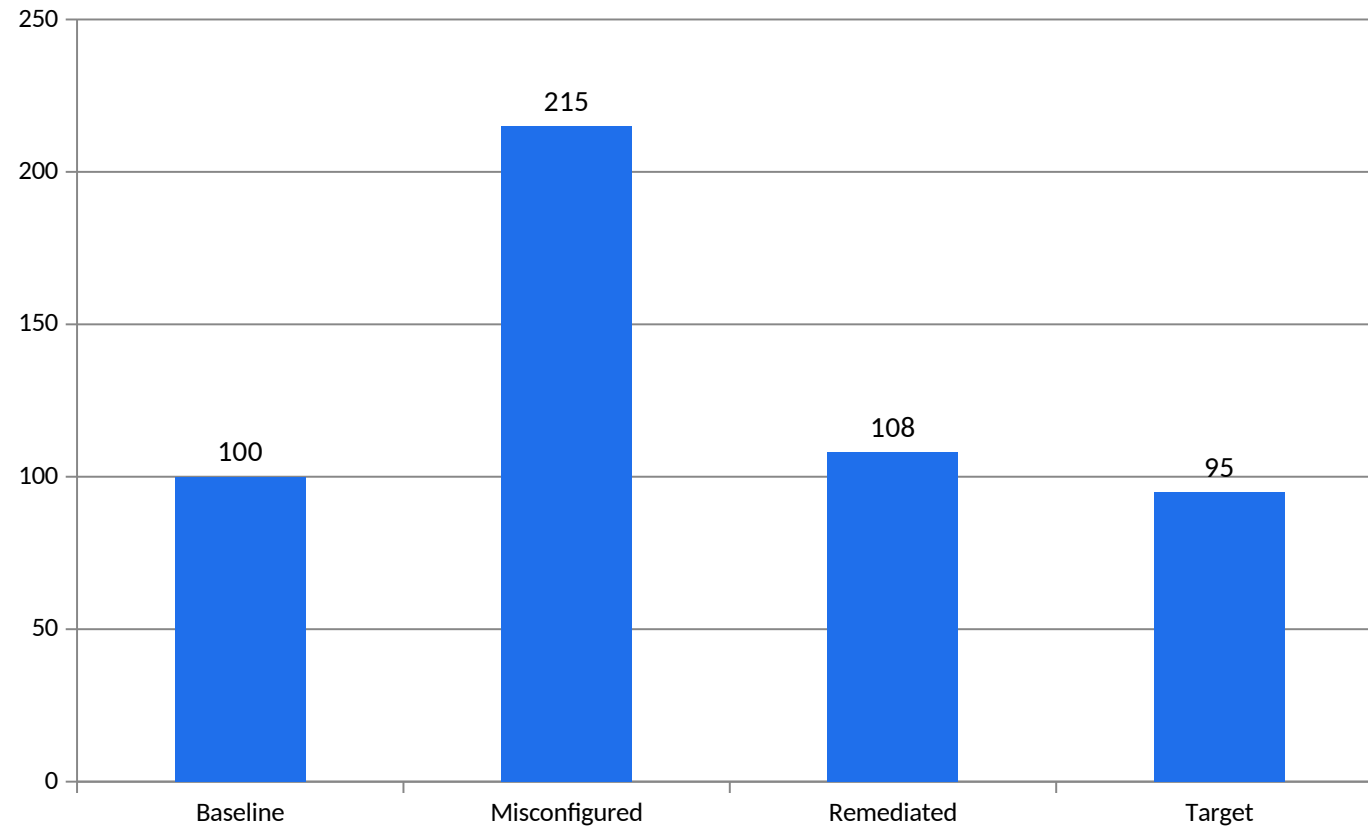
3

Capture stdout and exit status

4

Source: official compute documentation

Topic 4 evidence profile



WHAT THE DATA SHOWS

A controlled fault should move the signal away from baseline. The remediation is accepted only when it returns close to target and the exact configuration change is recorded.

Indicative teaching data. Learners replace it with measurements from the aligned activity.

Machine families and right-sizing · architecture

1

General Purpose

Boundary component 1. State and ownership must be explicit.

2

Compute Optimized

Boundary component 2. State and ownership must be explicit.

3

Memory Optimized

Boundary component 3. State and ownership must be explicit.

4

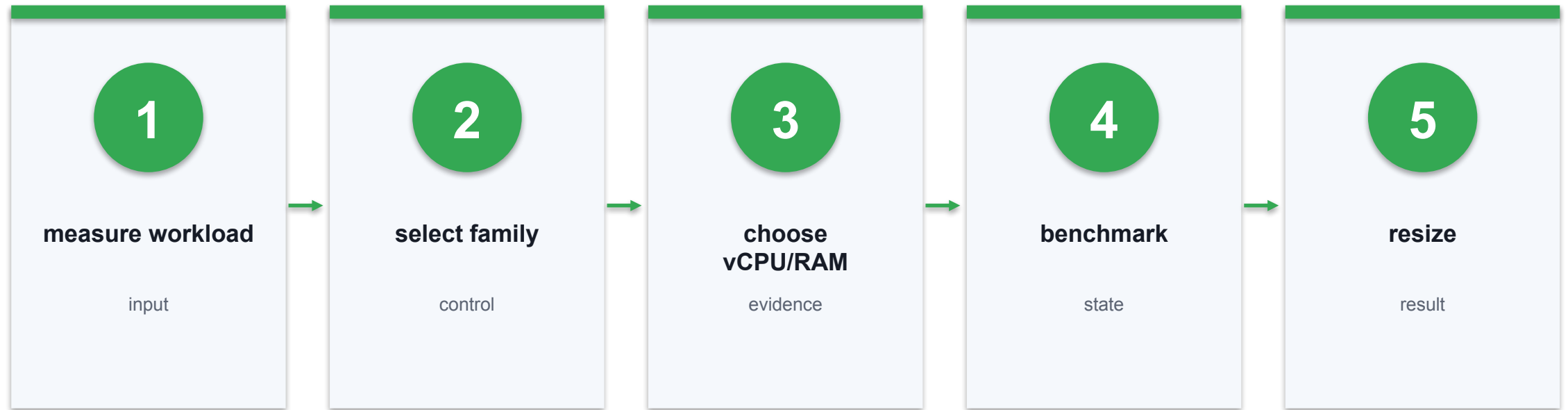
Accelerator

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Selection is justified by measured CPU, memory, latency and cost constraints.

Machine families and right-sizing · runtime trace



Selection is justified by measured CPU, memory, latency and cost constraints.

Machine families and right-sizing · concrete contract

```
gcloud compute machine-types describe e2-standard-2 01  
--zone=asia-southeast1-b
```

1

Scope: Virtual Machines

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official machine-types documentation

Machine families and right-sizing · failure and control

FAILURE MODES



CPU platform assumption



oversizing



quota not available in zone

VERIFICATION EVIDENCE



Expected state is explicit



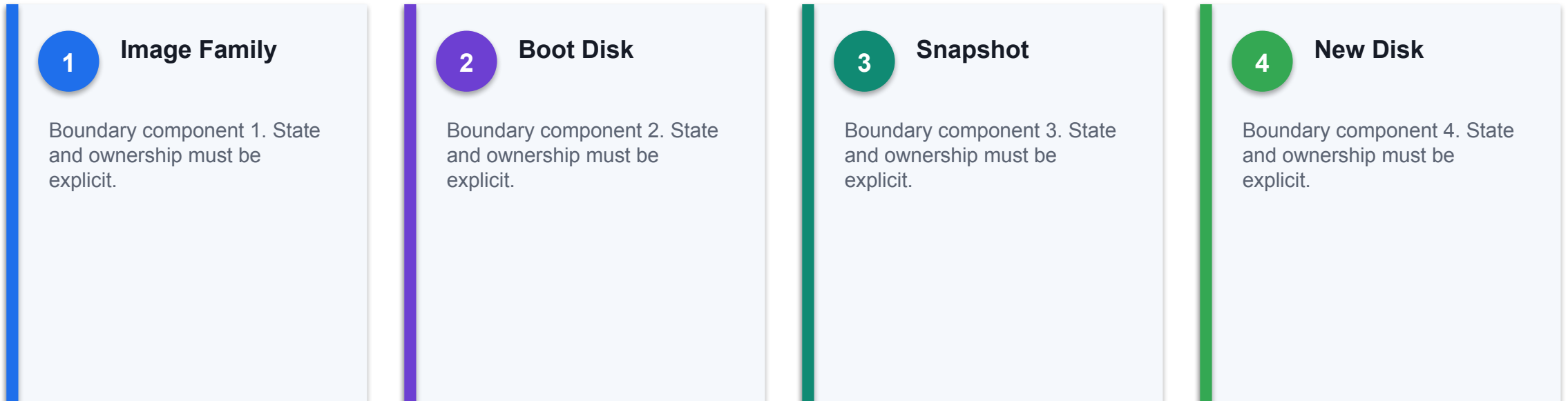
Observed output is captured



Timestamp and scope are recorded

Selection is justified by measured CPU, memory, latency and cost constraints.

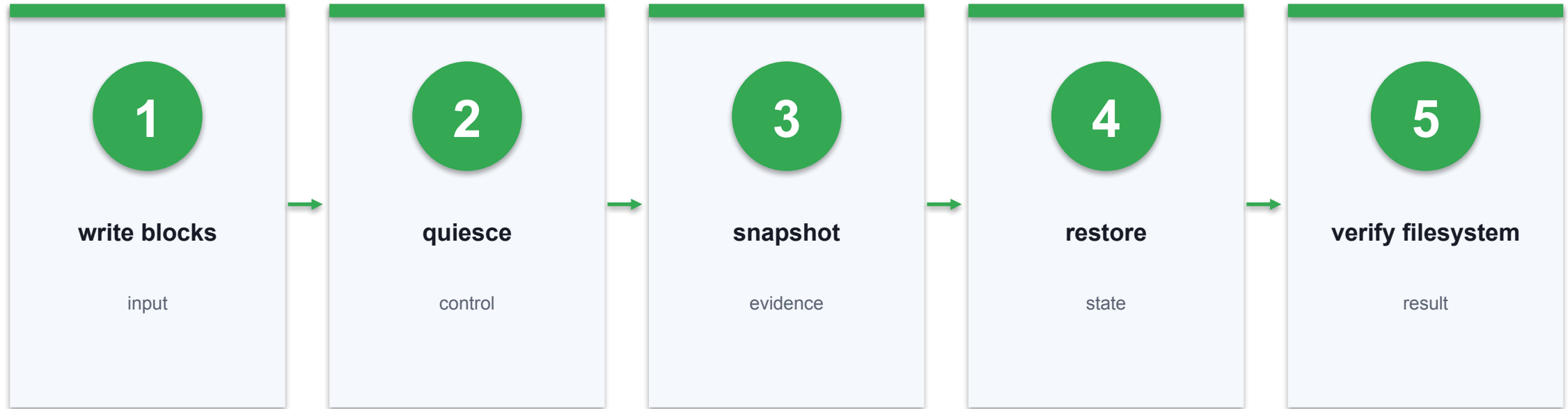
Images, snapshots and persistent disks · architecture



DECISION RULE

Scope first: Snapshot has a source disk, creation time and restorable status before a risky change.

Images, snapshots and persistent disks · runtime trace



Snapshot has a source disk, creation time and restorable status before a risky change.

Images, snapshots and persistent disks · concrete contract

```
gcloud compute disks snapshot web-1 --zone=asia-southeast1-b --snapshot-names=web-1-prechange 01
```

1

Scope: Virtual Machines

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official snapshots documentation

Images, snapshots and persistent disks · failure and control

FAILURE MODES



crash-consistent only



wrong source disk



snapshot schedule absent

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Snapshot has a source disk, creation time and restorable status before a risky change.

Startup scripts and metadata execution · architecture

1

Instance Metadata

Boundary component 1. State and ownership must be explicit.

2

Guest Agent

Boundary component 2. State and ownership must be explicit.

3

Startup Script

Boundary component 3. State and ownership must be explicit.

4

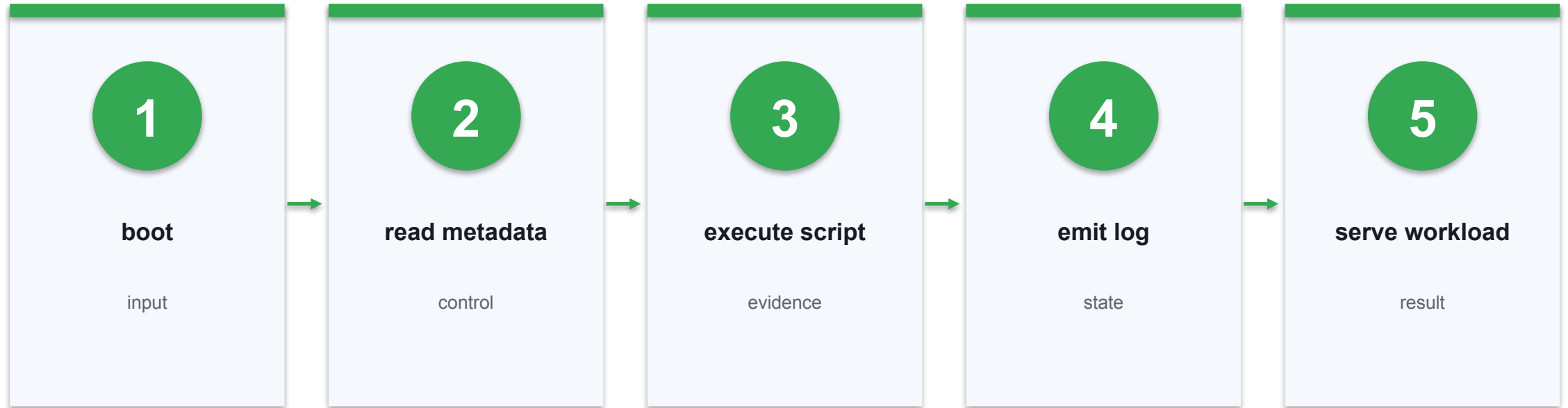
Serial Console

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Serial console or journal output proves script execution and idempotent configuration.

Startup scripts and metadata execution · runtime trace



Serial console or journal output proves script execution and idempotent configuration.

Startup scripts and metadata execution · concrete contract

```
gcloud compute instances add-metadata web-1 -- 01  
zone=asia-southeast1-b --metadata-from-file=startup-  
script=startup.sh
```

1

Scope: Virtual Machines

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official startup-scripts documentation

Startup scripts and metadata execution · failure and control

FAILURE MODES



script not executable



package repository unavailable



non-idempotent rerun

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Serial console or journal output proves script execution and idempotent configuration.

OS Login and SSH authorization · architecture

1

Iam Principal

Boundary component 1. State and ownership must be explicit.

2

Os Login Role

Boundary component 2. State and ownership must be explicit.

3

Posix Account

Boundary component 3. State and ownership must be explicit.

4

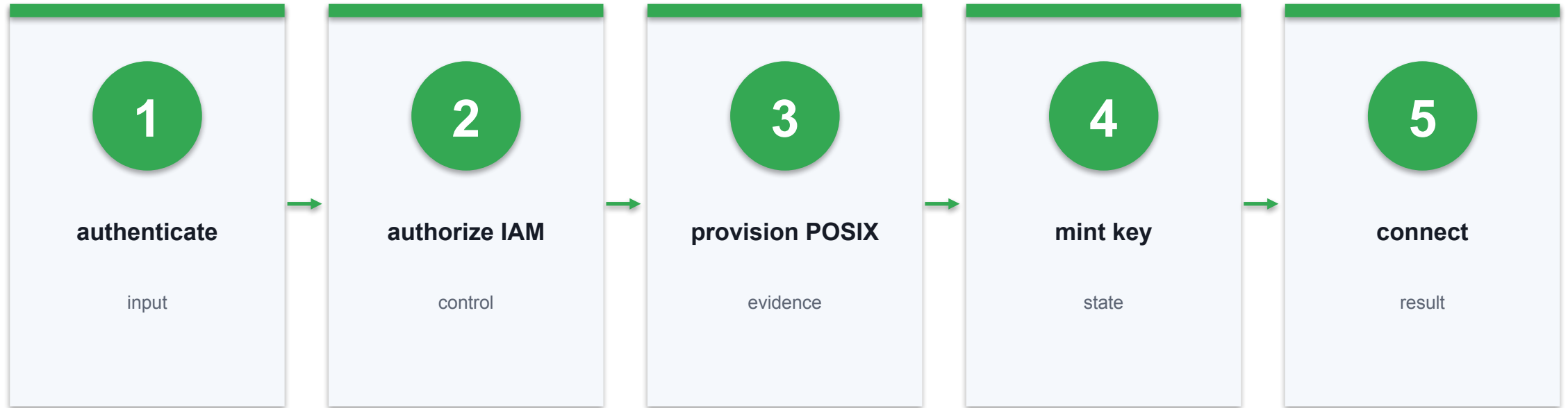
Ssh Certificate

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Access succeeds without exposing TCP 22 publicly and is tied to an identifiable principal.

OS Login and SSH authorization · runtime trace



Access succeeds without exposing TCP 22 publicly and is tied to an identifiable principal.

OS Login and SSH authorization · concrete contract

```
gcloud compute ssh web-1 --zone=asia-southeast1-b --01  
tunnel-through-iap
```

1

Scope: Virtual Machines

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official os-login documentation

OS Login and SSH authorization - failure and control

FAILURE MODES



OS Login disabled



IAP firewall missing



user lacks tunnel role

VERIFICATION EVIDENCE



Expected state is explicit



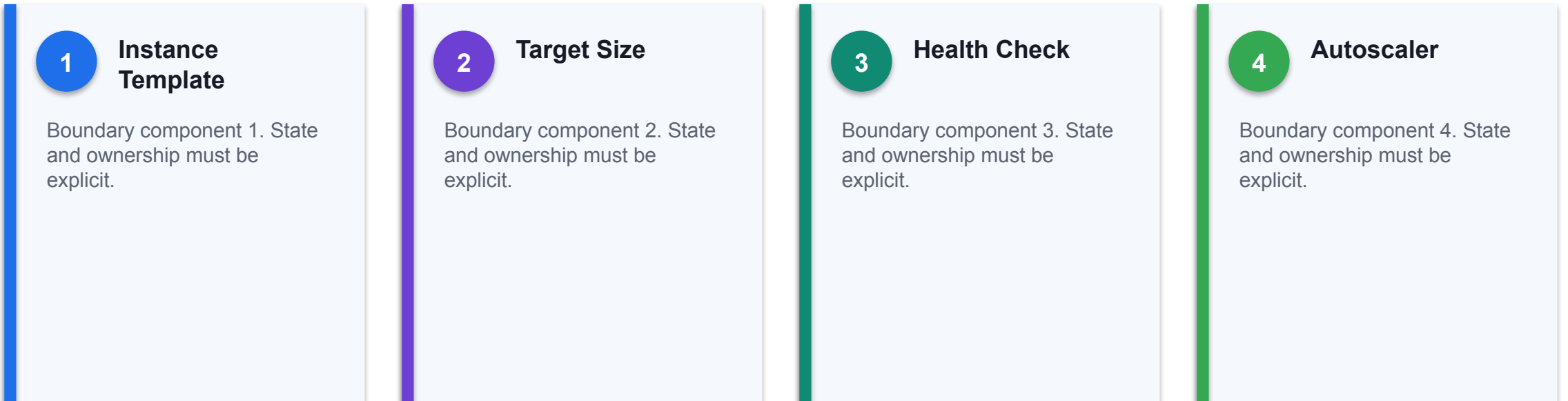
Observed output is captured



Timestamp and scope are recorded

Access succeeds without exposing TCP 22 publicly and is tied to an identifiable principal.

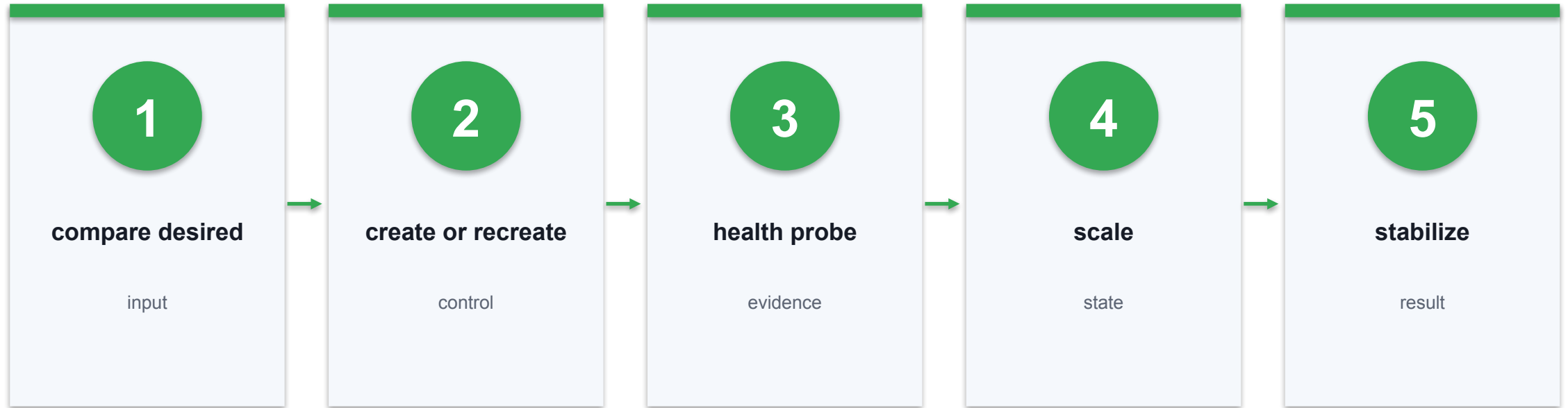
Managed instance group control loop · architecture



DECISION RULE

Scope first: All instances are current, healthy and derived from the intended template version.

Managed instance group control loop · runtime trace



All instances are current, healthy and derived from the intended template version.

Managed instance group control loop · concrete contract

```
gcloud compute instance-groups managed list-instances  
web-mig --region=asia-southeast1
```

1

Scope: Virtual Machines

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official mig documentation

Managed instance group control loop · failure and control

FAILURE MODES



template drift



health check blocked



autoscaler cool-down too short

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

All instances are current, healthy and derived from the intended template version.

Spot VM interruption-aware design · architecture

1 Spot Provision Model

Boundary component 1. State and ownership must be explicit.

2 Termination Signal

Boundary component 2. State and ownership must be explicit.

3 Checkpoint

Boundary component 3. State and ownership must be explicit.

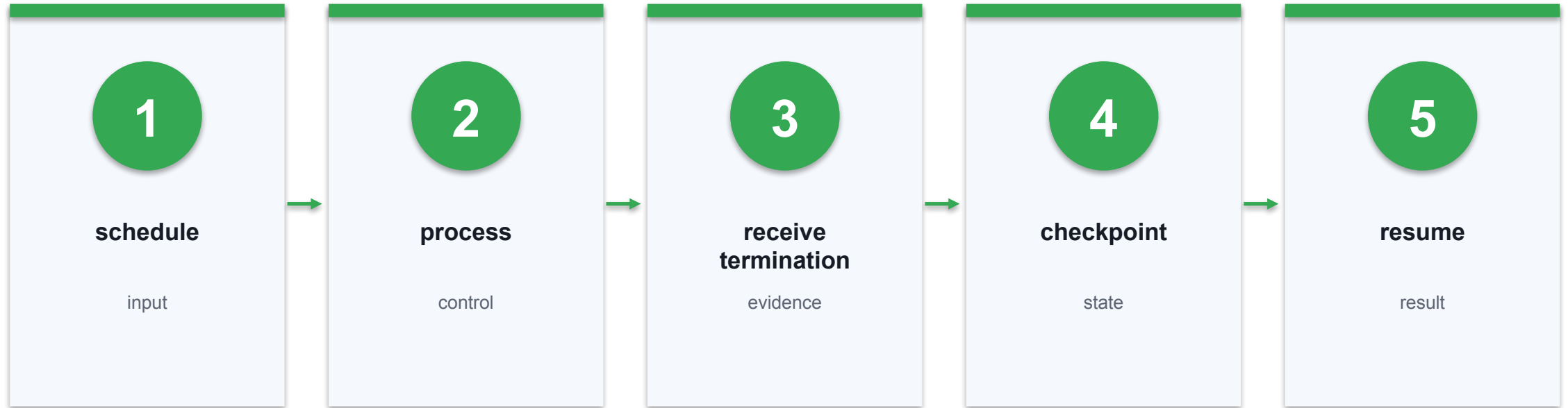
4 Durable Store

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Workload tolerates interruption and persists progress outside the instance.

Spot VM interruption-aware design · runtime trace



Workload tolerates interruption and persists progress outside the instance.

Spot VM interruption-aware design · concrete contract

```
gcloud compute instances describe batch-spot -- 01  
zone=asia-southeast1-b --  
format='value(scheduling.provisioningModel,scheduling  
.instanceTerminationAction) '
```

1

Scope: Virtual Machines

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official spot documentation

Spot VM interruption-aware design · failure and control

FAILURE MODES



state on local disk



no retry queue



capacity unavailable

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Workload tolerates interruption and persists progress outside the instance.

VM incident evidence bundle · architecture

1

Instance Status

Boundary component 1. State and ownership must be explicit.

2

Serial Logs

Boundary component 2. State and ownership must be explicit.

3

Guest Metrics

Boundary component 3. State and ownership must be explicit.

4

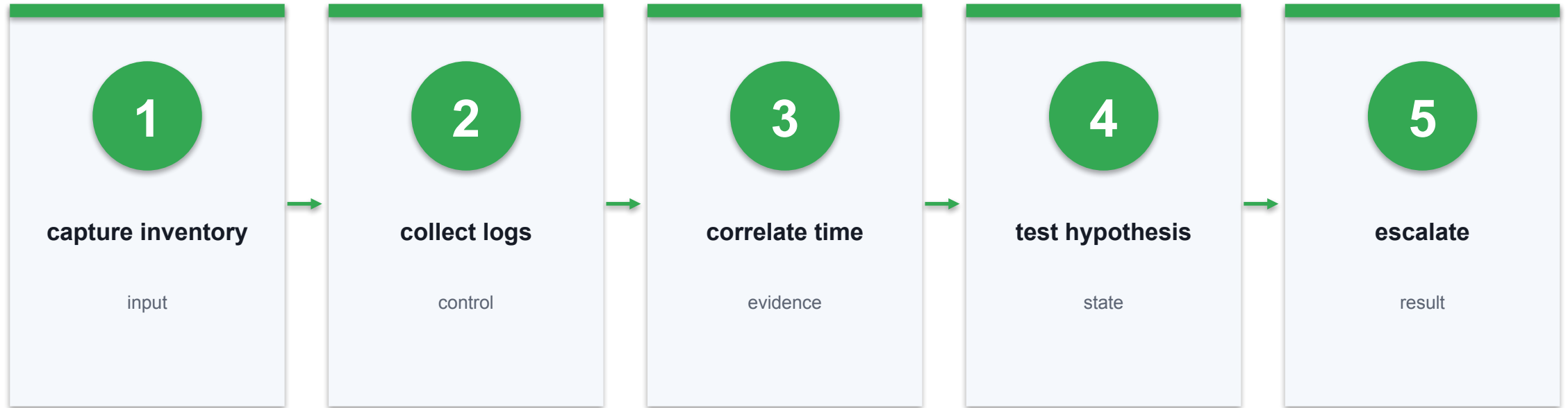
Audit Change

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Escalation includes timestamps, recent changes, expected/observed behaviour and exact commands.

VM incident evidence bundle · runtime trace



Escalation includes timestamps, recent changes, expected/observed behaviour and exact commands.

VM incident evidence bundle · concrete contract

```
gcloud compute instances get-serial-port-output web-01  
--zone=asia-southeast1-b --port=1 > serial.log
```

1

Scope: Virtual Machines

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official troubleshooting documentation

VM incident evidence bundle · failure and control

FAILURE MODES



reboot before evidence



timezone mismatch



missing reproduction

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Escalation includes timestamps, recent changes, expected/observed behaviour and exact commands.

Deploy and Recover a Compute Engine Web VM

SCENARIO

Deploy a private web VM with a user-managed service account, test the startup script and recover from a bad change.

YOU WILL PRODUCE

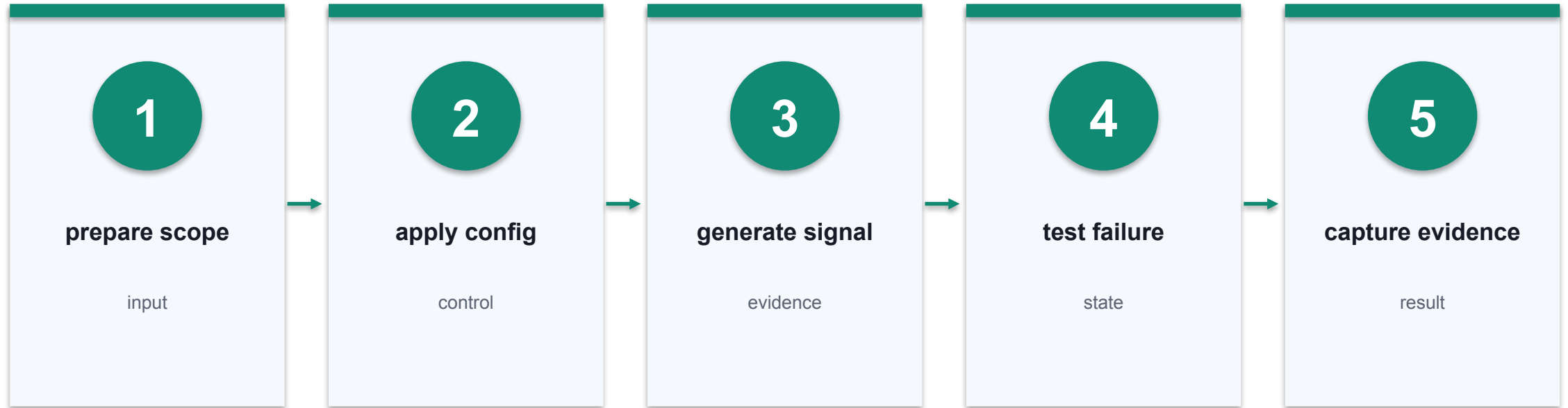
startup logs, a pre-change snapshot and recovery verification

`startup.sh`

`verify_vm.sh`

Full click-by-click instructions are in the Learner Guide and Activity PDF. This slide shows the work product, not the procedure.

Activity 4 system trace



Deploy a private web VM with a user-managed service account, test the startup script and recover from a bad change.

Activity 4 worked artifact

```
#!/usr/bin/env bash 01
set -euo pipefail 02
apt-get update 03
apt-get install -y nginx 04
echo 'ACE activity healthy' > /var/www/html/index.html 05
systemctl enable --now nginx 06
```

1

Use placeholders for project-specific values

2

Run only in the isolated training project

3

Capture both success and failure output

4

Keep secrets out of submitted evidence

Activity 4 acceptance gate

FAILURE MODES



wrong project or region



insufficient IAM or disabled API



observed output does not meet target

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Accept when startup logs, a pre-change snapshot and recovery verification are complete, timestamped and reproducible.

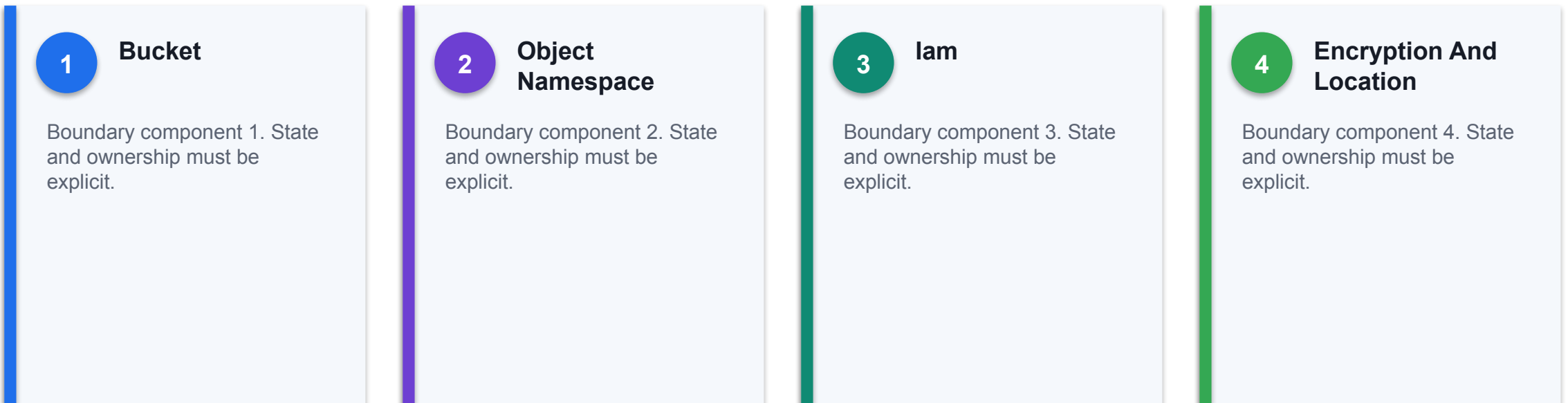
TOPIC 05 · A4 · K5

05

Storage and Databases

Plan, deploy and manage data solutions

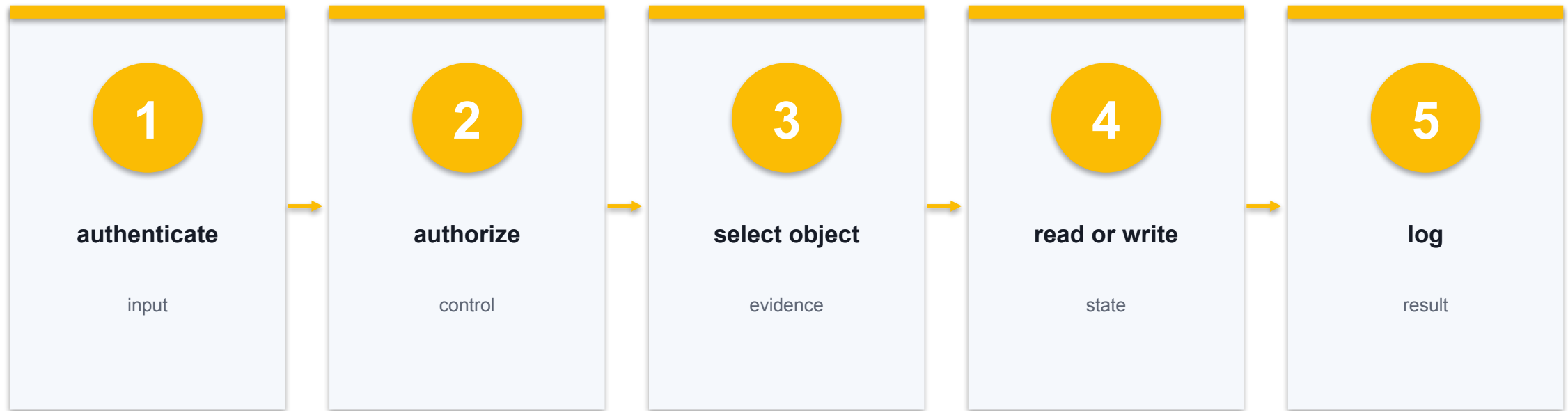
Cloud Storage request path · architecture



DECISION RULE

Scope first: Bucket location, uniform access and retention requirements are recorded before upload.

Cloud Storage request path · runtime trace



Bucket location, uniform access and retention requirements are recorded before upload.

Cloud Storage request path · concrete contract

```
gcloud storage buckets create gs://$BUCKET --  
location=asia-southeast1 --uniform-bucket-level-  
access
```

01

1

Scope: Storage and Databases

2

Mapped: A4 · K5

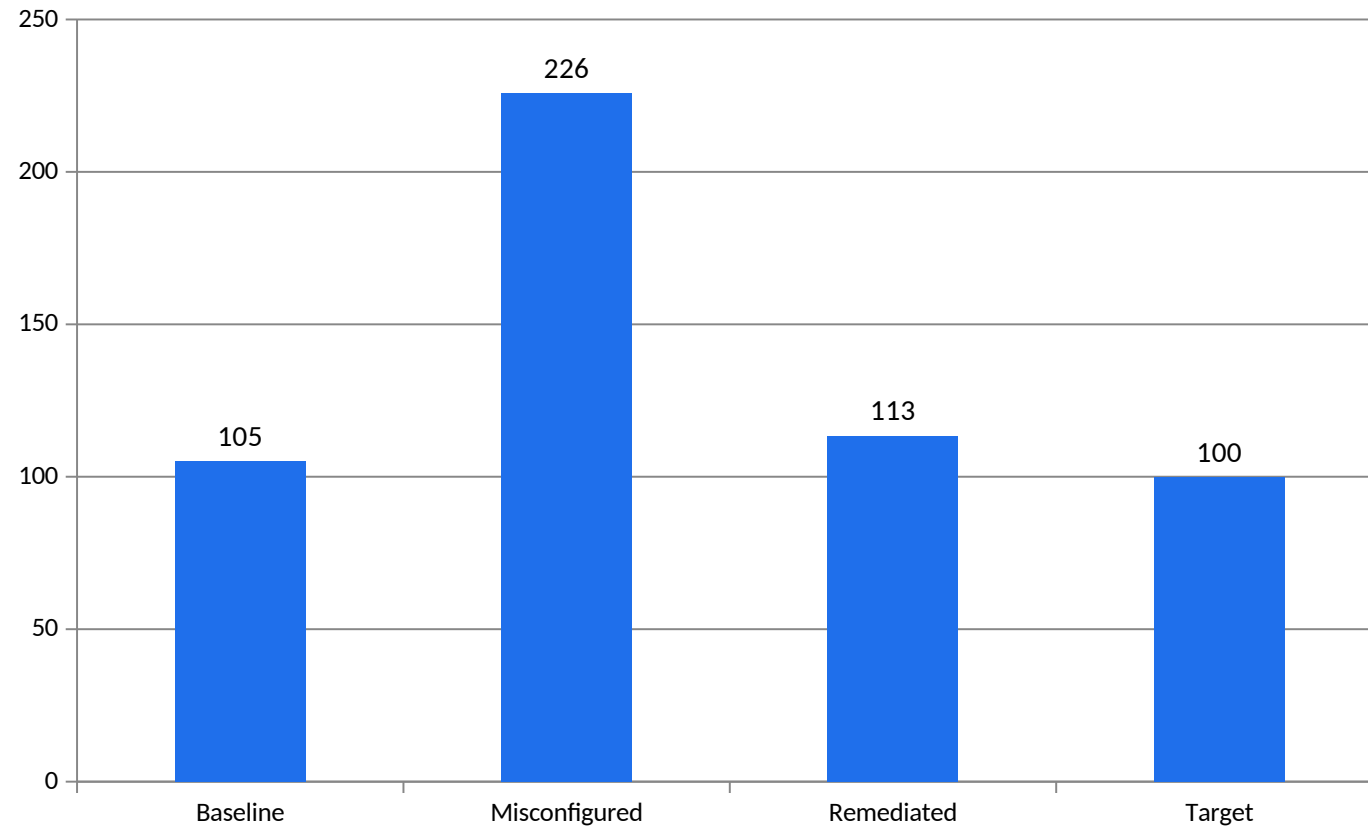
3

Capture stdout and exit status

4

Source: official storage documentation

Topic 5 evidence profile



WHAT THE DATA SHOWS

A controlled fault should move the signal away from baseline. The remediation is accepted only when it returns close to target and the exact configuration change is recorded.

Indicative teaching data. Learners replace it with measurements from the aligned activity.

Storage class decision rules · architecture

1

Standard

Boundary component 1. State and ownership must be explicit.

2

Nearline 30D

Boundary component 2. State and ownership must be explicit.

3

Coldline 90D

Boundary component 3. State and ownership must be explicit.

4

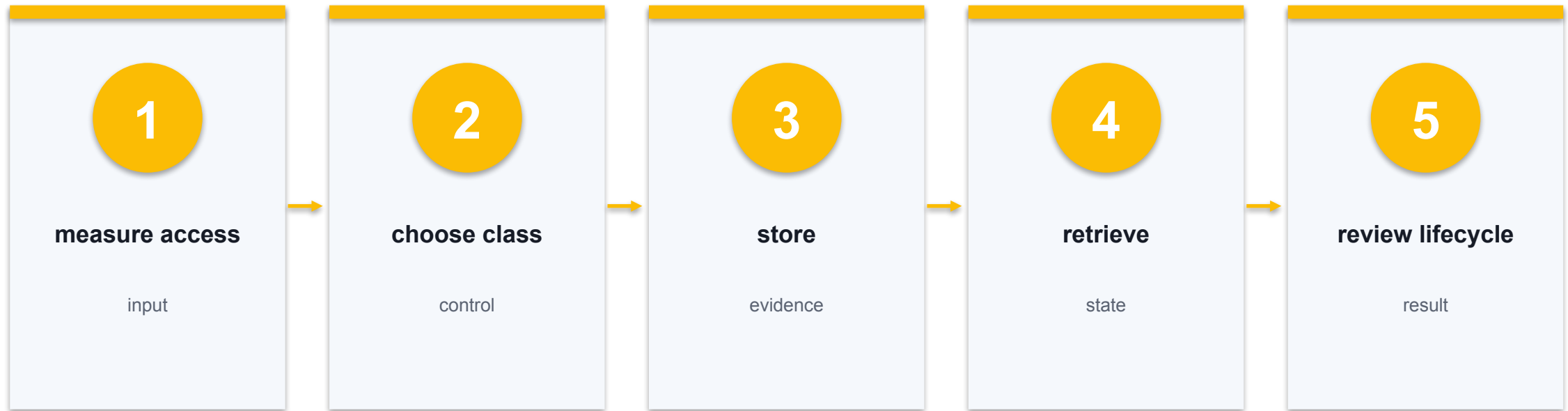
Archive 365D

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Class matches expected access frequency and minimum-duration economics.

Storage class decision rules · runtime trace



Class matches expected access frequency and minimum-duration economics.

Storage class decision rules · concrete contract

```
gcloud storage buckets describe gs://$BUCKET -- 01  
format='value(storageClass,location,type)'
```

1

Scope: Storage and Databases

2

Mapped: A4 · K5

3

Capture stdout and exit status

4

Source: official storage-classes documentation

Storage class decision rules - failure and control

FAILURE MODES



early deletion charge



retrieval fee ignored



class used as durability control

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Class matches expected access frequency and minimum-duration economics.

Lifecycle, versioning and retention · architecture

1

Lifecycle Rule

Boundary component 1. State and ownership must be explicit.

2

Object Age

Boundary component 2. State and ownership must be explicit.

3

Version State

Boundary component 3. State and ownership must be explicit.

4

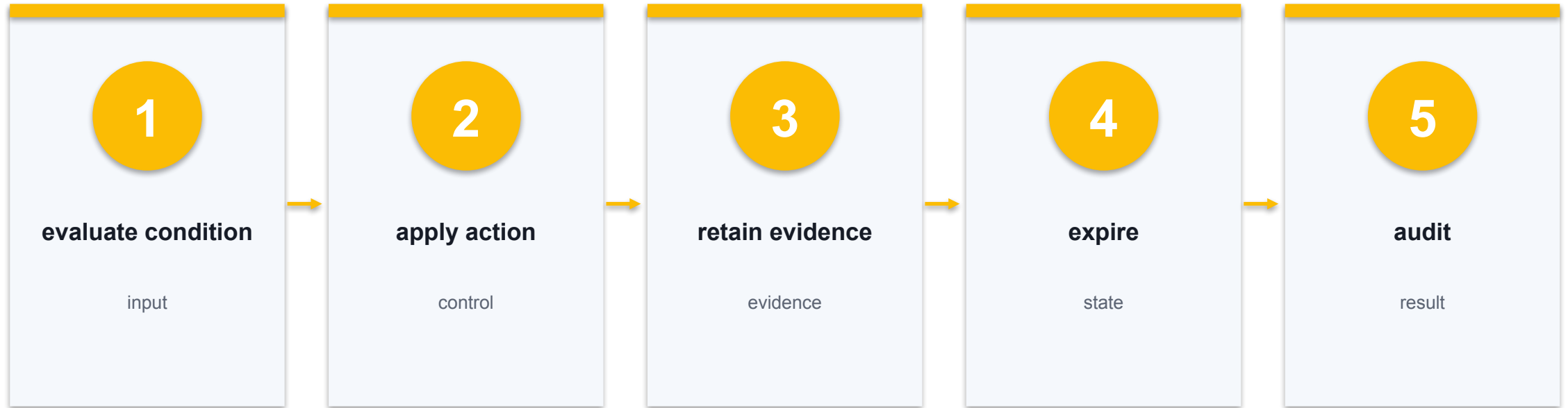
Retention Lock

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Policy JSON is valid and noncurrent versions follow the intended retention window.

Lifecycle, versioning and retention · runtime trace



Policy JSON is valid and noncurrent versions follow the intended retention window.

Lifecycle, versioning and retention · concrete contract

```
gcloud storage buckets update gs://$BUCKET -- 01
lifecycle-file=lifecycle.json
gcloud storage buckets update gs://$BUCKET -- 02
versioning
```

1

Scope: Storage and Databases

2

Mapped: A4 · K5

3

Capture stdout and exit status

4

Source: official lifecycle documentation

Lifecycle, versioning and retention · failure and control

FAILURE MODES



delete rule too broad



locked policy irreversible



soft delete cost overlooked

VERIFICATION EVIDENCE



Expected state is explicit



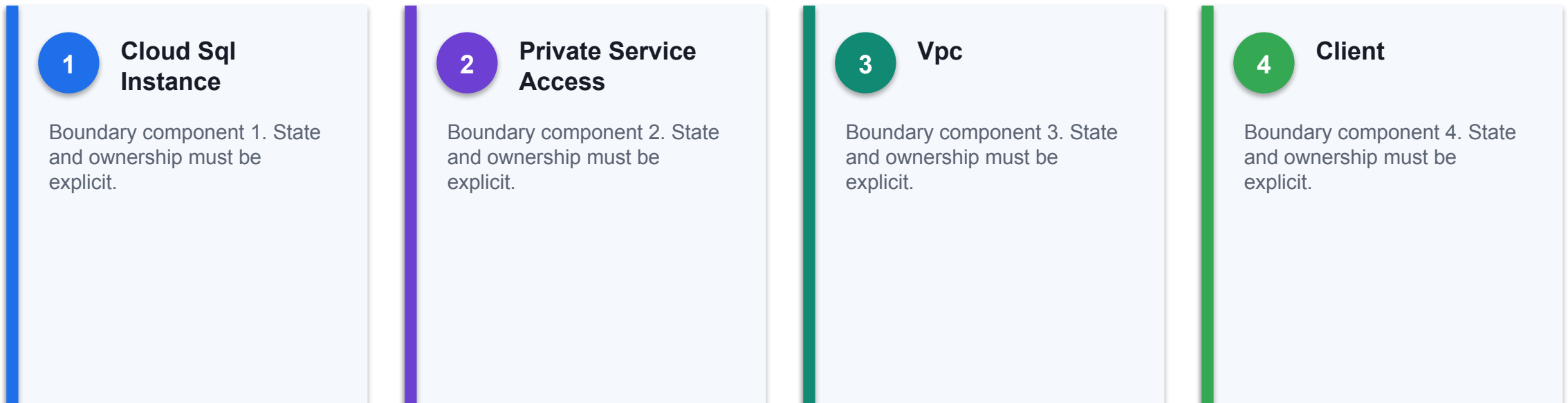
Observed output is captured



Timestamp and scope are recorded

Policy JSON is valid and noncurrent versions follow the intended retention window.

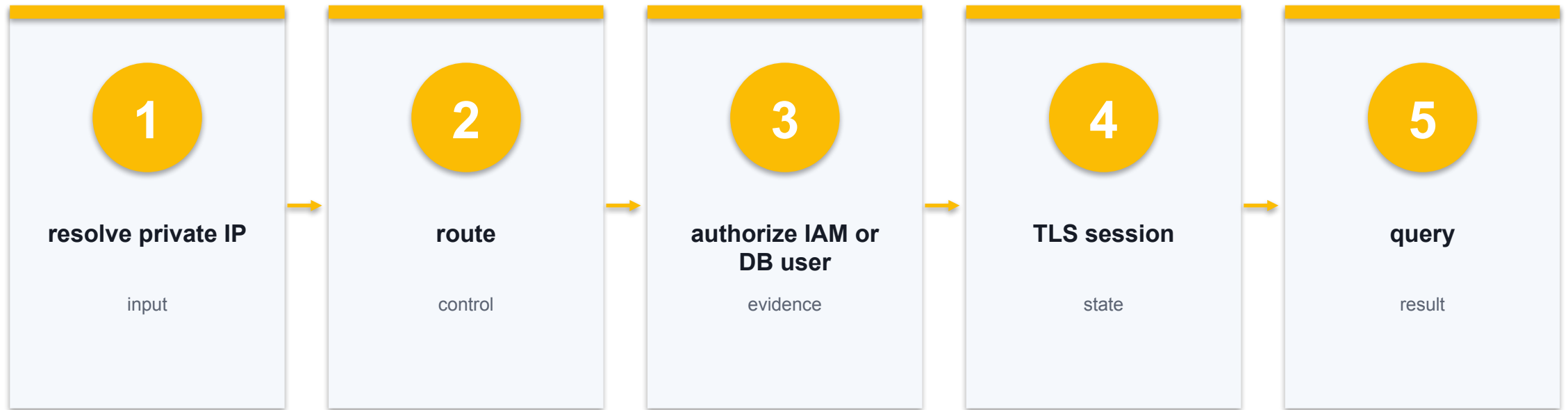
Cloud SQL private connectivity · architecture



DECISION RULE

Scope first: Client reaches the private address and database authentication is distinct from network reachability.

Cloud SQL private connectivity · runtime trace



Client reaches the private address and database authentication is distinct from network reachability.

Cloud SQL private connectivity · concrete contract

```
gcloud sql instances describe ace-db --  
format='value(ipAddresses,settings.availabilityType,s  
ettings.backupConfiguration.enabled) '01
```

1

Scope: Storage and Databases

2

Mapped: A4 · K5

3

Capture stdout and exit status

4

Source: official cloud-sql documentation

Cloud SQL private connectivity · failure and control

FAILURE MODES



network not peered



connection limit



wrong database user

VERIFICATION EVIDENCE



Expected state is explicit



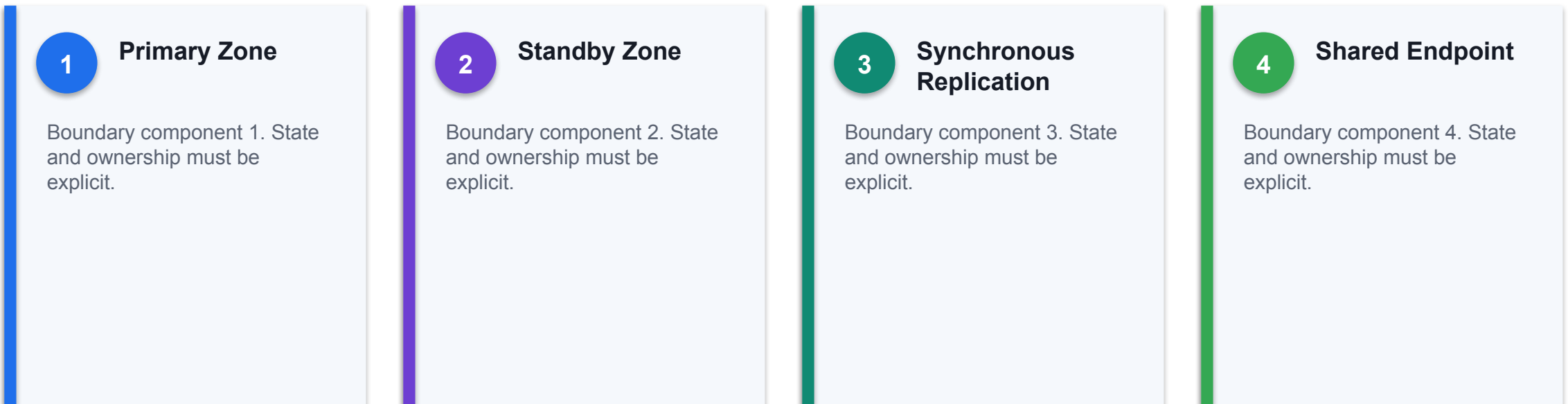
Observed output is captured



Timestamp and scope are recorded

Client reaches the private address and database authentication is distinct from network reachability.

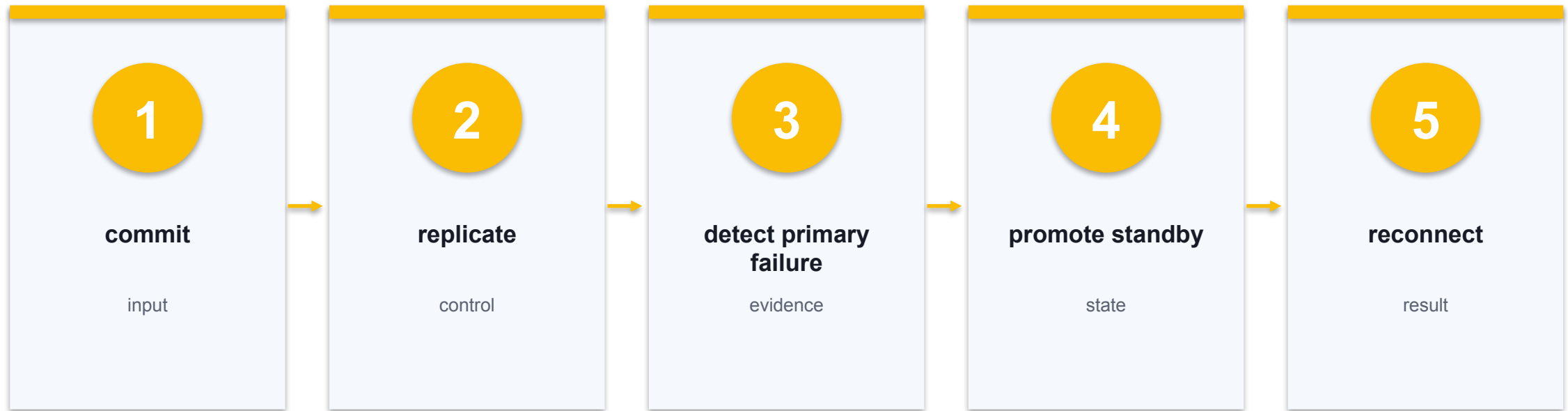
Regional HA and failover · architecture



DECISION RULE

Scope first: REGIONAL availability and reconnect behaviour are proven by a controlled failover test.

Regional HA and failover · runtime trace



REGIONAL availability and reconnect behaviour are proven by a controlled failover test.

Regional HA and failover · concrete contract

```
gcloud sql instances describe ace-db --  
format='value(settings.availabilityType,gceZone,second01  
aryGceZone)'
```

1

Scope: Storage and Databases

2

Mapped: A4 · K5

3

Capture stdout and exit status

4

Source: official cloud-sql-ha documentation

Regional HA and failover · failure and control

FAILURE MODES



zonal instance



application no retry



long transaction during failover

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

REGIONAL availability and reconnect behaviour are proven by a controlled failover test.

Database observability and query diagnosis · architecture

1

Query Insights

Boundary component 1. State and ownership must be explicit.

2

Cpu And Memory Metrics

Boundary component 2. State and ownership must be explicit.

3

Connections

Boundary component 3. State and ownership must be explicit.

4

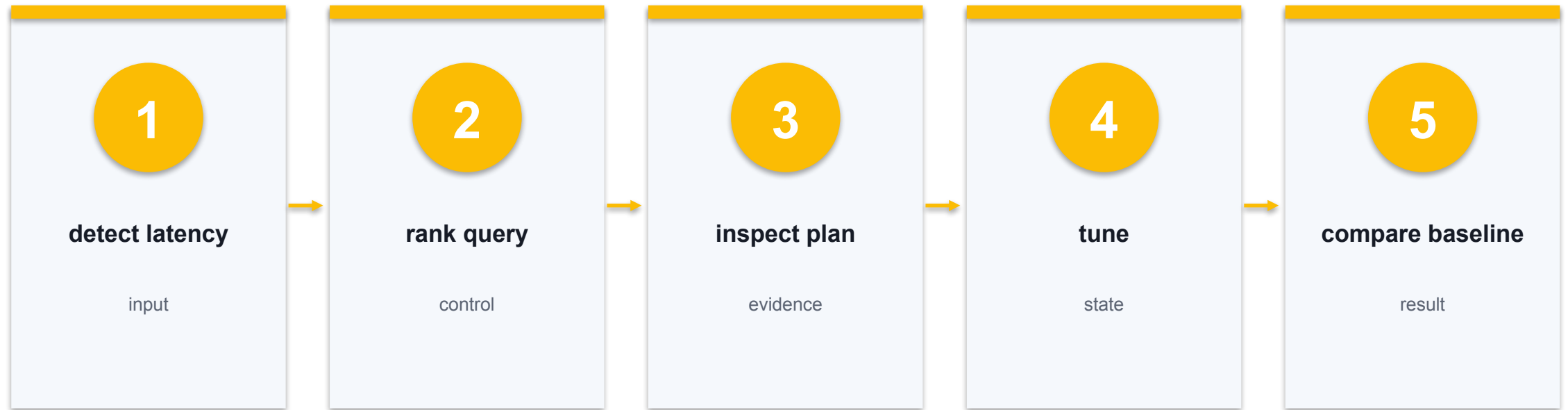
Slow Query

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Before/after latency, rows scanned and connection pressure support the chosen fix.

Database observability and query diagnosis · runtime trace



Before/after latency, rows scanned and connection pressure support the chosen fix.

Database observability and query diagnosis · concrete contract

```
gcloud monitoring time-series list --  
filter='metric.type="cloudsql.googleapis.com/database  
/cpu/utilization"' --limit=5
```

01

1

Scope: Storage and Databases

2

Mapped: A4 · K5

3

Capture stdout and exit status

4

Source: official query-insights documentation

Database observability and query diagnosis · failure and control

FAILURE MODES



scale hides bad query



pool leak



index write overhead

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Before/after latency, rows scanned and connection pressure support the chosen fix.

Managed database selection · architecture

1

Cloud Sql

Boundary component 1. State and ownership must be explicit.

2

Spanner

Boundary component 2. State and ownership must be explicit.

3

Bigtable

Boundary component 3. State and ownership must be explicit.

4

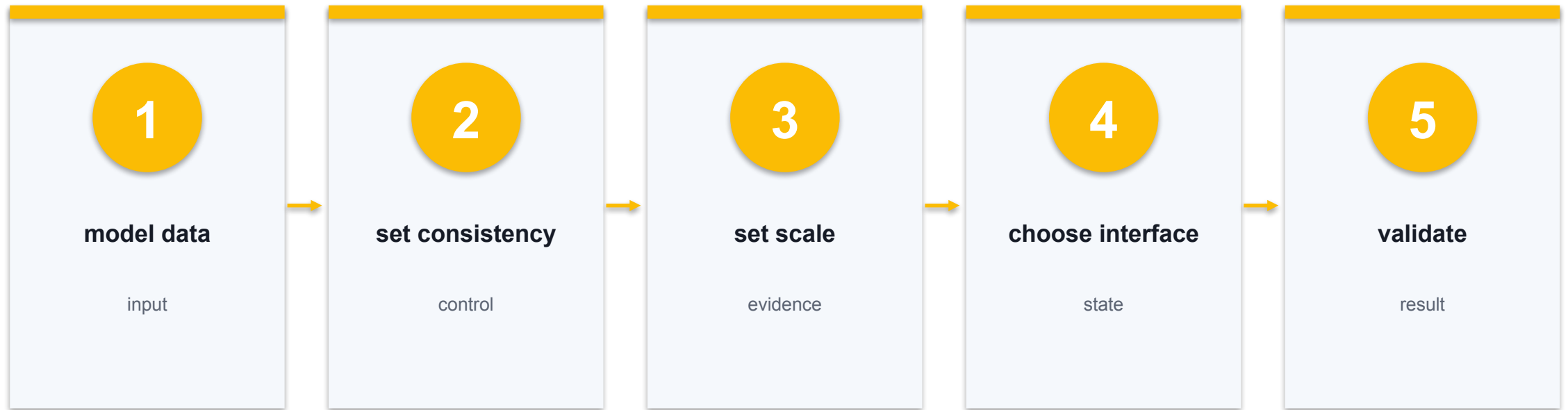
Firestore

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Choice is based on transaction model, consistency, access pattern and operational boundary.

Managed database selection · runtime trace



Choice is based on transaction model, consistency, access pattern and operational boundary.

Managed database selection · concrete contract

```
gcloud sql tiers list --filter='region:asia-  
southeast1' --format='table(tier, RAM, diskQuota) '
```

01

1

Scope: Storage and Databases

2

Mapped: A4 · K5

3

Capture stdout and exit status

4

Source: official databases documentation

Managed database selection · failure and control

FAILURE MODES



product by brand familiarity



relational joins on Bigtable



global scale assumed free

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Choice is based on transaction model, consistency, access pattern and operational boundary.

Backup, restore and data validation · architecture

1

Automated Backup

Boundary component 1. State and ownership must be explicit.

2

Point-In-Time Recovery

Boundary component 2. State and ownership must be explicit.

3

Restore Target

Boundary component 3. State and ownership must be explicit.

4

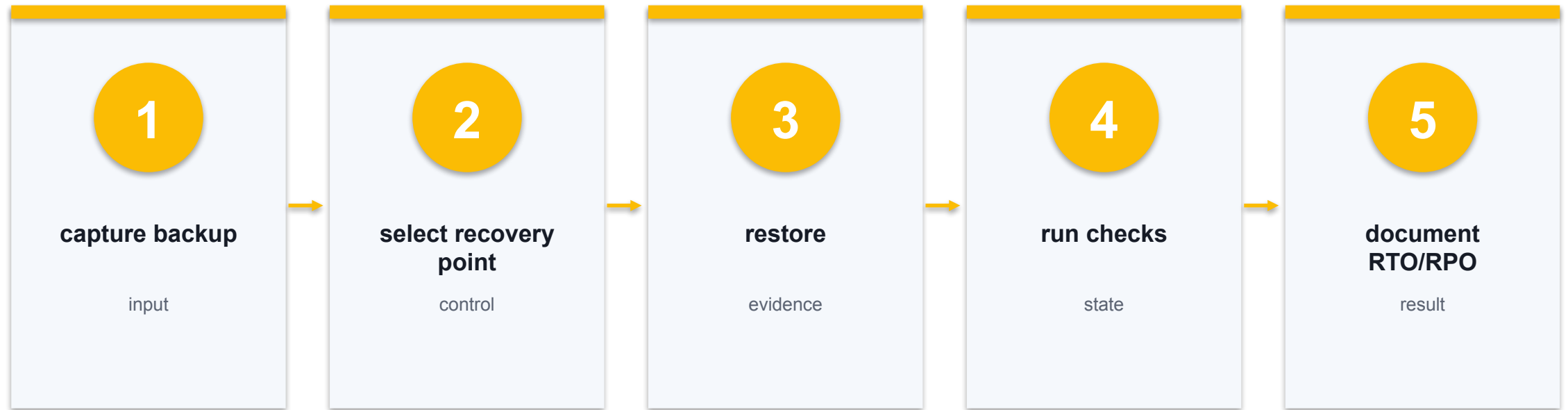
Validation Query

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Restored copy passes row-count, checksum and application connection tests.

Backup, restore and data validation · runtime trace



Restored copy passes row-count, checksum and application connection tests.

Backup, restore and data validation · concrete contract

```
gcloud sql backups list --instance=ace-db      01
gcloud sql operations list --instance=ace-db -- 02
limit=10
```

1

Scope: Storage and Databases

2

Mapped: A4 · K5

3

Capture stdout and exit status

4

Source: official backup-restore documentation

Backup, restore and data validation · failure and control

FAILURE MODES



backup exists but restore untested



PITR log gap



restore over production

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Restored copy passes row-count, checksum and application connection tests.

Operate Cloud Storage and Diagnose Cloud SQL

SCENARIO

Apply storage lifecycle controls, load mock customer data and diagnose a simulated Cloud SQL connection bottleneck.

YOU WILL PRODUCE

lifecycle policy, query evidence and database incident report

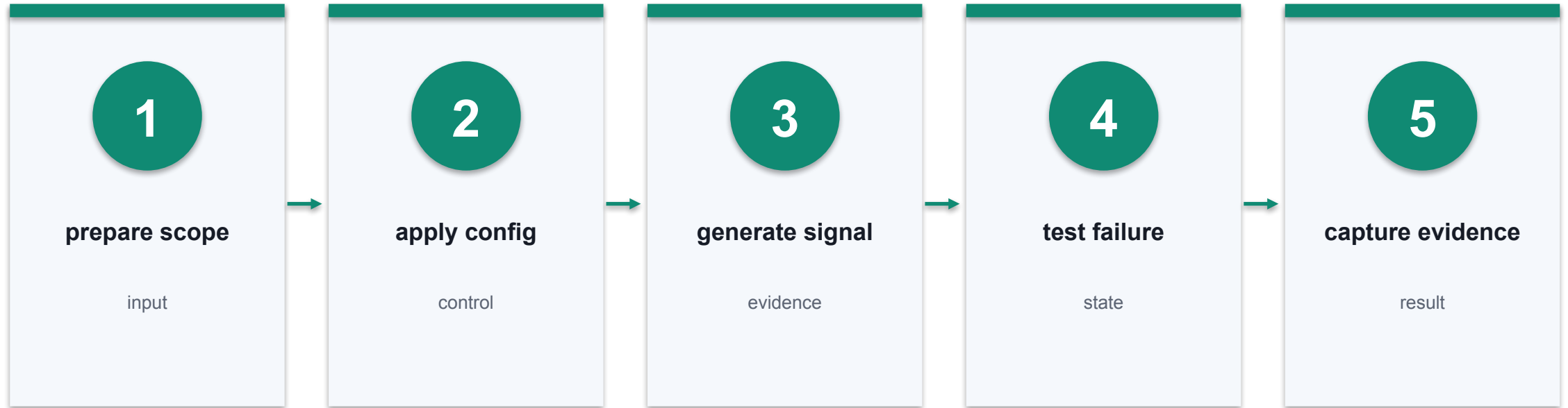
customers.csv

lifecycle.json

diagnose.sql

Full click-by-click instructions are in the Learner Guide and Activity PDF. This slide shows the work product, not the procedure.

Activity 5 system trace



Apply storage lifecycle controls, load mock customer data and diagnose a simulated Cloud SQL connection bottleneck.

Activity 5 worked artifact

```
customer_id,region,tier,last_order_date    01
C001,SG,gold,2026-09-01                  02
C002,MY,silver,2026-08-17                03
C003,SG,bronze,2026-06-12                04
C004,ID,gold,2026-09-10                  05
```

1

Use placeholders for project-specific values

2

Run only in the isolated training project

3

Capture both success and failure output

4

Keep secrets out of submitted evidence

Activity 5 acceptance gate

FAILURE MODES



wrong project or region



insufficient IAM or disabled API



observed output does not meet target

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



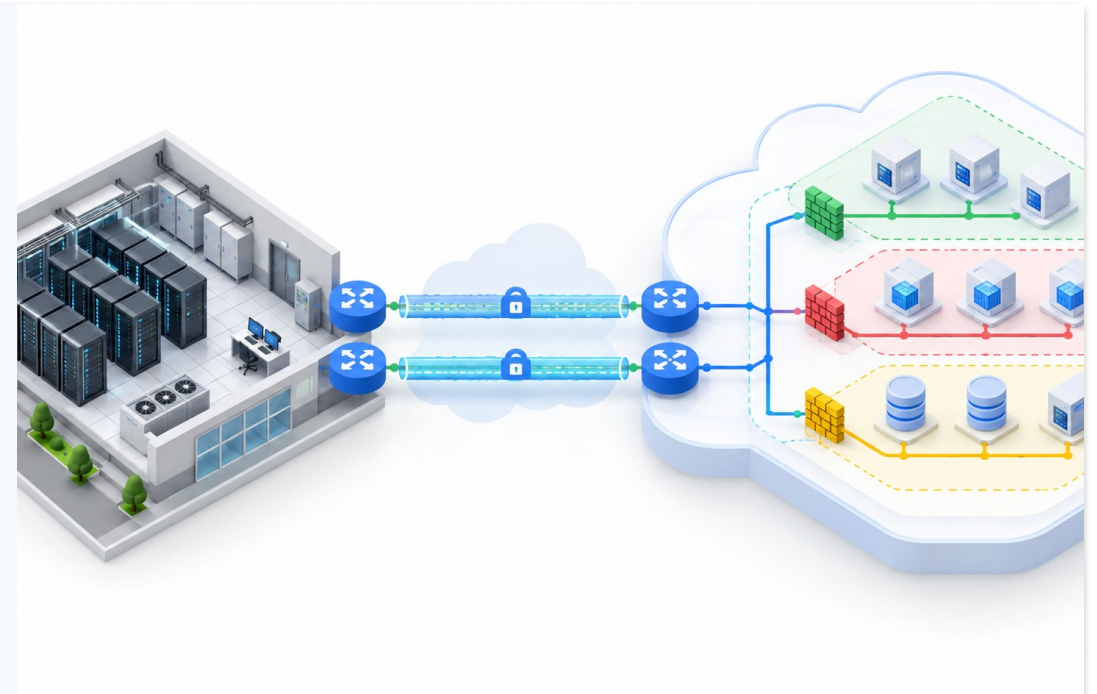
Timestamp and scope are recorded

Accept when lifecycle policy, query evidence and database incident report are complete, timestamped and reproducible.

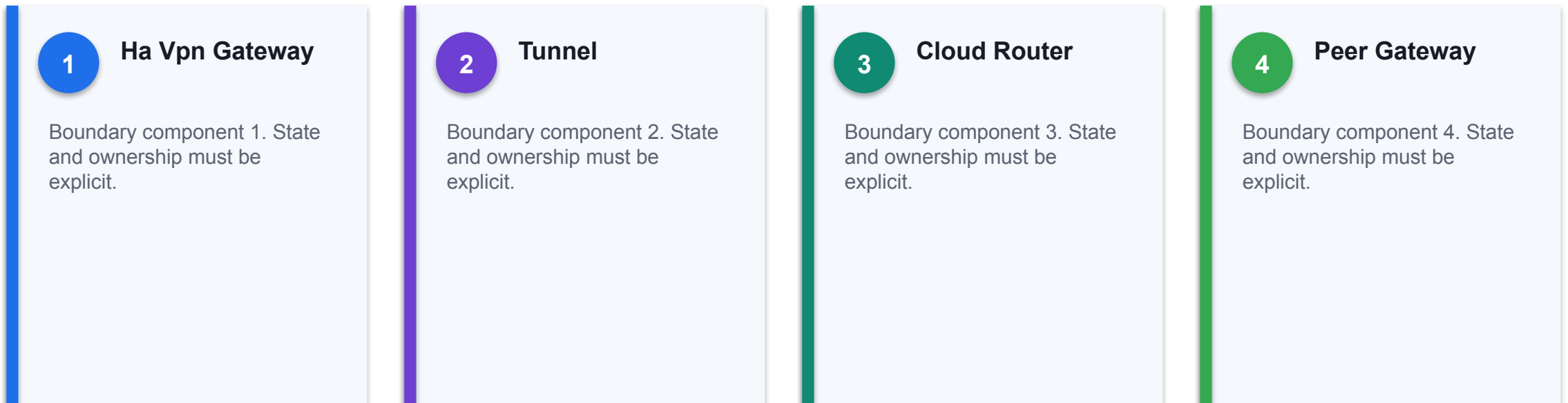
TOPIC 06 · A4 · K5

Interconnecting Networks

Implement hybrid and multi-project connectivity



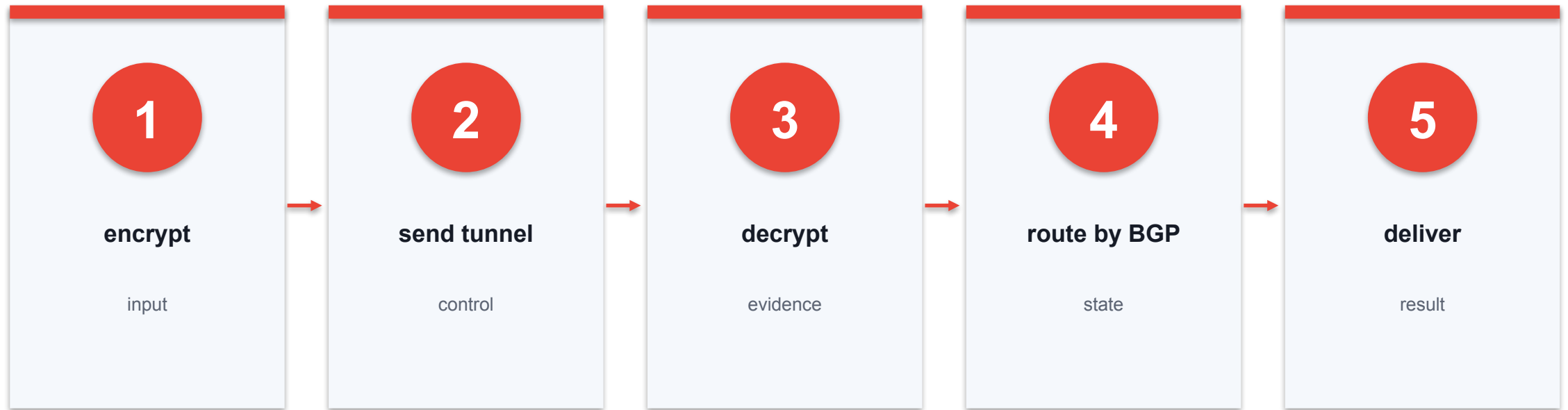
Cloud VPN packet path · architecture



DECISION RULE

Scope first: Both HA interfaces and BGP sessions meet the intended redundancy design.

Cloud VPN packet path · runtime trace



Both HA interfaces and BGP sessions meet the intended redundancy design.

Cloud VPN packet path · concrete contract

```
gcloud compute vpn-tunnels list --  
format='table(name,region,status,peerIp,router) '
```

01

1

Scope: Interconnecting Networks

2

Mapped: A4 · K5

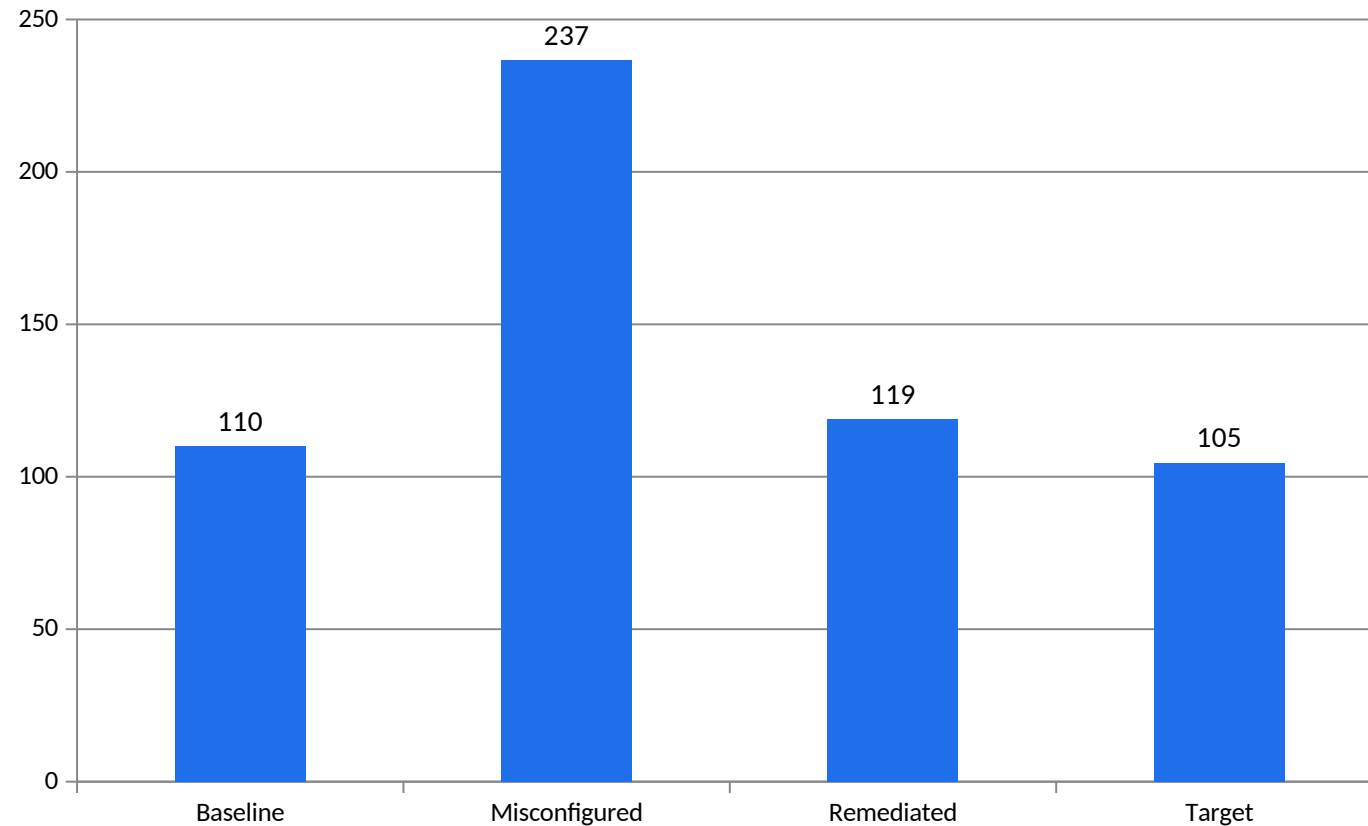
3

Capture stdout and exit status

4

Source: official vpn documentation

Topic 6 evidence profile



WHAT THE DATA SHOWS

A controlled fault should move the signal away from baseline. The remediation is accepted only when it returns close to target and the exact configuration change is recorded.

Indicative teaching data. Learners replace it with measurements from the aligned activity.

BGP dynamic route exchange · architecture

1

Cloud Router

Boundary component 1. State and ownership must be explicit.

2

Bgp Peer

Boundary component 2. State and ownership must be explicit.

3

Advertised Prefix

Boundary component 3. State and ownership must be explicit.

4

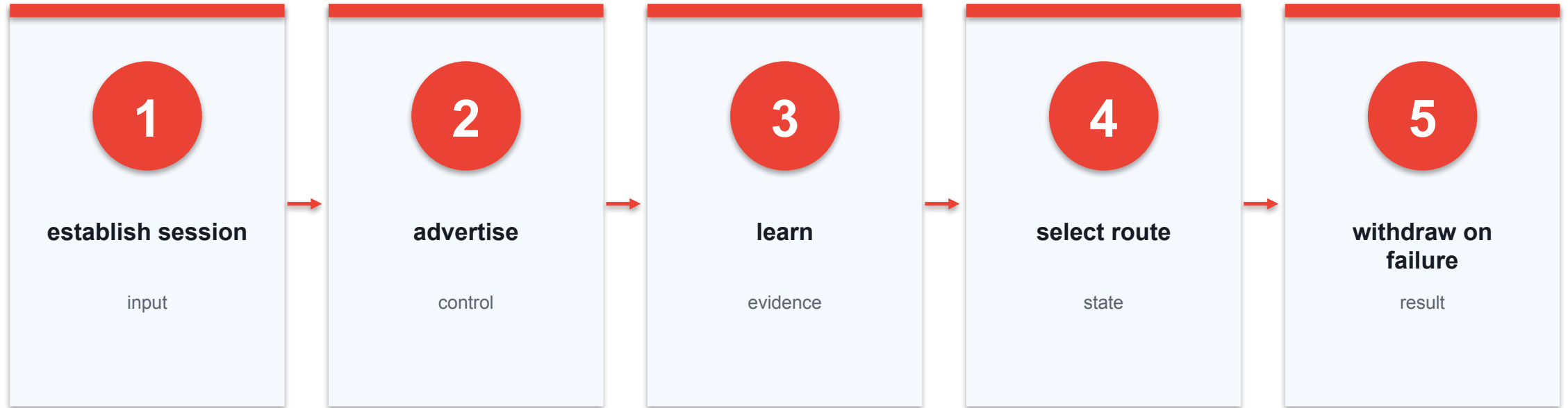
Learned Route

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Peer status is UP and learned/advertised prefixes match the approved plan.

BGP dynamic route exchange · runtime trace



Peer status is UP and learned/advertised prefixes match the approved plan.

BGP dynamic route exchange · concrete contract

```
gcloud compute routers get-status ace-router -- 01  
region=asia-southeast1 --format=json
```

1

Scope: Interconnecting Networks

2

Mapped: A4 · K5

3

Capture stdout and exit status

4

Source: official cloud-router documentation

BGP dynamic route exchange · failure and control

FAILURE MODES



ASN conflict



MD5/session mismatch



custom advertisement omits subnet

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Peer status is UP and learned/advertised prefixes match the approved plan.

Interconnect product decision · architecture

1

Dedicated Interconnect

Boundary component 1. State and ownership must be explicit.

2

Partner Interconnect

Boundary component 2. State and ownership must be explicit.

3

Cross-Cloud Interconnect

Boundary component 3. State and ownership must be explicit.

4

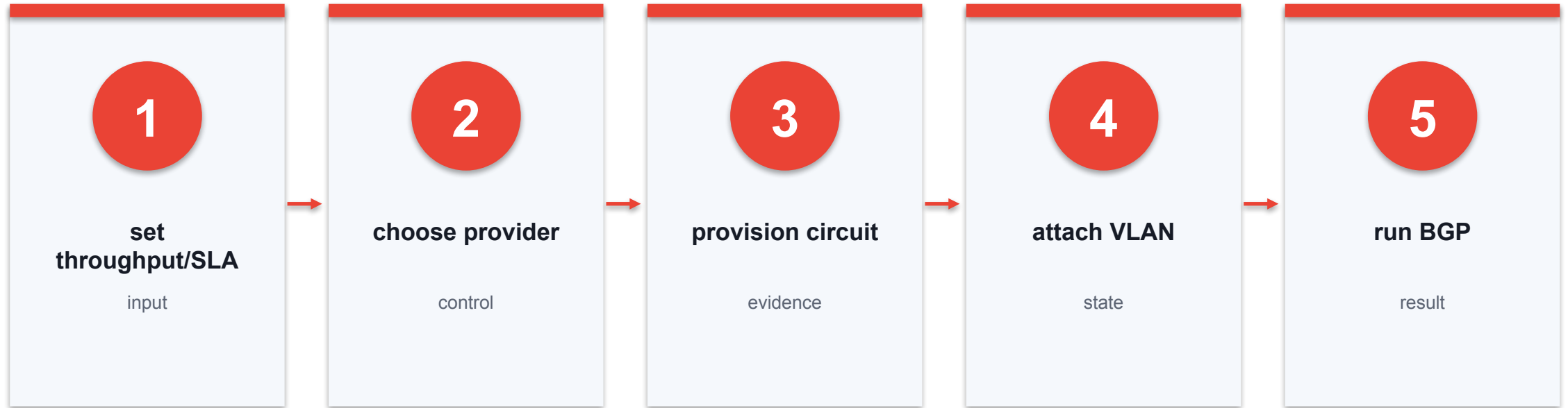
Vlan Attachment

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Selection reflects throughput, location, provider, encryption and SLA requirements.

Interconnect product decision · runtime trace



Selection reflects throughput, location, provider, encryption and SLA requirements.

Interconnect product decision · concrete contract

```
gcloud compute interconnects attachments list -- 01  
format='table(name,region,state,type,router)'
```

1

Scope: Interconnecting Networks

2

Mapped: A4 · K5

3

Capture stdout and exit status

4

Source: official interconnect documentation

Interconnect product decision · failure and control

FAILURE MODES



no redundant metro



VLAN attachment inactive



BGP not established

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Selection reflects throughput, location, provider, encryption and SLA requirements.

VPC Network Peering semantics · architecture

1

Peer Vpc A

Boundary component 1. State and ownership must be explicit.

2

Peer Vpc B

Boundary component 2. State and ownership must be explicit.

3

Subnet Routes

Boundary component 3. State and ownership must be explicit.

4

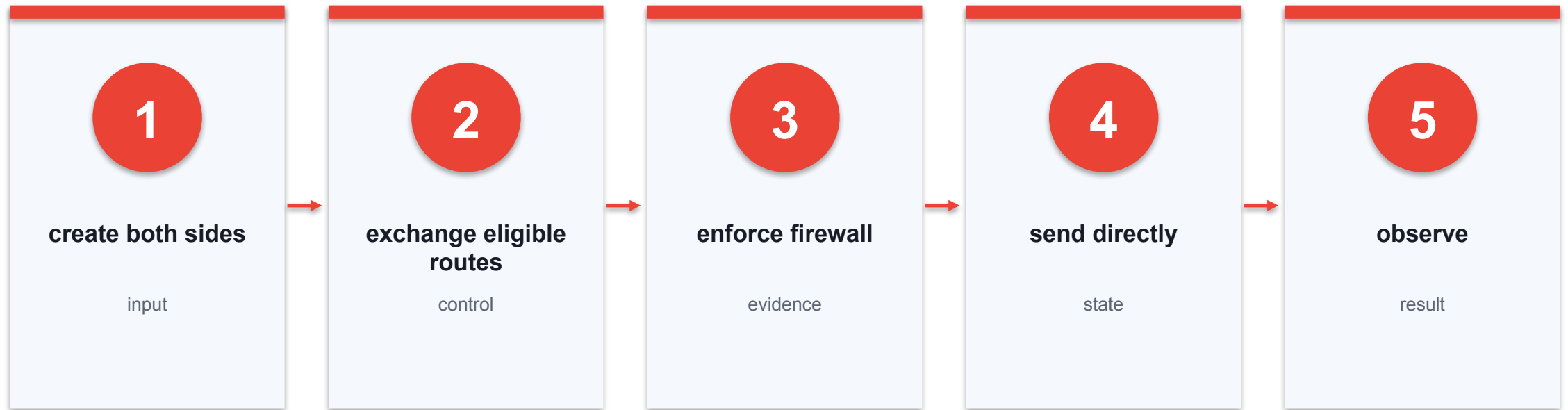
Non-Transitive Boundary

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Both sides are ACTIVE and no design depends on transitive routing.

VPC Network Peering semantics · runtime trace



Both sides are ACTIVE and no design depends on transitive routing.

VPC Network Peering semantics · concrete contract

```
gcloud compute networks peerings list --network=ace-01  
vpc
```

1

Scope: Interconnecting Networks

2

Mapped: A4 · K5

3

Capture stdout and exit status

4

Source: official peering documentation

VPC Network Peering semantics · failure and control

FAILURE MODES



overlapping ranges



one side missing



firewall not automatically shared

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Both sides are ACTIVE and no design depends on transitive routing.

Shared VPC versus peering · architecture

1

Central Host Project

Boundary component 1. State and ownership must be explicit.

2

Service Projects

Boundary component 2. State and ownership must be explicit.

3

Separate Admin Roles

Boundary component 3. State and ownership must be explicit.

4

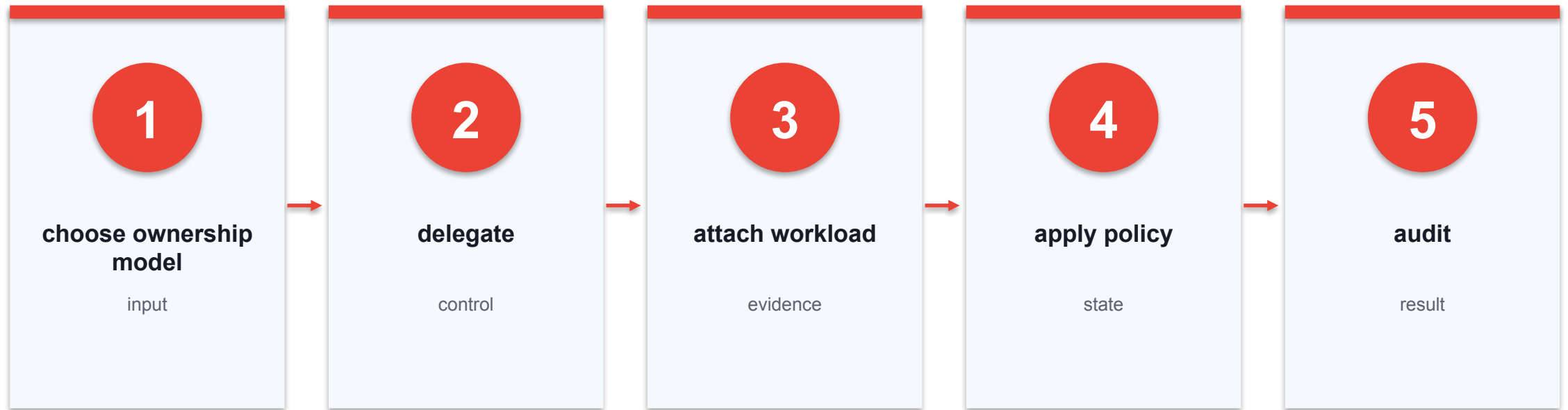
Shared Subnets

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Chosen model matches whether central administrators must own the network dataplane.

Shared VPC versus peering · runtime trace



Chosen model matches whether central administrators must own the network dataplane.

Shared VPC versus peering · concrete contract

```
gcloud compute shared-vpc associated-projects list 01  
$HOST_PROJECT
```

1

Scope: Interconnecting Networks

2

Mapped: A4 · K5

3

Capture stdout and exit status

4

Source: official shared-vpc documentation

Shared VPC versus peering · failure and control

FAILURE MODES



using peering for centralized subnet ownership



host service agent missing



project association drift

VERIFICATION EVIDENCE



Expected state is explicit



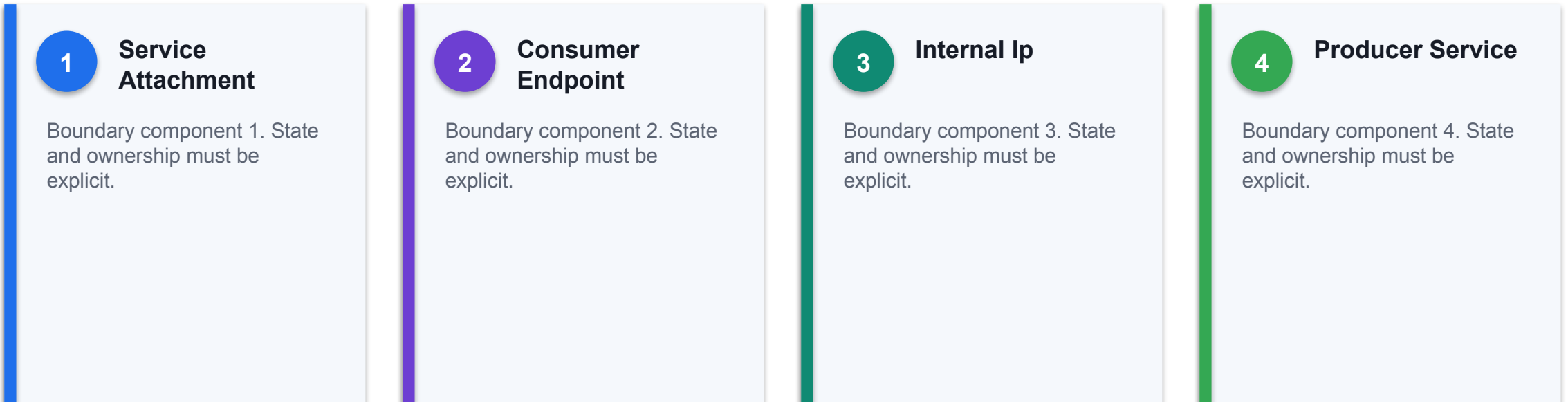
Observed output is captured



Timestamp and scope are recorded

Chosen model matches whether central administrators must own the network dataplane.

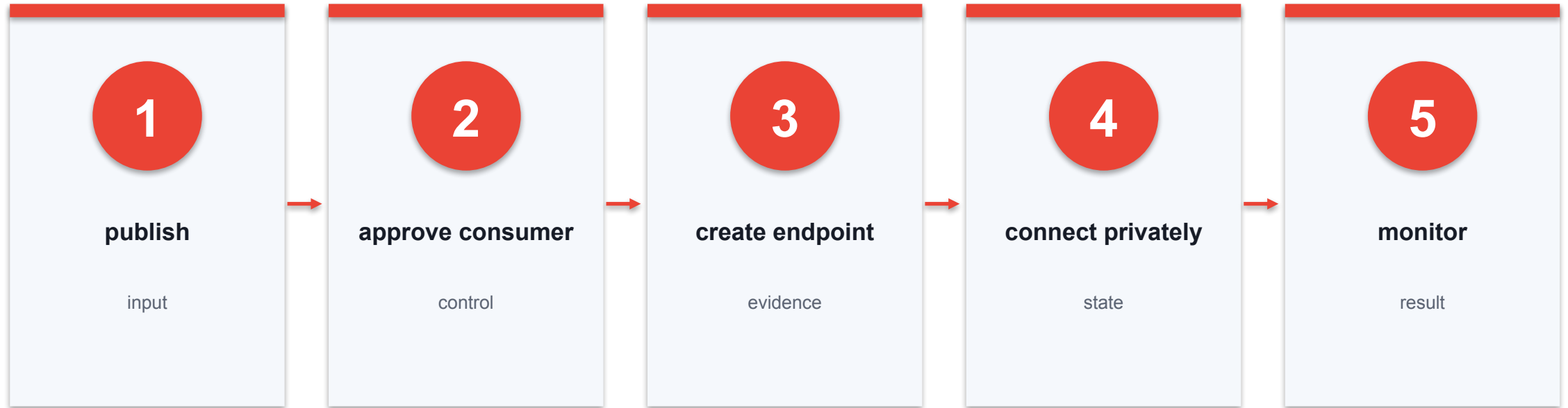
Private Service Connect service boundary · architecture



DECISION RULE

Scope first: Consumer uses a private endpoint without importing the producer's routes.

Private Service Connect service boundary · runtime trace



Consumer uses a private endpoint without importing the producer's routes.

Private Service Connect service boundary · concrete contract

```
gcloud compute forwarding-rules list --  
filter='target:serviceAttachments' --  
format='table(name,IPAddress,network,region) '
```

01

1

Scope: Interconnecting Networks

2

Mapped: A4 · K5

3

Capture stdout and exit status

4

Source: official psc documentation

Private Service Connect service boundary · failure and control

FAILURE MODES



connection not accepted



wrong region



DNS still resolves public endpoint

VERIFICATION EVIDENCE



Expected state is explicit



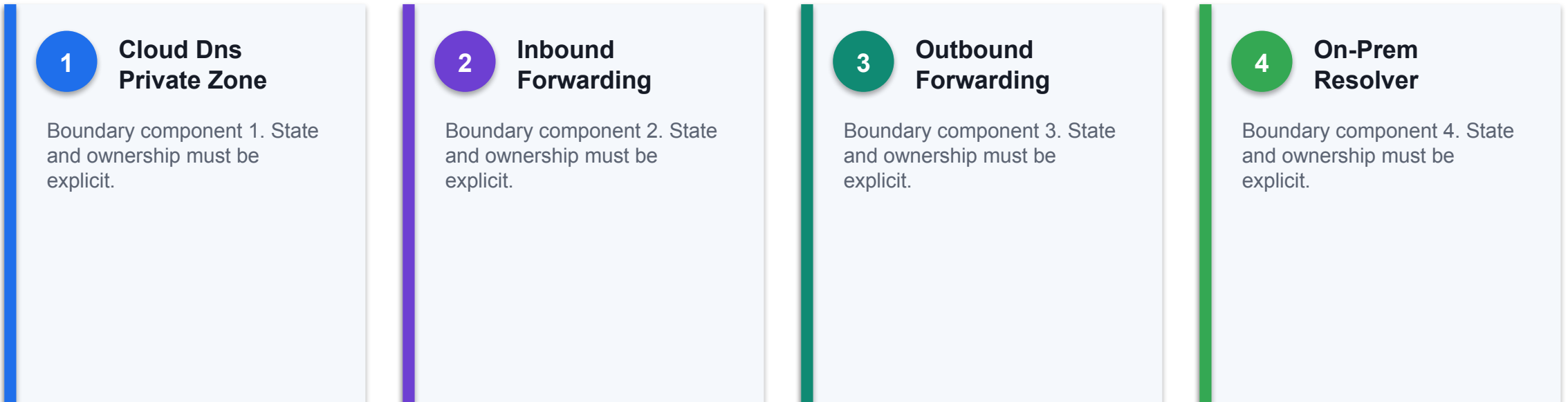
Observed output is captured



Timestamp and scope are recorded

Consumer uses a private endpoint without importing the producer's routes.

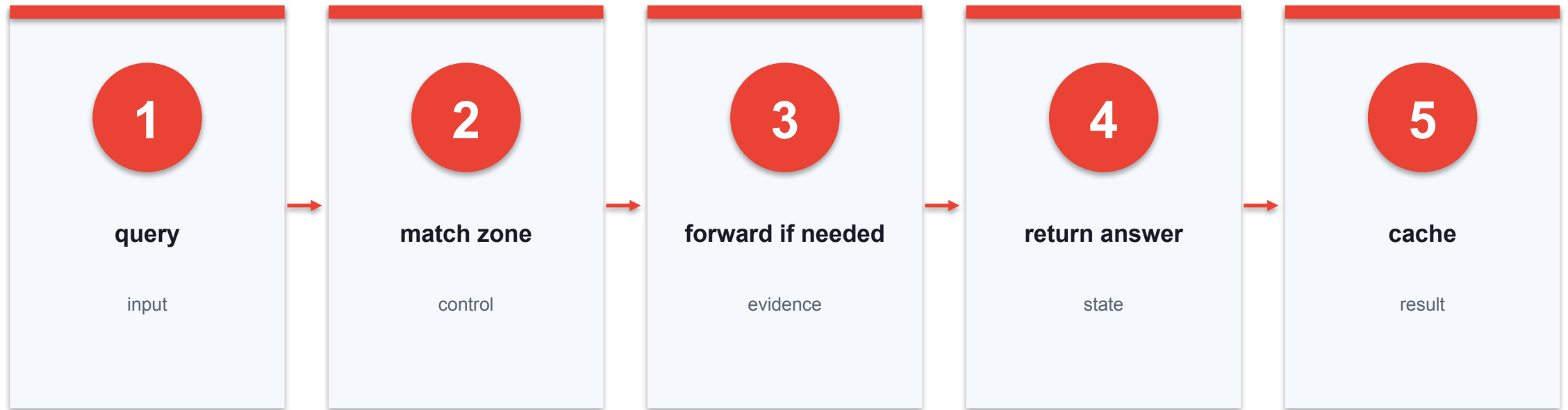
Hybrid DNS resolution · architecture



DECISION RULE

Scope first: Resolution path and authoritative zone are testable from both cloud and on-prem sources.

Hybrid DNS resolution · runtime trace



Resolution path and authoritative zone are testable from both cloud and on-prem sources.

Hybrid DNS resolution · concrete contract

```
gcloud dns managed-zones list --  
format='table(name,dnsName,visibility)'
```

01

1

Scope: Interconnecting Networks

2

Mapped: A4 · K5

3

Capture stdout and exit status

4

Source: official dns documentation

Hybrid DNS resolution · failure and control

FAILURE MODES



overlapping zone suffix



forwarder unreachable



stale cache

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Resolution path and authoritative zone are testable from both cloud and on-prem sources.

Hybrid connectivity fault isolation · architecture

1

Interface

Boundary component 1. State and ownership must be explicit.

2

Bgp Session

Boundary component 2. State and ownership must be explicit.

3

Route

Boundary component 3. State and ownership must be explicit.

4

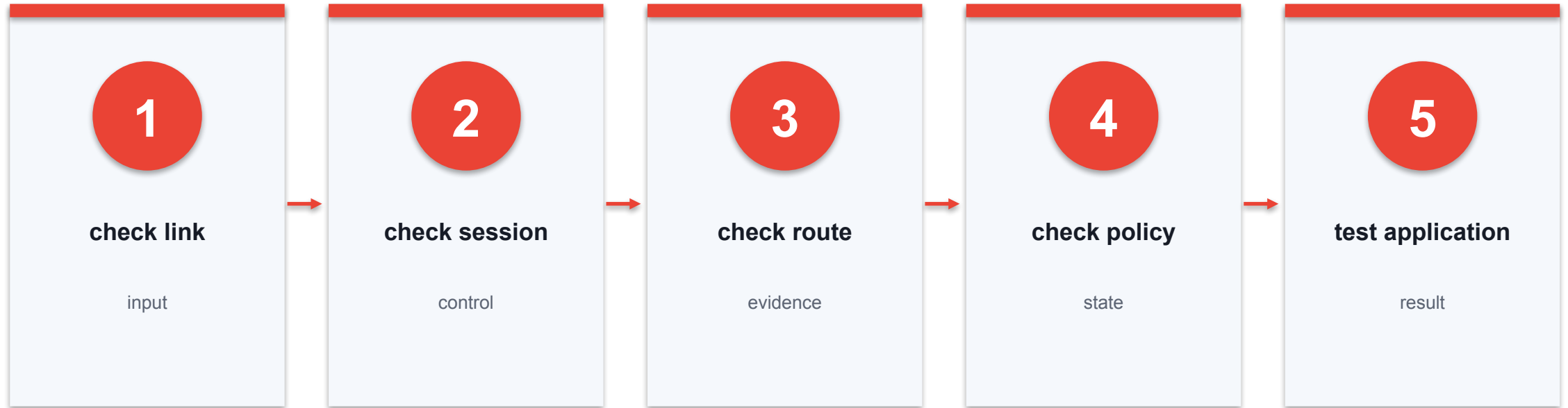
Firewall

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Evidence isolates control-plane, routing, policy or application failure before escalation.

Hybrid connectivity fault isolation · runtime trace



Evidence isolates control-plane, routing, policy or application failure before escalation.

Hybrid connectivity fault isolation · concrete contract

```
gcloud compute routers get-status ace-router -- 01  
region=asia-southeast1  
gcloud network-management connectivity-tests list 02
```

1

Scope: Interconnecting Networks

2

Mapped: A4 · K5

3

Capture stdout and exit status

4

Source: official network-diagnostics
documentation

Hybrid connectivity fault isolation · failure and control

FAILURE MODES



testing only ping



asymmetric route



MTU black hole

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Evidence isolates control-plane, routing, policy or application failure before escalation.

Model Hybrid Connectivity and BGP Failure

SCENARIO

Evaluate VPN versus Interconnect, inspect mock BGP routes and isolate a hybrid routing failure.

YOU WILL PRODUCE

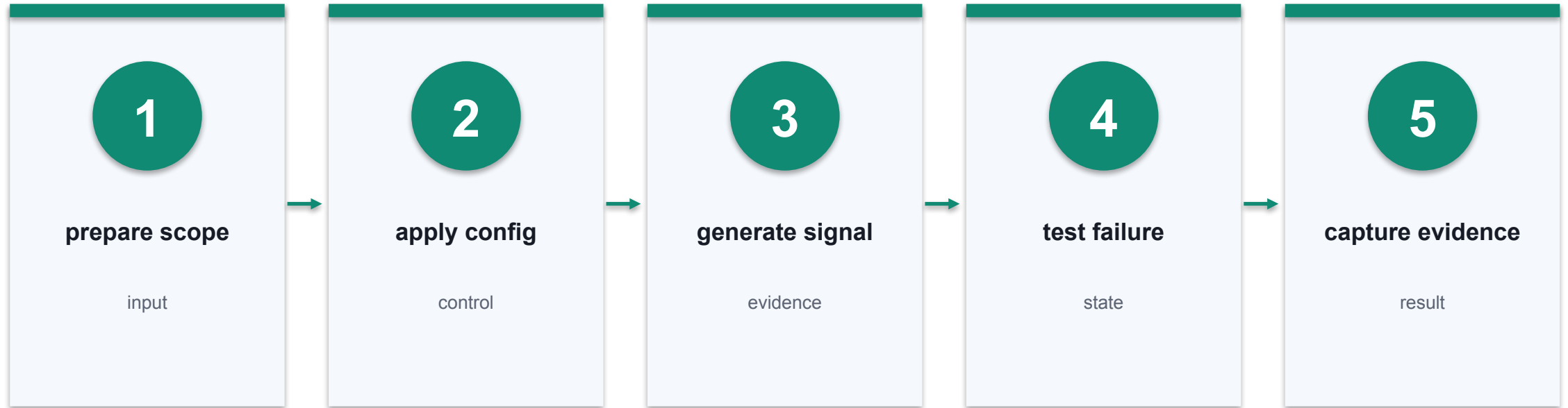
connectivity decision, route analysis and corrected BGP configuration

bgp_routes.csv

router_config.yaml

Full click-by-click instructions are in the Learner Guide and Activity PDF. This slide shows the work product, not the procedure.

Activity 6 system trace



Evaluate VPN versus Interconnect, inspect mock BGP routes and isolate a hybrid routing failure.

Activity 6 worked artifact

```
prefix,next_hop,as_path,priority,status    01
10.20.0.0/16,cloud-vpn,64514,100,learned  02
172.16.0.0/16,onprem-router,65010,100,advertised  03
10.20.10.0/24,cloud-vpn,64514,90,learned    04
```

1

Use placeholders for project-specific values

2

Run only in the isolated training project

3

Capture both success and failure output

4

Keep secrets out of submitted evidence

Activity 6 acceptance gate

FAILURE MODES



wrong project or region



insufficient IAM or disabled API



observed output does not meet target

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Accept when connectivity decision, route analysis and corrected BGP configuration are complete, timestamped and reproducible.

Lunch and resource check

Before you step away

Stop test traffic

Check active resources

Save evidence

Do not leave tokens on screen

Day 2 evidence recap

1

Topic 4

Virtual Machines

2

Topic 5

Storage and Databases

3

Topic 6

Interconnecting Networks

4

Evidence

Record expected versus observed output.

DECISION RULE

A configuration is not complete until its observable state matches the acceptance criterion.

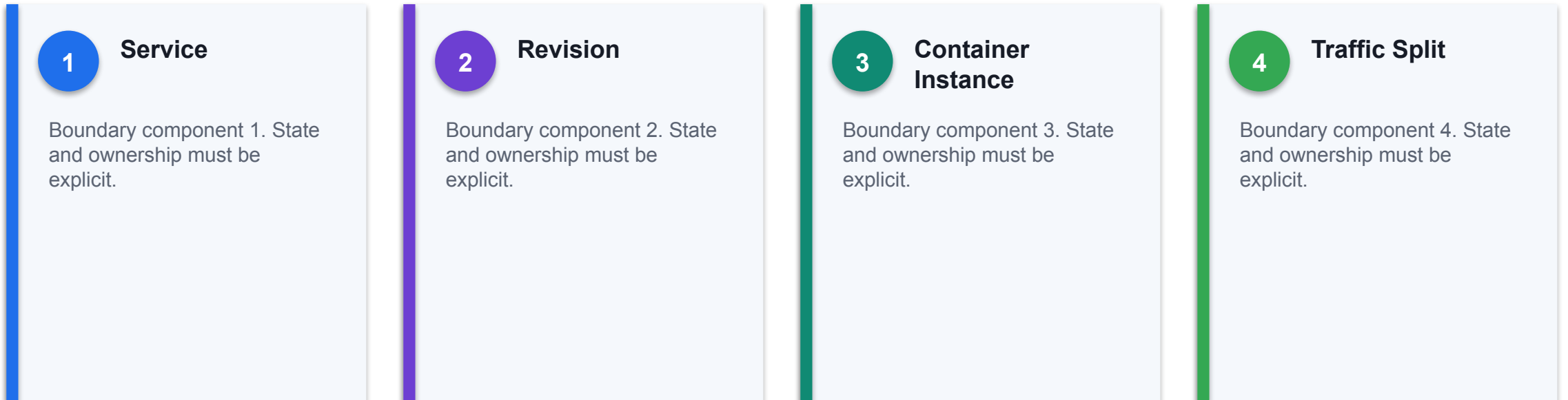
TOPIC 07 · A3 · K4

07

Managed Services

Deploy serverless and data services

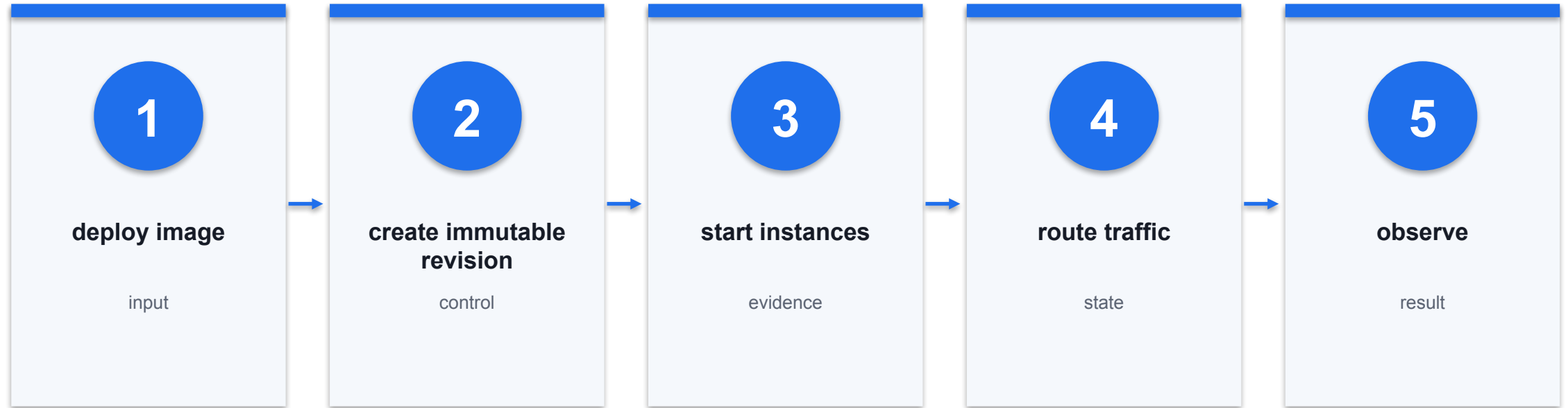
Cloud Run revision and traffic model · architecture



DECISION RULE

Scope first: Revision name, image digest, service identity and ingress are captured after deployment.

Cloud Run revision and traffic model · runtime trace



Revision name, image digest, service identity and ingress are captured after deployment.

Cloud Run revision and traffic model · concrete contract

```
gcloud run deploy ace-api --source=. --region=asia-01  
southeast1 --no-allow-unauthenticated
```

1

Scope: Managed Services

2

Mapped: A3 · K4

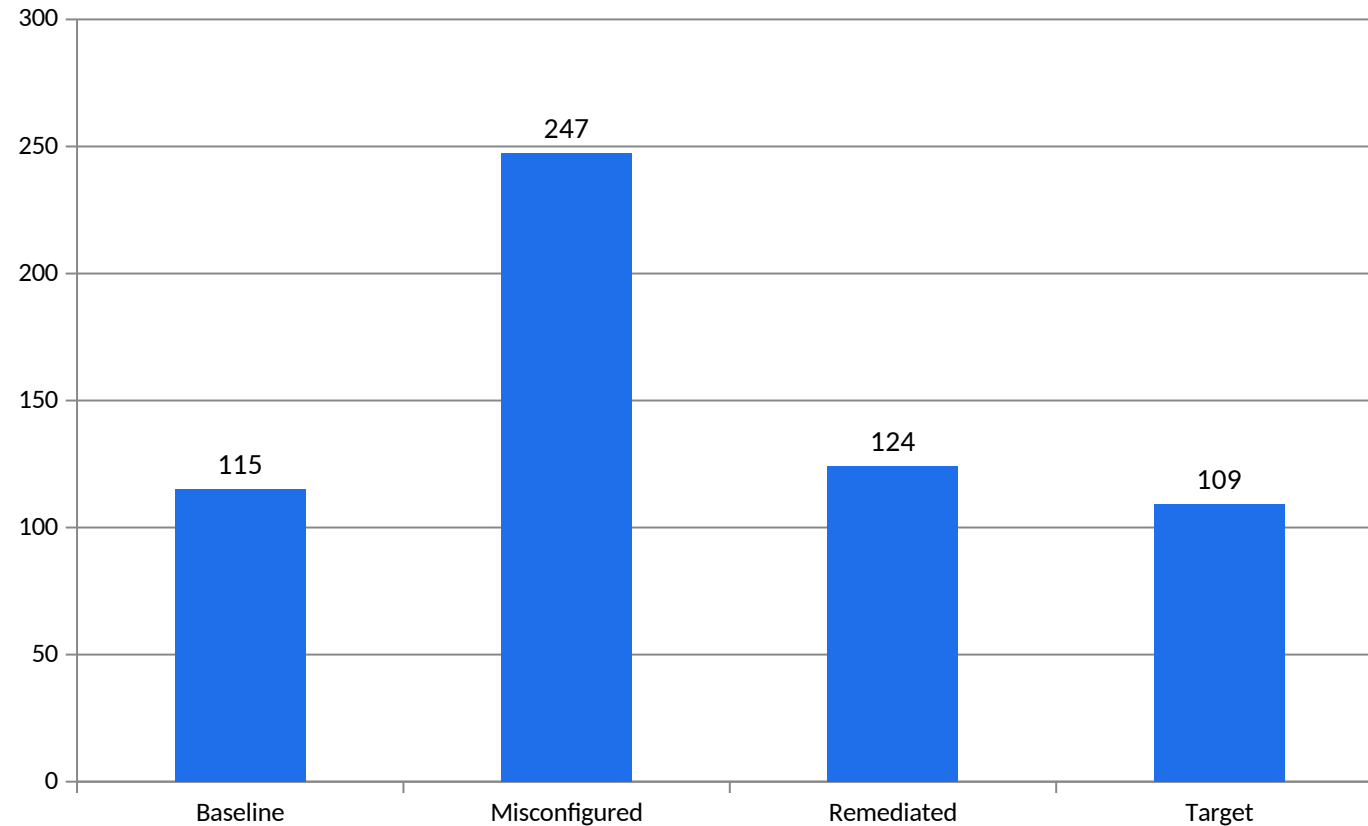
3

Capture stdout and exit status

4

Source: official cloud-run documentation

Topic 7 evidence profile



WHAT THE DATA SHOWS

A controlled fault should move the signal away from baseline. The remediation is accepted only when it returns close to target and the exact configuration change is recorded.

Indicative teaching data. Learners replace it with measurements from the aligned activity.

Cloud Run autoscaling controls · architecture

1

Cpu

Boundary component 1. State and ownership must be explicit.

2

Concurrency

Boundary component 2. State and ownership must be explicit.

3

Min Instances

Boundary component 3. State and ownership must be explicit.

4

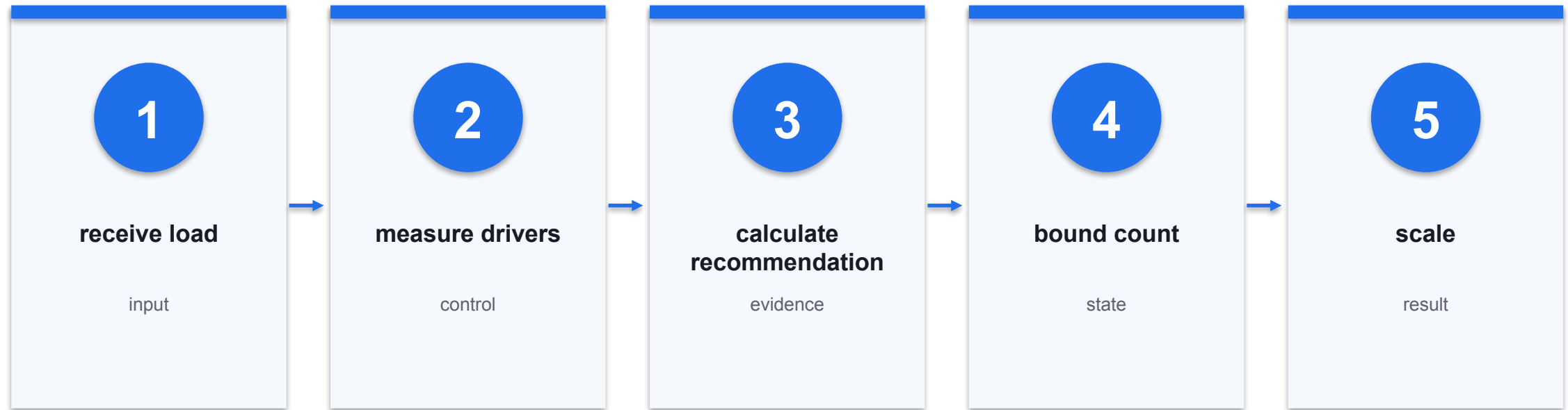
Max Instances

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Instance count and latency respond to load without overwhelming the database connection budget.

Cloud Run autoscaling controls · runtime trace



Instance count and latency respond to load without overwhelming the database connection budget.

Cloud Run autoscaling controls · concrete contract

```
gcloud run services update ace-api --region=asia-southeast1 --concurrency=40 --min=0 --max=10 01
```

1

Scope: Managed Services

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official cloud-run-autoscaling documentation

Cloud Run autoscaling controls · failure and control

FAILURE MODES



single-thread vCPU hotspot



max too high for DB



cold-start latency

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Instance count and latency respond to load without overwhelming the database connection budget.

Pub/Sub delivery contract · architecture

1

Topic

Boundary component 1. State and ownership must be explicit.

2

Subscription

Boundary component 2. State and ownership must be explicit.

3

Message

Boundary component 3. State and ownership must be explicit.

4

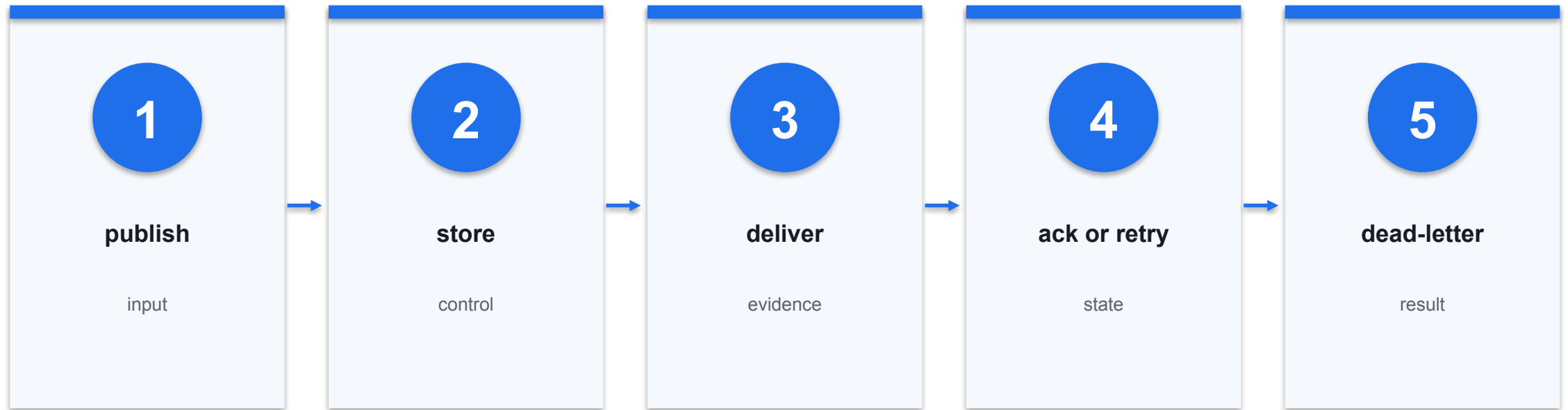
Ack Deadline

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Subscriber handles duplicate delivery idempotently and backlog metrics remain within target.

Pub/Sub delivery contract · runtime trace



Subscriber handles duplicate delivery idempotently and backlog metrics remain within target.

Pub/Sub delivery contract · concrete contract

```
gcloud pubsub topics create orders                                01
gcloud pubsub subscriptions create orders-worker -- 02
topic=orders --ack-deadline=30
```

1

Scope: Managed Services

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official pubsub documentation

Pub/Sub delivery contract · failure and control

FAILURE MODES



no ack



poison message loop



publisher permission missing

VERIFICATION EVIDENCE



Expected state is explicit



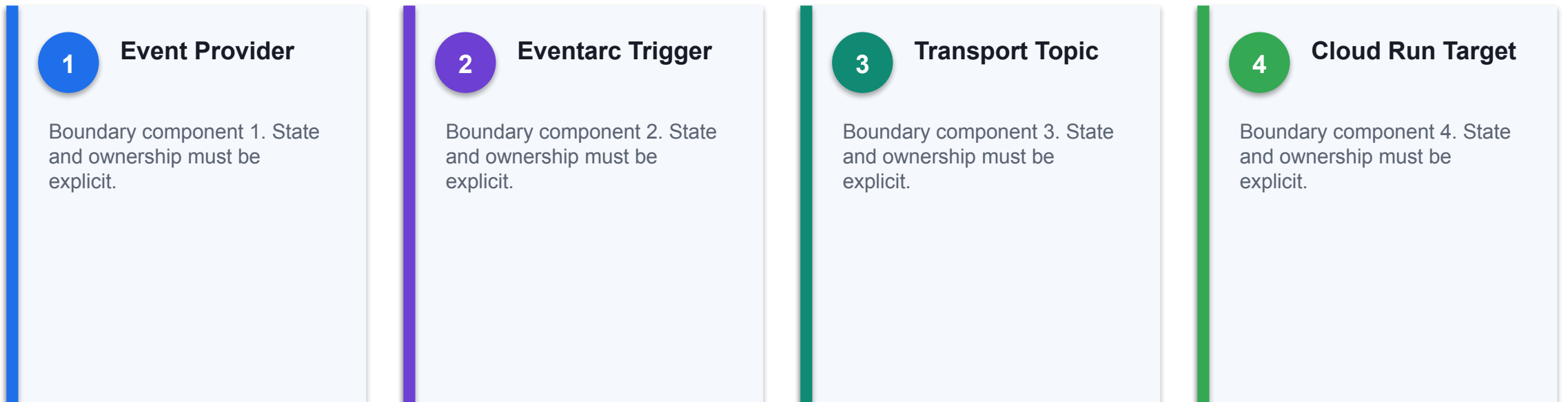
Observed output is captured



Timestamp and scope are recorded

Subscriber handles duplicate delivery idempotently and backlog metrics remain within target.

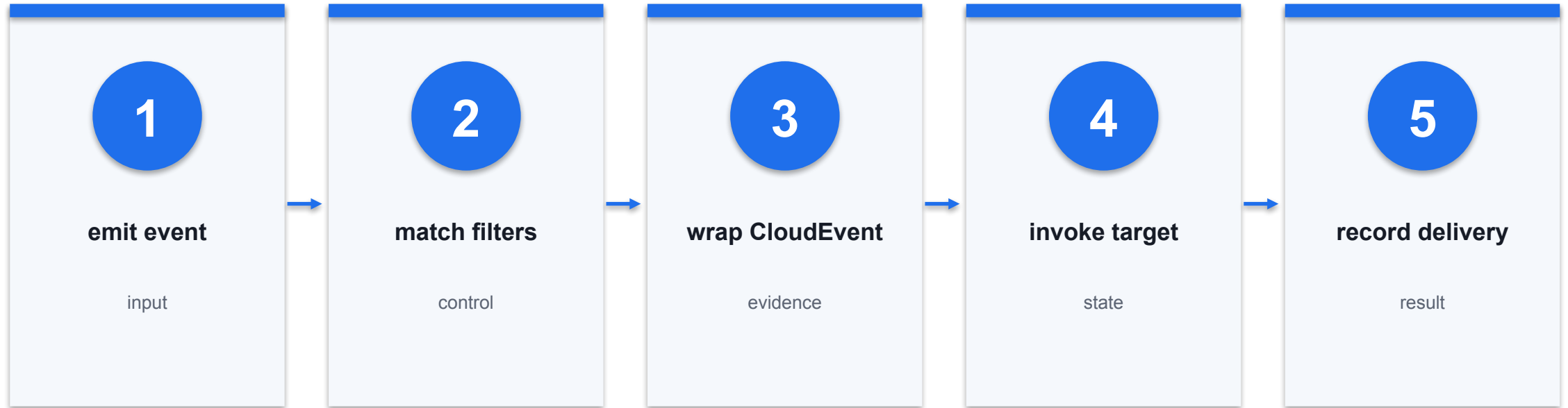
Eventarc event routing · architecture



DECISION RULE

Scope first: Trigger filters match the intended event type and target region.

Eventarc event routing · runtime trace



Trigger filters match the intended event type and target region.

Eventarc event routing · concrete contract

```
gcloud eventarc triggers list --location=asia-01  
southeast1 --  
format='table(name,eventFilters,destination) '
```

1

Scope: Managed Services

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official eventarc documentation

Eventarc event routing · failure and control

FAILURE MODES



region mismatch



service agent role missing



event filter typo

VERIFICATION EVIDENCE



Expected state is explicit



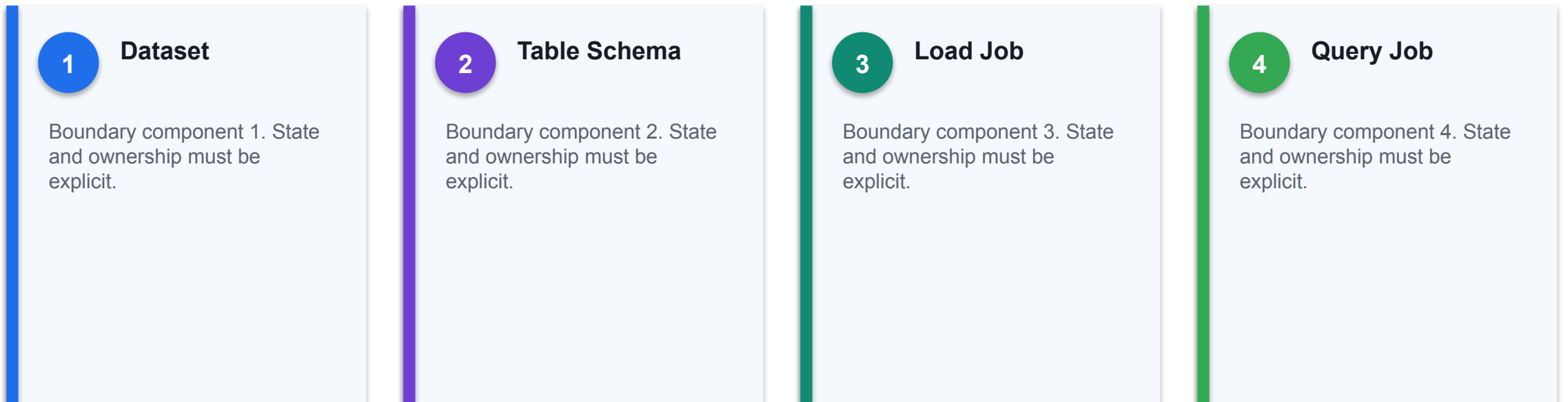
Observed output is captured



Timestamp and scope are recorded

Trigger filters match the intended event type and target region.

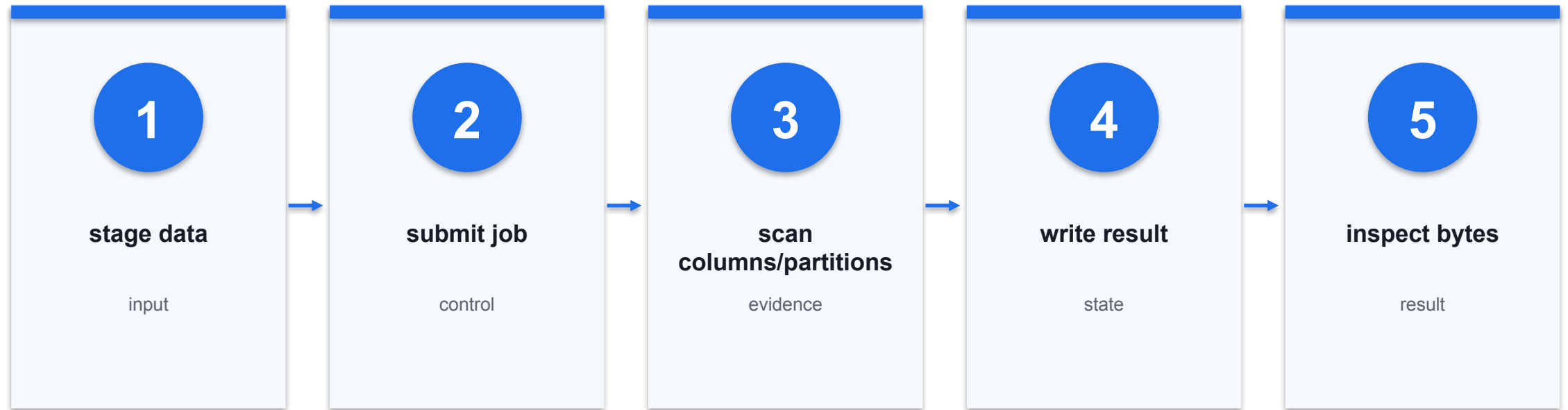
BigQuery load and query job model · architecture



DECISION RULE

Scope first: Job status is DONE with no error and bytes processed are reviewed before scale-up.

BigQuery load and query job model · runtime trace



Job status is DONE with no error and bytes processed are reviewed before scale-up.

BigQuery load and query job model · concrete contract

```
bq load --source_format=CSV --autodetect ops.events 01
gs://$BUCKET/events.csv
bq query --use_legacy_sql=false 'SELECT          02
severity,COUNT(*) n FROM `ops.events` GROUP BY
severity'
```

1

Scope: Managed Services

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official bigquery documentation

BigQuery load and query job model · failure and control

FAILURE MODES



schema autodetect drift



unbounded scan



dataset location mismatch

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Job status is DONE with no error and bytes processed are reviewed before scale-up.

Dataflow pipeline boundaries · architecture

1

Source

Boundary component 1. State and ownership must be explicit.

2

Apache Beam Transforms

Boundary component 2. State and ownership must be explicit.

3

Runner Workers

Boundary component 3. State and ownership must be explicit.

4

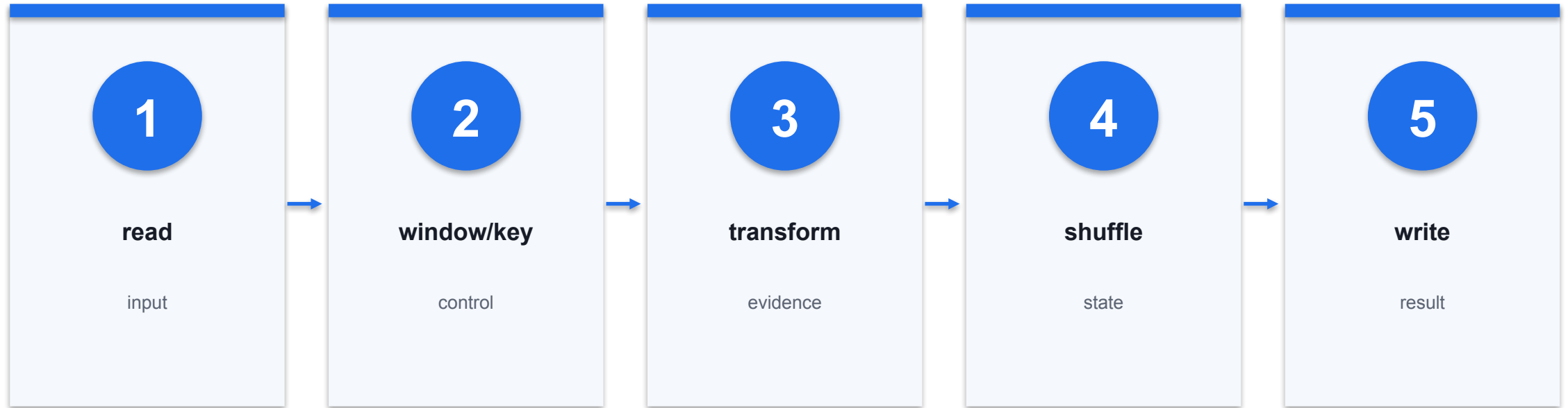
Sink

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Job graph, worker logs, watermark and failed records show pipeline health.

Dataflow pipeline boundaries · runtime trace



Job graph, worker logs, watermark and failed records show pipeline health.

Dataflow pipeline boundaries · concrete contract

```
gcloud dataflow jobs list --region=asia-southeast1 --1  
status=active
```

1

Scope: Managed Services

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official dataflow documentation

Dataflow pipeline boundaries · failure and control

FAILURE MODES



hot key



serialization failure



sink quota

VERIFICATION EVIDENCE



Expected state is explicit



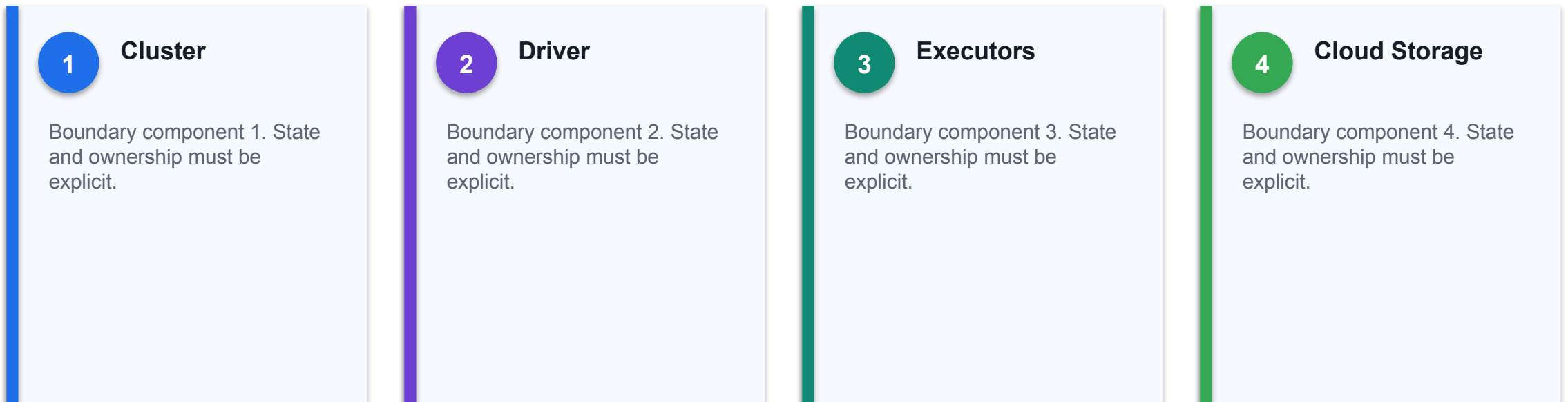
Observed output is captured



Timestamp and scope are recorded

Job graph, worker logs, watermark and failed records show pipeline health.

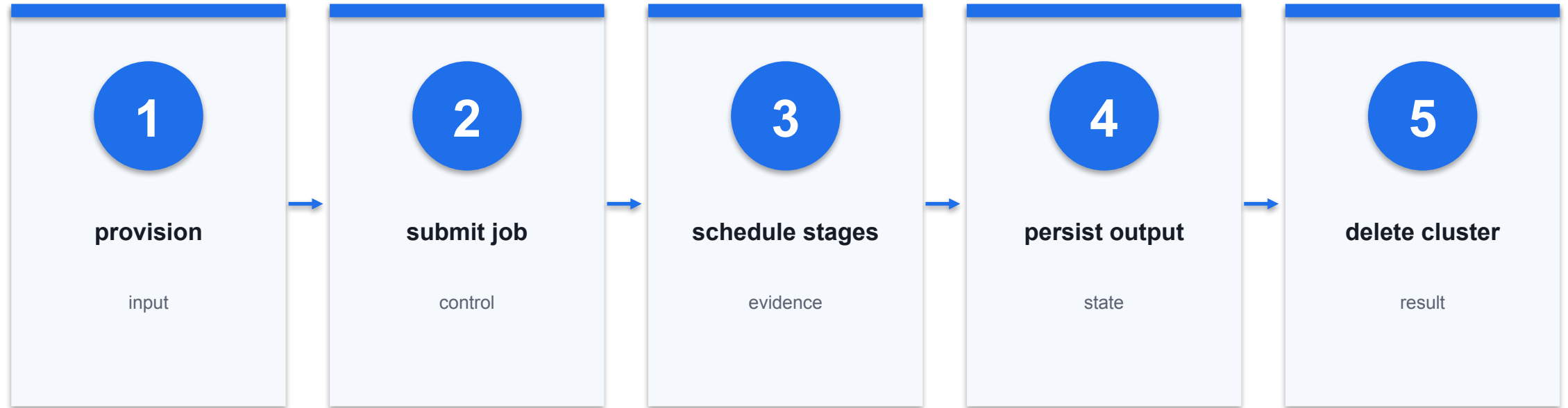
Dataproc cluster and job separation · architecture



DECISION RULE

Scope first: Ephemeral compute is separated from durable input/output and job state is recorded.

Dataproc cluster and job separation · runtime trace



Ephemeral compute is separated from durable input/output and job state is recorded.

Dataproc cluster and job separation · concrete contract

```
gcloud dataproc jobs list --region=asia-southeast1 --format='table(reference.jobId,status.state,placement.clusterName)'
```

1

Scope: Managed Services

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official dataproc documentation

Dataproc cluster and job separation · failure and control

FAILURE MODES



cluster left running



initialization action fails



shuffle disk pressure

VERIFICATION EVIDENCE



Expected state is explicit



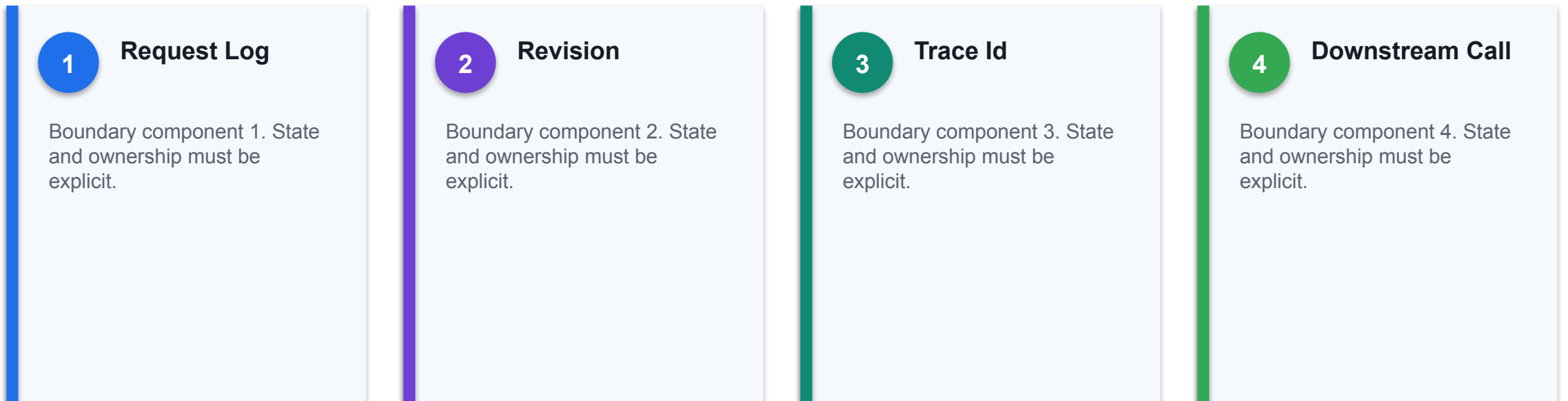
Observed output is captured



Timestamp and scope are recorded

Ephemeral compute is separated from durable input/output and job state is recorded.

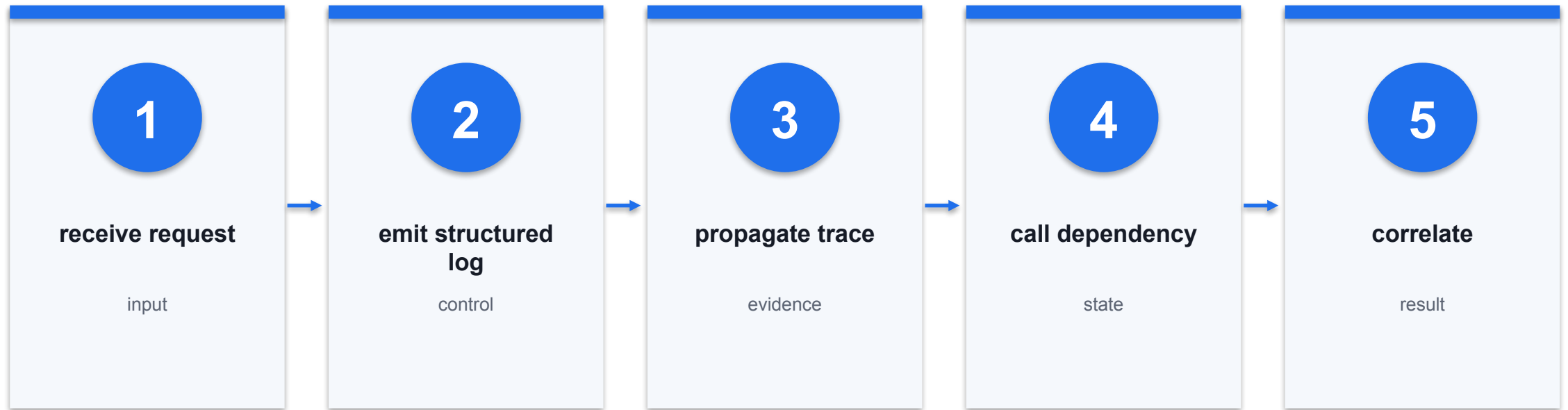
Serverless incident trace · architecture



DECISION RULE

Scope first: One trace identifier links entry, application error and downstream latency.

Serverless incident trace · runtime trace



One trace identifier links entry, application error and downstream latency.

Serverless incident trace · concrete contract

```
gcloud logging read                                01  
'resource.type="cloud_run_revision" AND  
resource.labels.service_name="ace-api"' --limit=30 --  
format=json
```

1

Scope: Managed Services

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official observability documentation

Serverless incident trace · failure and control

FAILURE MODES



unstructured logs



trace header dropped



wrong revision receives traffic

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

One trace identifier links entry, application error and downstream latency.

Build a Pub Sub to Cloud Run Event Pipeline

SCENARIO

Deploy a small event receiver, publish mock orders and diagnose retry behaviour without losing messages.

YOU WILL PRODUCE

Cloud Run revision, Pub/Sub subscription and processed event log

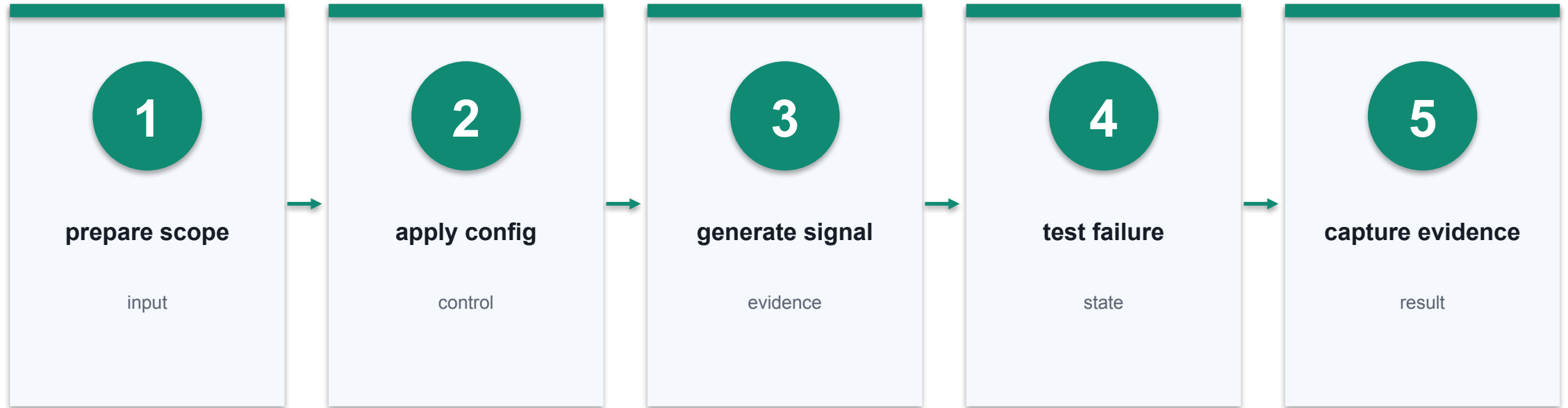
app.py

requirements.txt

orders.jsonl

Full click-by-click instructions are in the Learner Guide and Activity PDF. This slide shows the work product, not the procedure.

Activity 7 system trace



Deploy a small event receiver, publish mock orders and diagnose retry behaviour without losing messages.

Activity 7 worked artifact

```
import base64, json                                01
from flask import Flask, request                    02
app = Flask(__name__)                               03
@app.post('/')                                       04
def receive():                                     05
    envelope = request.get_json(force=True)         06
    msg = envelope.get('message', {})               07
    payload = json.loads(base64.b64decode(msg.get('data', '') or 08
    'e30='))
    print(json.dumps({'message_id': msg.get('messageId'), 'payload':09
    payload}))
    return ('', 204)                                10
```

1

Use placeholders for project-specific values

2

Run only in the isolated training project

3

Capture both success and failure output

4

Keep secrets out of submitted evidence

Activity 7 acceptance gate

FAILURE MODES



wrong project or region



insufficient IAM or disabled API



observed output does not meet target

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Accept when Cloud Run revision, Pub/Sub subscription and processed event log are complete, timestamped and reproducible.

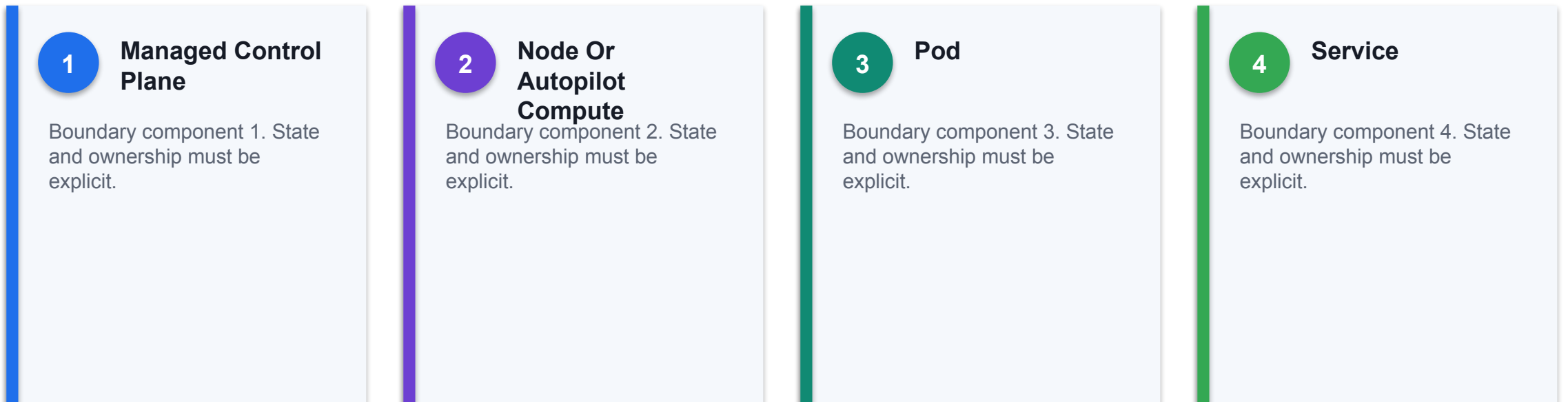
TOPIC 08 · A3 · K4

08

Kubernetes

Deploy and operate GKE

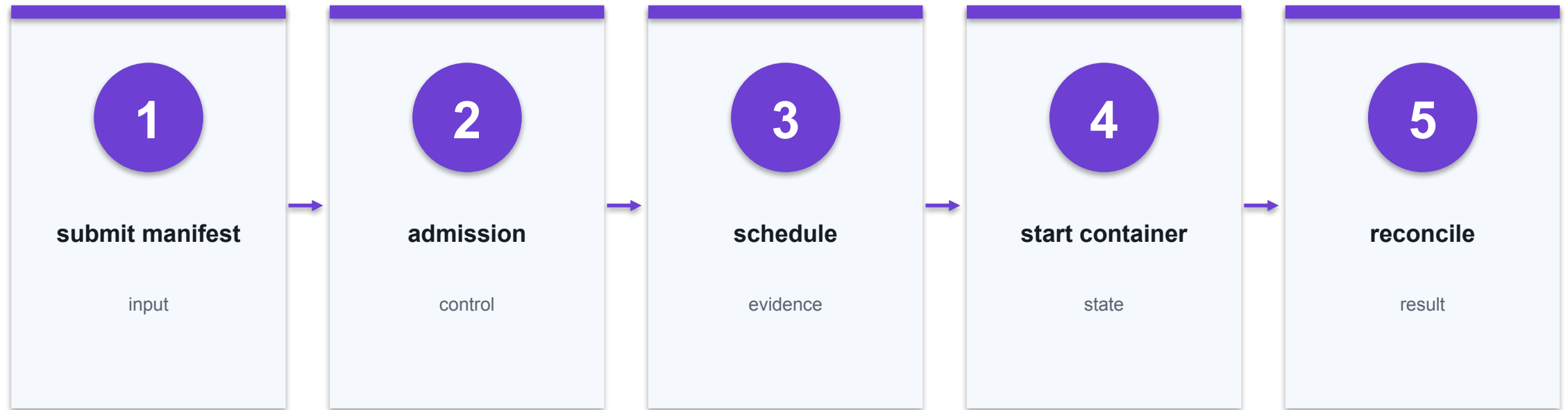
GKE control plane and workload boundary · architecture



DECISION RULE

Scope first: Cluster endpoint is reachable and credentials bind kubectl to the intended context.

GKE control plane and workload boundary · runtime trace



Cluster endpoint is reachable and credentials bind kubectl to the intended context.

GKE control plane and workload boundary · concrete contract

```
gcloud container clusters create-auto ace-gke -- 01  
region=asia-southeast1  
gcloud container clusters get-credentials ace-gke --02  
region=asia-southeast1
```

1

Scope: Kubernetes

2

Mapped: A3 · K4

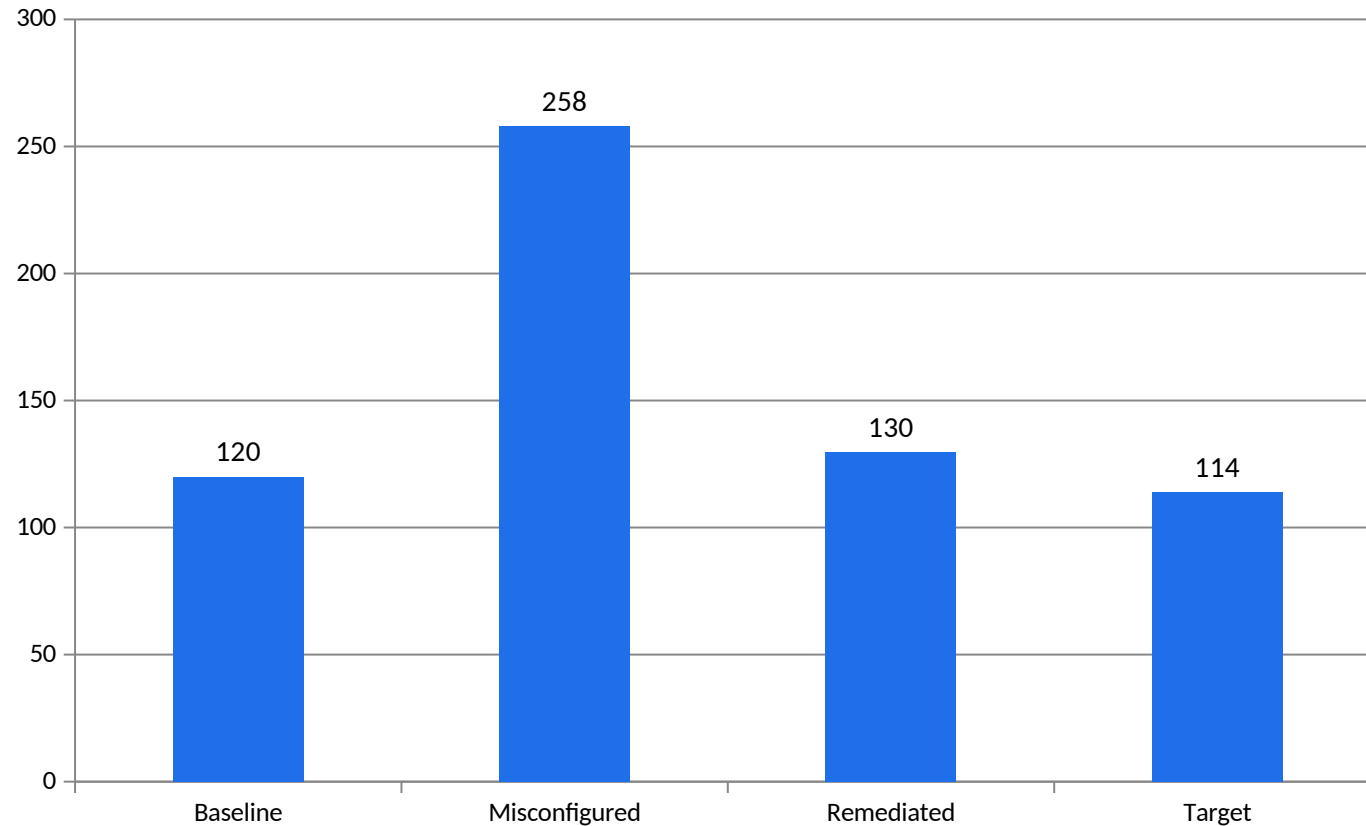
3

Capture stdout and exit status

4

Source: official gke-autopilot documentation

Topic 8 evidence profile

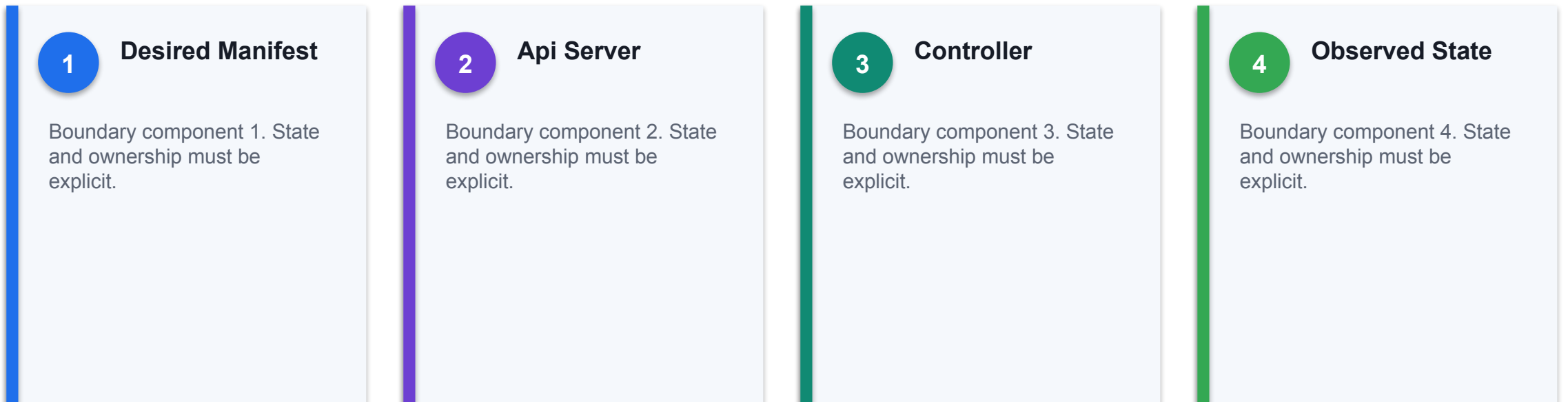


WHAT THE DATA SHOWS

A controlled fault should move the signal away from baseline. The remediation is accepted only when it returns close to target and the exact configuration change is recorded.

Indicative teaching data. Learners replace it with measurements from the aligned activity.

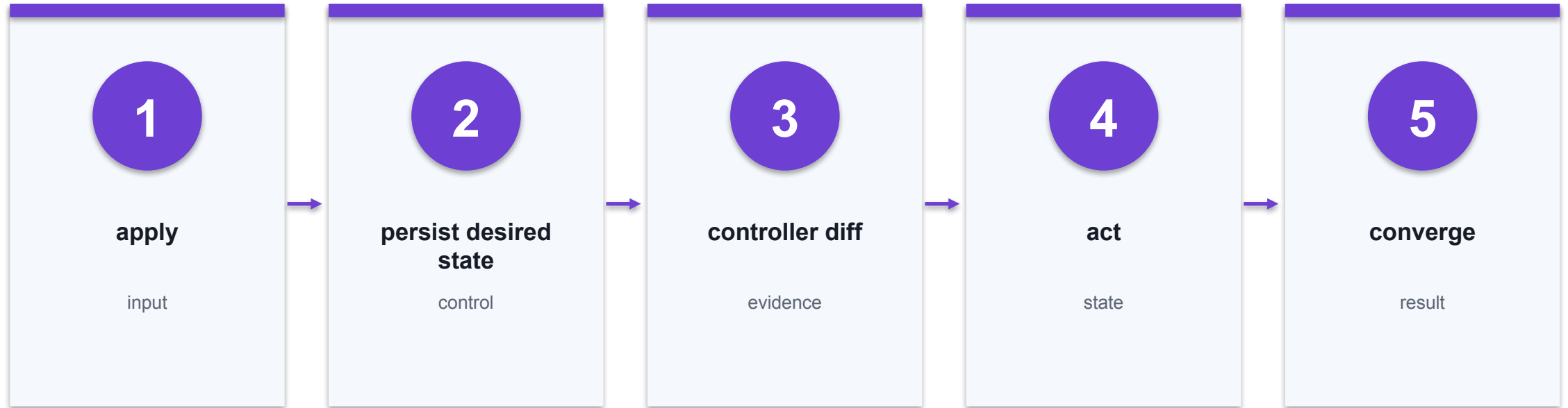
Kubernetes declarative reconciliation · architecture



DECISION RULE

Scope first: Deployment AVAILABLE replicas equal desired replicas and Pods are Ready.

Kubernetes declarative reconciliation · runtime trace



Deployment **AVAILABLE** replicas equal desired replicas and Pods are Ready.

Kubernetes declarative reconciliation · concrete contract

```
kubectl apply -f deployment.yaml 01  
kubectl get deploy,pods,svc -o wide 02
```

1

Scope: Kubernetes

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official kubernetes documentation

Kubernetes declarative reconciliation · failure and control

FAILURE MODES



selector mismatch



image pull error



readiness probe fails

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Deployment AVAILABLE replicas equal desired replicas and Pods are Ready.

Resource requests, limits and scheduling · architecture

1

Pod Request

Boundary component 1. State and ownership must be explicit.

2

Scheduler

Boundary component 2. State and ownership must be explicit.

3

Node Capacity

Boundary component 3. State and ownership must be explicit.

4

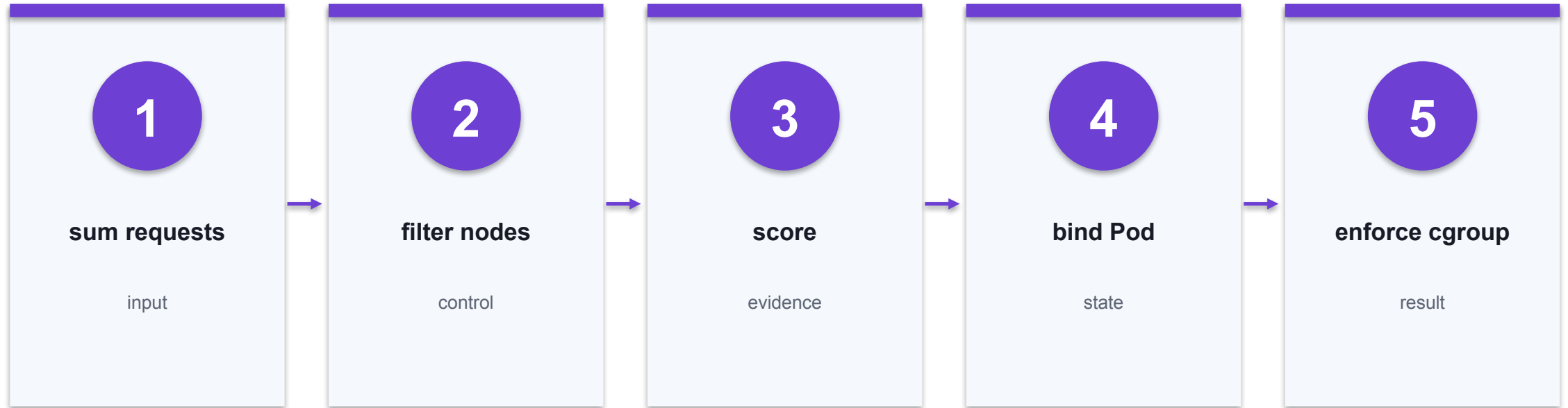
Container Limit

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Requests reflect observed workload and explain both scheduling and autoscaling decisions.

Resource requests, limits and scheduling · runtime trace



Requests reflect observed workload and explain both scheduling and autoscaling decisions.

Resource requests, limits and scheduling · concrete contract

```
kubectl describe pod -l app=ace-web 01  
kubectl top pods 02
```

1

Scope: Kubernetes

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official gke-resources documentation

Resource requests, limits and scheduling · failure and control

FAILURE MODES



Pending due to insufficient CPU



OOMKilled



no metrics

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Requests reflect observed workload and explain both scheduling and autoscaling decisions.

Service discovery and traffic · architecture

1

Service Selector

Boundary component 1. State and ownership must be explicit.

2

Endpointslice

Boundary component 2. State and ownership must be explicit.

3

Cluster Ip

Boundary component 3. State and ownership must be explicit.

4

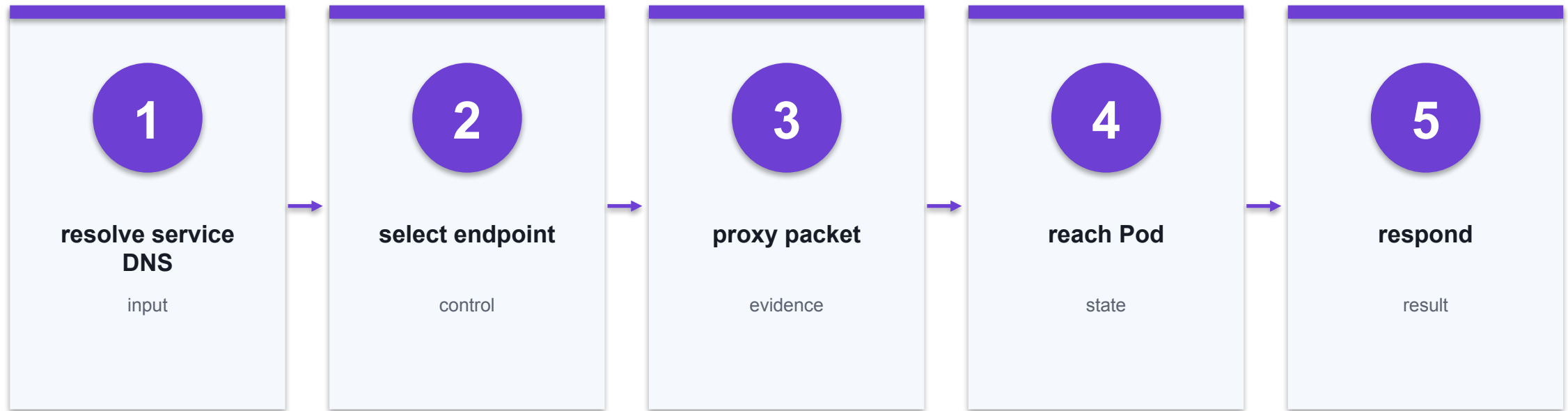
Pod Targetport

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Selector labels produce ready endpoints and port/targetPort match the container.

Service discovery and traffic · runtime trace



Selector labels produce ready endpoints and port/targetPort match the container.

Service discovery and traffic · concrete contract

```
kubectl get svc,endpointslices -o wide 01  
kubectl describe svc ace-web 02
```

1

Scope: Kubernetes

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official gke-networking documentation

Service discovery and traffic · failure and control

FAILURE MODES



zero endpoints



wrong targetPort



NetworkPolicy blocks flow

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Selector labels produce ready endpoints and port/targetPort match the container.

Artifact Registry image pull · architecture

1

Repository

Boundary component 1. State and ownership must be explicit.

2

Image Digest

Boundary component 2. State and ownership must be explicit.

3

Node Identity

Boundary component 3. State and ownership must be explicit.

4

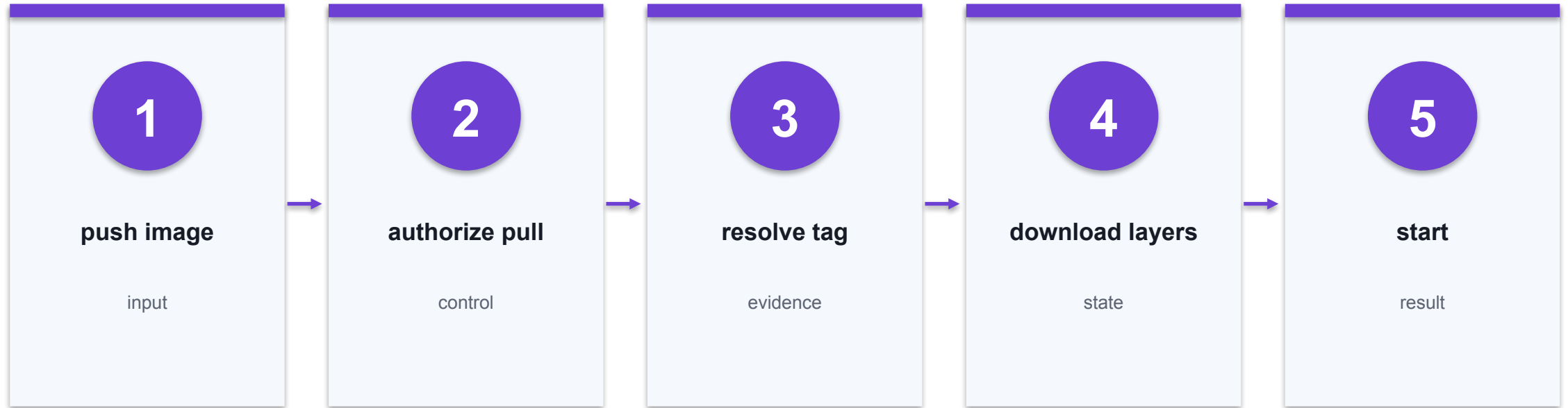
Pod Spec

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Running Pod imageID records the immutable digest expected from the build.

Artifact Registry image pull · runtime trace



Running Pod imageID records the immutable digest expected from the build.

Artifact Registry image pull · concrete contract

```
gcloud artifacts docker images list asia-southeast1-01  
docker.pkg.dev/$PROJECT_ID/ace/apps --include-tags
```

1

Scope: Kubernetes

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official artifact-registry documentation

Artifact Registry image pull · failure and control

FAILURE MODES



repository region typo



node lacks reader role



mutable tag drift

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Running Pod imageID records the immutable digest expected from the build.

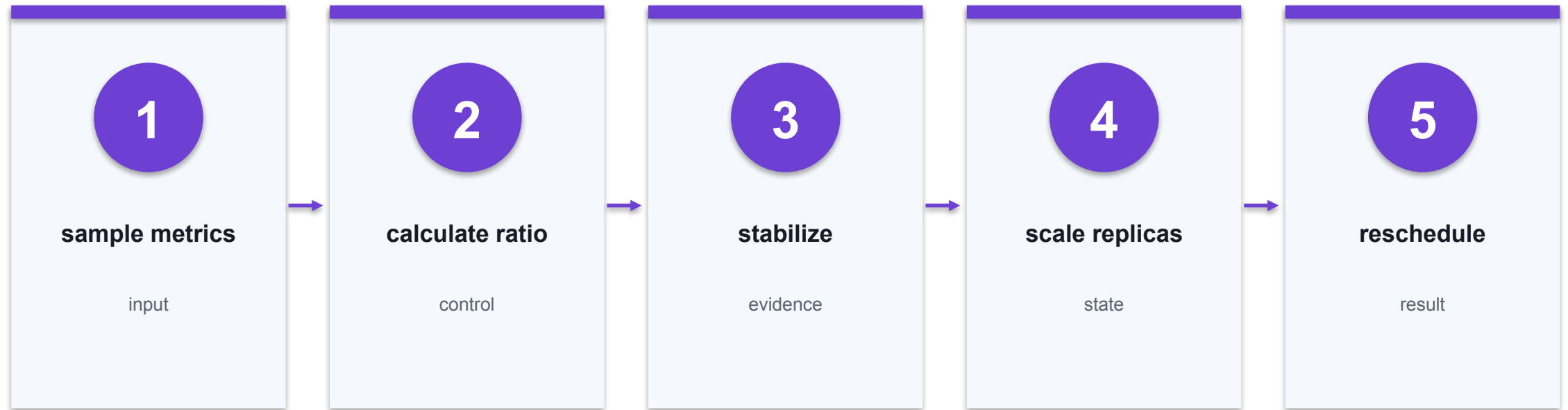
Horizontal Pod Autoscaler control loop · architecture



DECISION RULE

Scope first: HPA shows current/target metric and increases replicas during the load window.

Horizontal Pod Autoscaler control loop · runtime trace



HPA shows current/target metric and increases replicas during the load window.

Horizontal Pod Autoscaler control loop · concrete contract

```
kubectl autoscale deployment ace-web --cpu-percent=60  
--min=2 --max=10  
kubectl get hpa -w
```

1

Scope: Kubernetes

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official hpa documentation

Horizontal Pod Autoscaler control loop · failure and control

FAILURE MODES



CPU requests missing



metrics unavailable



load too short

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

HPA shows current/target metric and increases replicas during the load window.

Cluster autoscaling and Autopilot · architecture

1 Unschedulable Pod

Boundary component 1. State and ownership must be explicit.

2 Requested Resources

Boundary component 2. State and ownership must be explicit.

3 Autopilot Provisioner

Boundary component 3. State and ownership must be explicit.

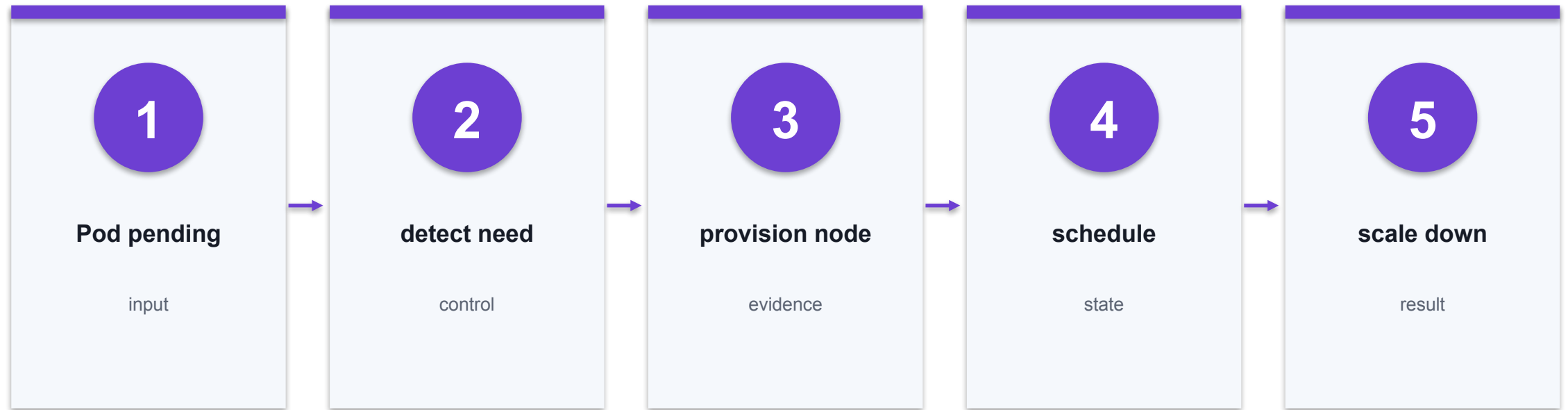
4 New Capacity

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Events distinguish Pod HPA from underlying capacity provisioning.

Cluster autoscaling and Autopilot · runtime trace



Events distinguish Pod HPA from underlying capacity provisioning.

Cluster autoscaling and Autopilot · concrete contract

```
kubectl get events --sort-by=.lastTimestamp | tail -30
```

1

Scope: Kubernetes

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official gke-autopilot documentation

Cluster autoscaling and Autopilot · failure and control

FAILURE MODES



unsupported workload constraint



quota exhausted



first workload provisioning delay

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Events distinguish Pod HPA from underlying capacity provisioning.

GKE troubleshooting evidence · architecture

1

Pod Status

Boundary component 1. State and ownership must be explicit.

2

Events

Boundary component 2. State and ownership must be explicit.

3

Container Logs

Boundary component 3. State and ownership must be explicit.

4

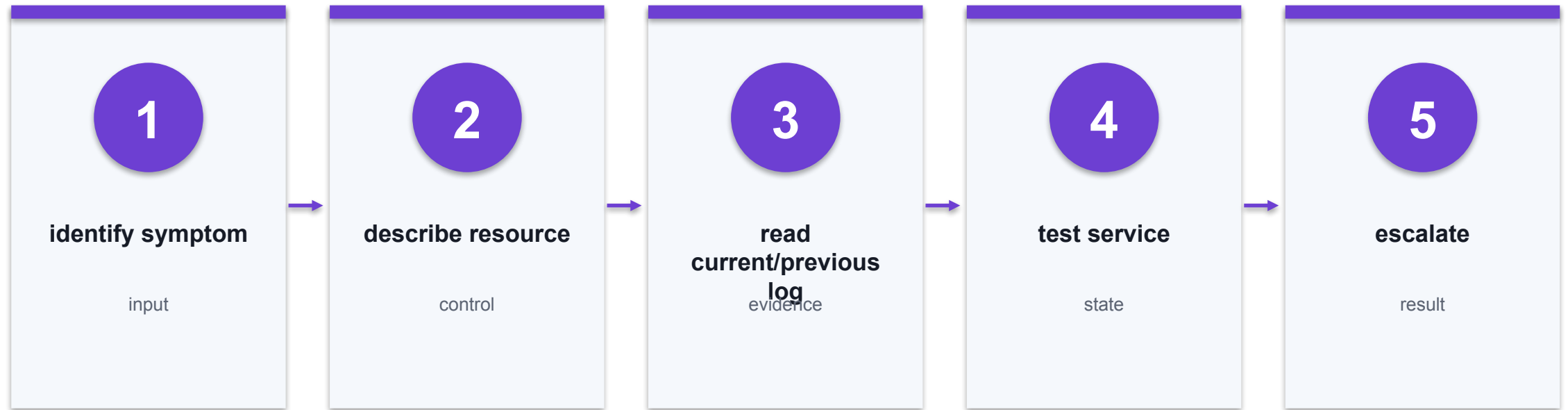
Cluster Metrics

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Incident bundle preserves YAML, events, logs, metrics and the exact failed test.

GKE troubleshooting evidence · runtime trace



Incident bundle preserves YAML, events, logs, metrics and the exact failed test.

GKE troubleshooting evidence · concrete contract

```
kubectl describe pod $POD 01
kubectl logs $POD --previous 02
kubectl get events --sort-by=.lastTimestamp 03
```

1

Scope: Kubernetes

2

Mapped: A3 · K4

3

Capture stdout and exit status

4

Source: official gke-troubleshooting documentation

GKE troubleshooting evidence · failure and control

FAILURE MODES



Pod recreated before capture



namespace omitted



only application log collected

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Incident bundle preserves YAML, events, logs, metrics and the exact failed test.

Deploy and Test a GKE Autopilot Workload

SCENARIO

Deploy a containerized web workload with probes and resource requests, add HPA and diagnose why replicas do not scale.

YOU WILL PRODUCE

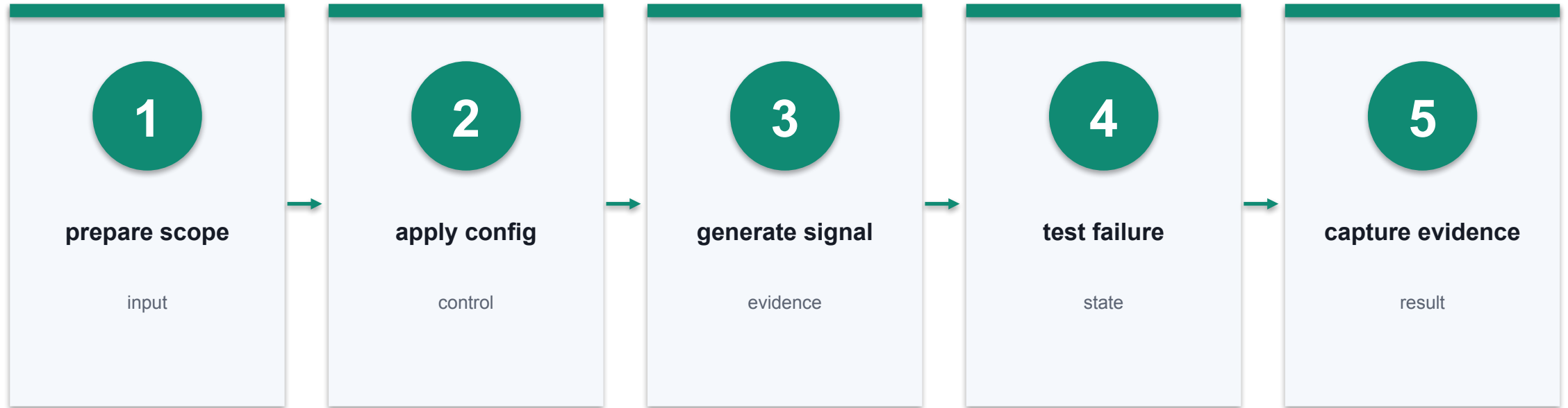
Kubernetes manifests, load-test CSV and HPA evidence

deployment.yaml

hpa.yaml

Full click-by-click instructions are in the Learner Guide and Activity PDF. This slide shows the work product, not the procedure.

Activity 8 system trace



Deploy a containerized web workload with probes and resource requests, add HPA and diagnose why replicas do not scale.

Activity 8 worked artifact

```

apiVersion: apps/v1                                01
kind: Deployment                                    02
metadata: {name: ace-web}                            03
spec:                                                04
  replicas: 2                                         05
  selector: {matchLabels: {app: ace-web}}            06
  template:                                          07
    metadata: {labels: {app: ace-web}}                08
    spec:                                            09
      containers:                                    10
        - name: web                                  11
          image: us-docker.pkg.dev/google-samples/containers/gke/hello12
app:1.0
          resources:                                  13
            requests: {cpu: 250m, memory: 256Mi}      14
            limits: {cpu: 500m, memory: 512Mi}        15
            readinessProbe: {httpGet: {path: /, port: 8080},
initialDelaySeconds: 5}                             16

```

1

Use placeholders for project-specific values

2

Run only in the isolated training project

3

Capture both success and failure output

4

Keep secrets out of submitted evidence

Activity 8 acceptance gate

FAILURE MODES



wrong project or region



insufficient IAM or disabled API



observed output does not meet target

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Accept when Kubernetes manifests, load-test CSV and HPA evidence are complete, timestamped and reproducible.

TOPIC 09 · A5 · K2

Resource Management and Monitoring

Ensure successful operation



Asset inventory and labels · architecture

1

Cloud Asset Inventory

Boundary component 1. State and ownership must be explicit.

2

Resource Name

Boundary component 2. State and ownership must be explicit.

3

Labels

Boundary component 3. State and ownership must be explicit.

4

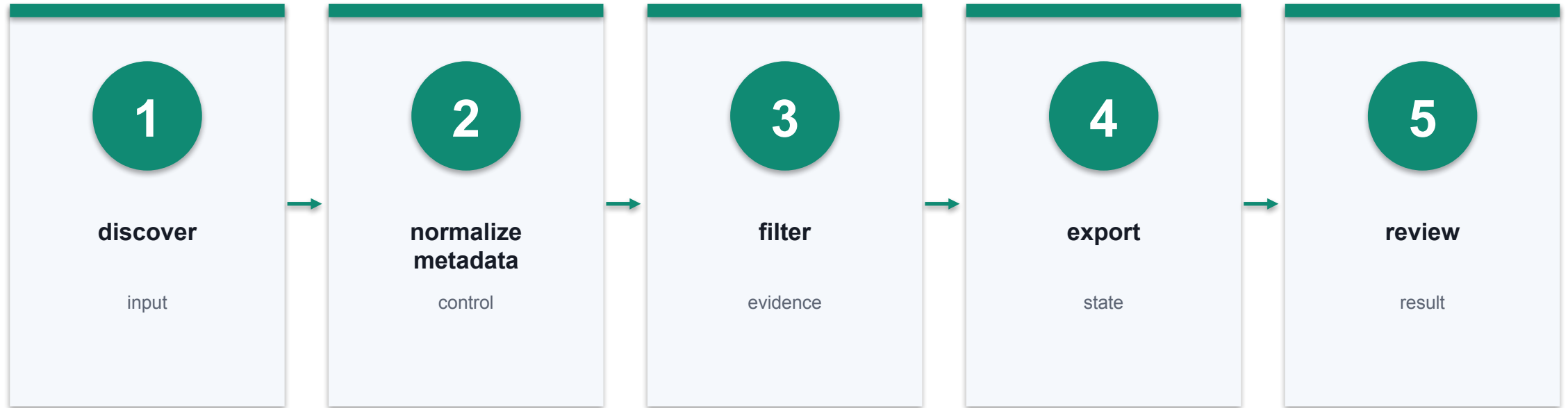
Search Query

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Inventory finds all expected resources and exposes unlabelled or wrong-location assets.

Asset inventory and labels · runtime trace



Inventory finds all expected resources and exposes unlabelled or wrong-location assets.

Asset inventory and labels · concrete contract

```
gcloud asset search-all-resources  
--scope=projects/$PROJECT_NUMBER --  
query='labels.env=training' --  
format='table(assetType,name,location) '
```

01

1

Scope: Resource Management and Monitoring

2

Mapped: A5 · K2

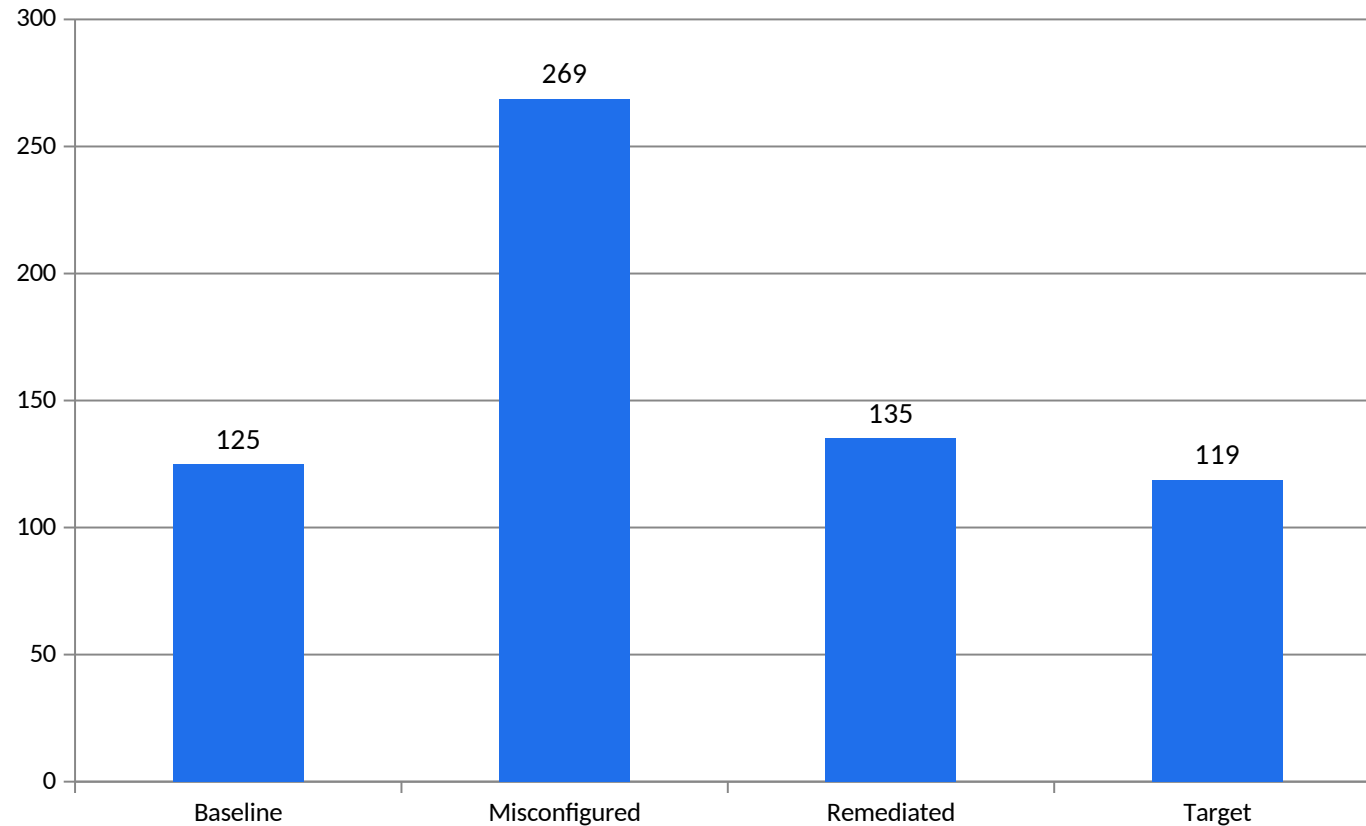
3

Capture stdout and exit status

4

Source: official asset-inventory documentation

Topic 9 evidence profile



WHAT THE DATA SHOWS

A controlled fault should move the signal away from baseline. The remediation is accepted only when it returns close to target and the exact configuration change is recorded.

Indicative teaching data. Learners replace it with measurements from the aligned activity.

Quota versus system limit · architecture

1

Quota Metric

Boundary component 1. State and ownership must be explicit.

2

Regional Dimension

Boundary component 2. State and ownership must be explicit.

3

Consumer Override

Boundary component 3. State and ownership must be explicit.

4

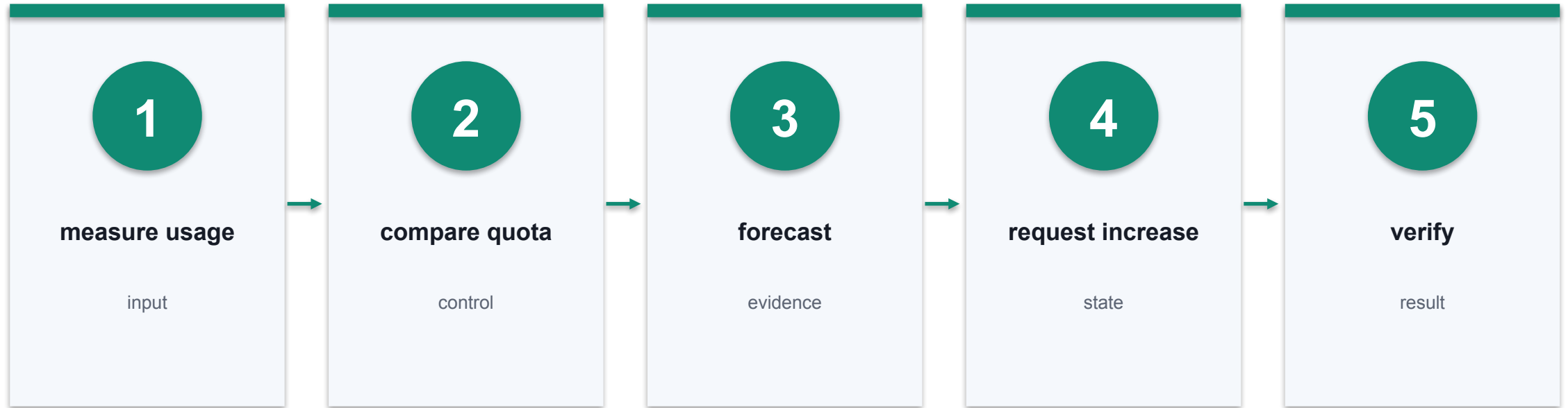
Hard Limit

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Capacity plan identifies the quota and region before deployment pressure.

Quota versus system limit · runtime trace



Capacity plan identifies the quota and region before deployment pressure.

Quota versus system limit · concrete contract

```
gcloud compute regions describe asia-southeast1 -- 01  
format='table(quotas.metric,quotas.usage,quotas.limit  
) '
```

1

Scope: Resource Management and Monitoring

2

Mapped: A5 · K2

3

Capture stdout and exit status

4

Source: official quotas documentation

Quota versus system limit · failure and control

FAILURE MODES



quota confused with capacity



request too late



per-project versus per-region

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Capacity plan identifies the quota and region before deployment pressure.

Metrics, logs and traces · architecture

1

Time Series

Boundary component 1. State and ownership must be explicit.

2

Log Entry

Boundary component 2. State and ownership must be explicit.

3

Trace Span

Boundary component 3. State and ownership must be explicit.

4

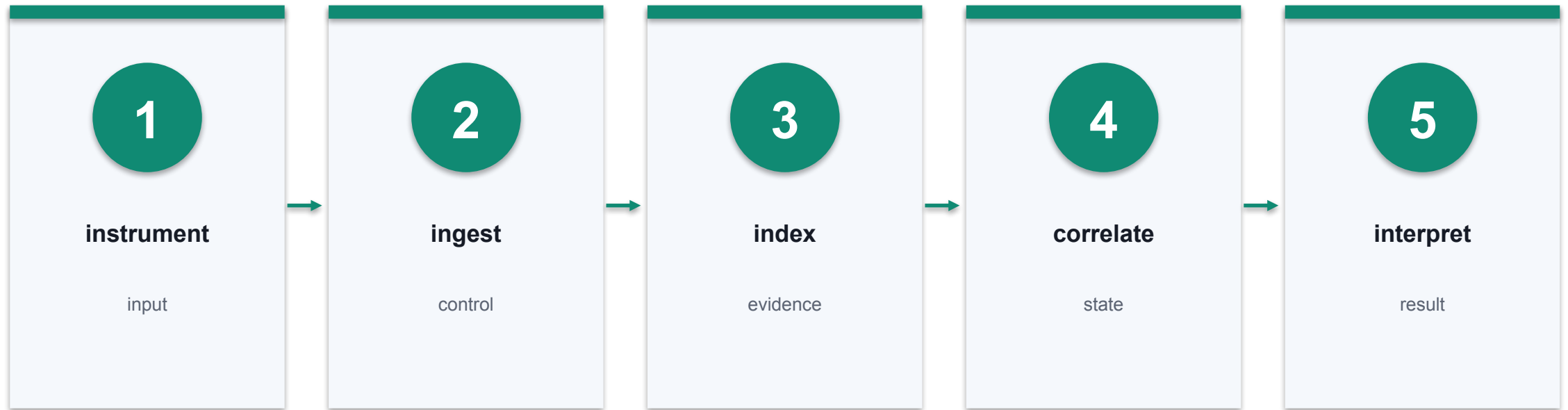
Resource Labels

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Signal includes resource type, metric/log name, timestamp and useful labels.

Metrics, logs and traces · runtime trace



Signal includes resource type, metric/log name, timestamp and useful labels.

Metrics, logs and traces · concrete contract

```
gcloud monitoring metrics list --  
filter='metric.type:compute.googleapis.com' --  
limit=10
```

01

1

Scope: Resource Management and Monitoring

2

Mapped: A5 · K2

3

Capture stdout and exit status

4

Source: official observability documentation

Metrics, logs and traces · failure and control

FAILURE MODES



metric has wrong alignment



logs lack context



trace sampling gap

VERIFICATION EVIDENCE



Expected state is explicit



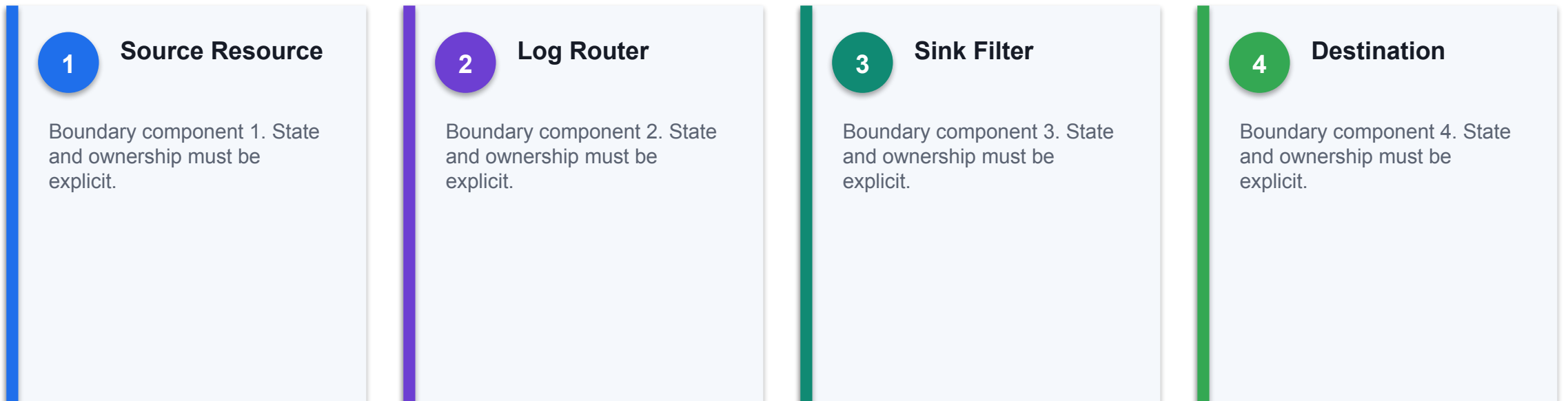
Observed output is captured



Timestamp and scope are recorded

Signal includes resource type, metric/log name, timestamp and useful labels.

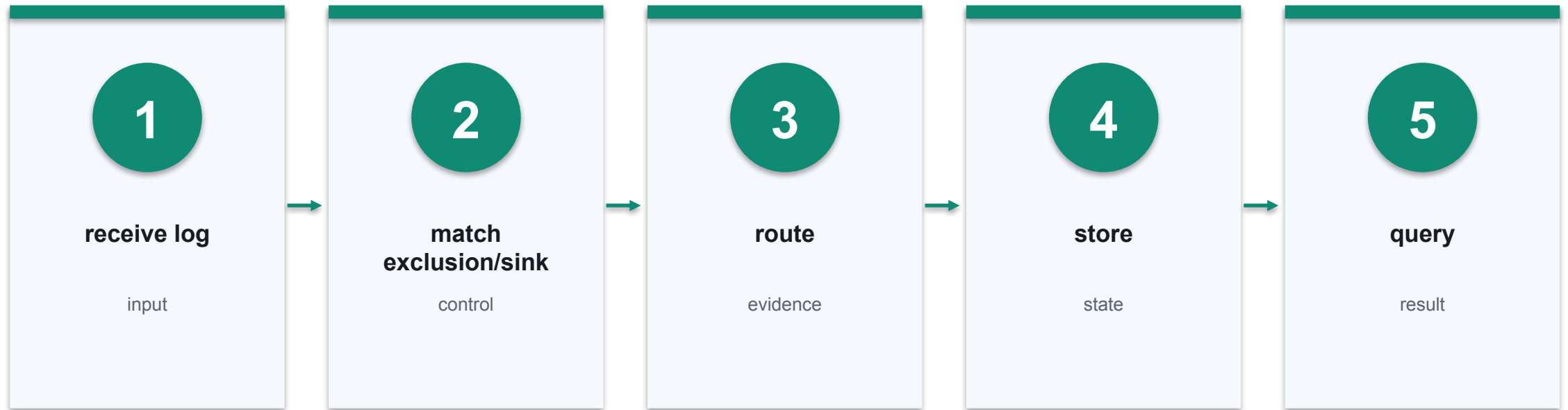
Cloud Logging router · architecture



DECISION RULE

Scope first: Sink destination and filter are explicit; required audit logs remain retained.

Cloud Logging router · runtime trace



Sink destination and filter are explicit; required audit logs remain retained.

Cloud Logging router · concrete contract

```
gcloud logging sinks list --  
format='table(name,destination,filter)'
```

01

1

Scope: Resource Management and Monitoring

2

Mapped: A5 · K2

3

Capture stdout and exit status

4

Source: official logging-routing documentation

Cloud Logging router · failure and control

FAILURE MODES



recursive routing



writer identity lacks destination role



exclusion drops evidence

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Sink destination and filter are explicit; required audit logs remain retained.

Log-based metric contract · architecture

1

Log Filter

Boundary component 1. State and ownership must be explicit.

2

Counter Metric

Boundary component 2. State and ownership must be explicit.

3

Time Series

Boundary component 3. State and ownership must be explicit.

4

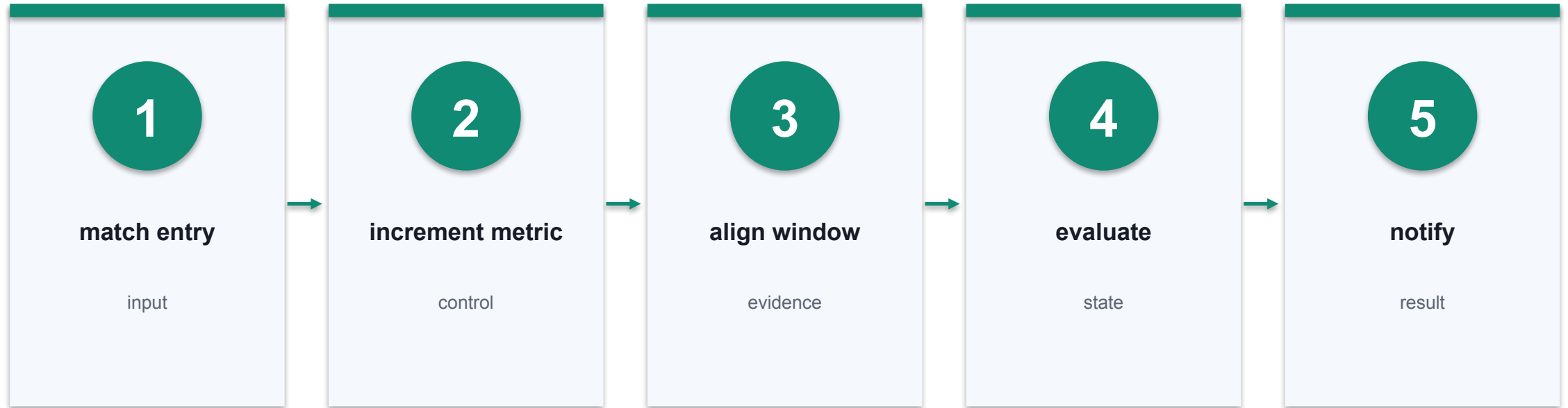
Alert Policy

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Test error increments the metric and can be tied to a known trace or revision.

Log-based metric contract · runtime trace



Test error increments the metric and can be tied to a known trace or revision.

Log-based metric contract · concrete contract

```
gcloud logging metrics create app-errors -- 01
description='ACE app errors' --log-
filter='severity>=ERROR AND
resource.type="cloud_run_revision"'
```

1

Scope: Resource Management and Monitoring

2

Mapped: A5 · K2

3

Capture stdout and exit status

4

Source: official log-metrics documentation

Log-based metric contract · failure and control

FAILURE MODES



filter never matches



high-cardinality label



ingestion delay misread

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Test error increments the metric and can be tied to a known trace or revision.

Alert policy evaluation · architecture

1

Metric Filter

Boundary component 1. State and ownership must be explicit.

2

Alignment Window

Boundary component 2. State and ownership must be explicit.

3

Threshold

Boundary component 3. State and ownership must be explicit.

4

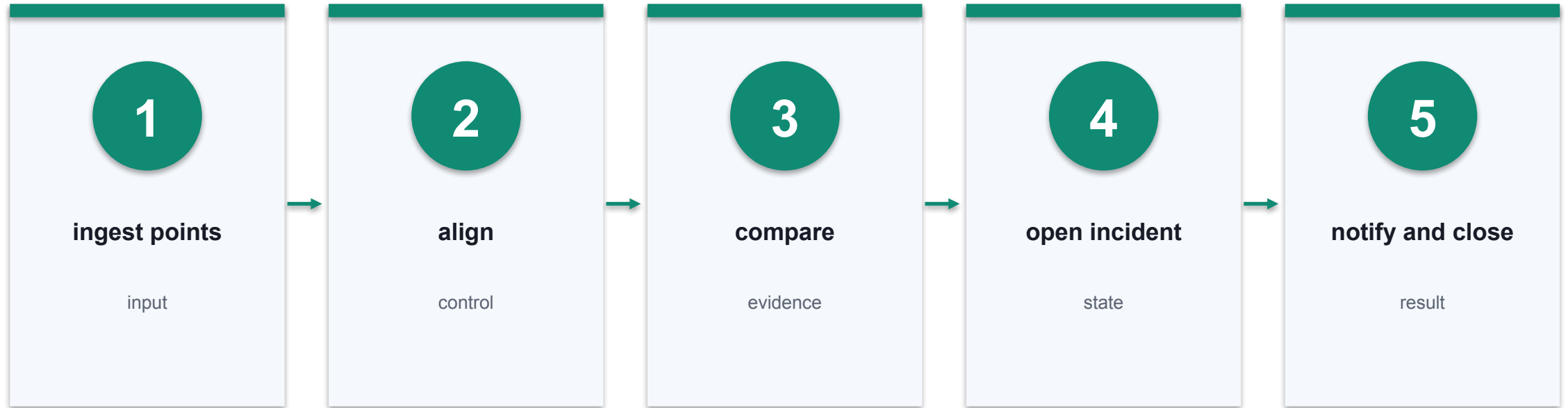
Notification Channel

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Synthetic breach opens one incident and recovery closes it without alert flapping.

Alert policy evaluation · runtime trace



Synthetic breach opens one incident and recovery closes it without alert flapping.

Alert policy evaluation · concrete contract

```
gcloud alpha monitoring policies list --  
format='table(displayName,enabled,conditions)'
```

01

1

Scope: Resource Management and Monitoring

2

Mapped: A5 · K2

3

Capture stdout and exit status

4

Source: official alerts documentation

Alert policy evaluation · failure and control

FAILURE MODES



window too short



missing data behaviour



channel unverified

VERIFICATION EVIDENCE



Expected state is explicit



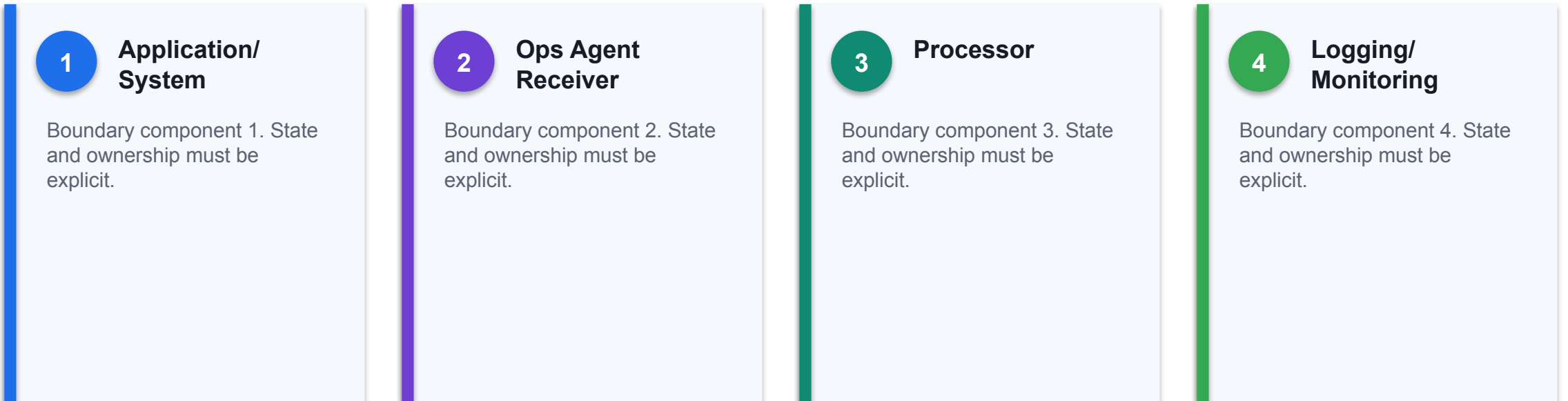
Observed output is captured



Timestamp and scope are recorded

Synthetic breach opens one incident and recovery closes it without alert flapping.

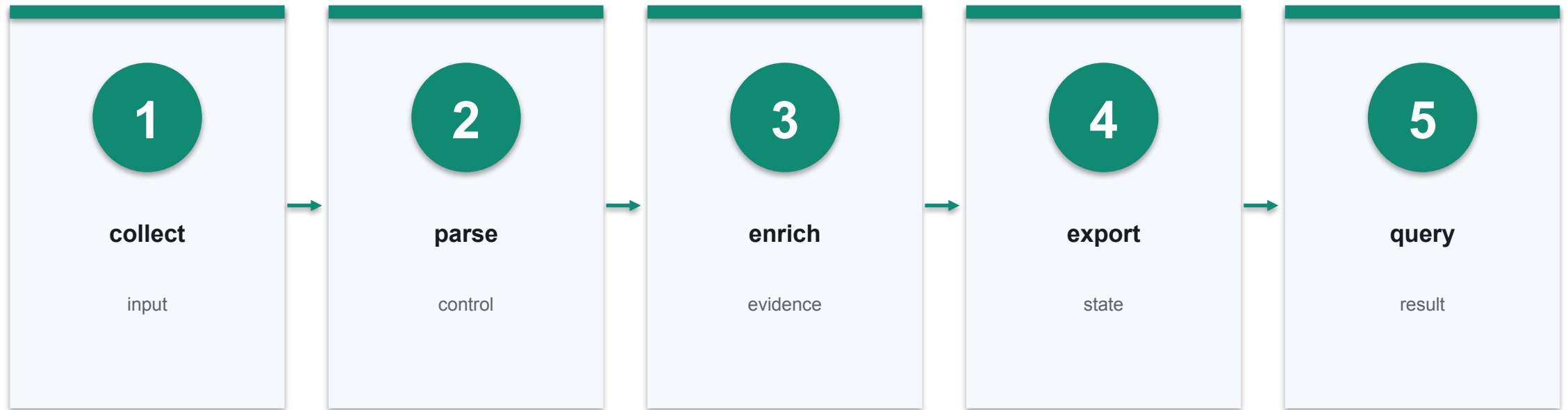
Ops Agent telemetry path · architecture



DECISION RULE

Scope first: Agent service is healthy and expected log/metric names appear with resource labels.

Ops Agent telemetry path · runtime trace



Agent service is healthy and expected log/metric names appear with resource labels.

Ops Agent telemetry path · concrete contract

```
sudo systemctl status google-cloud-ops-agent 01  
sudo /opt/stackdriver/stack-config/config-checker --02  
config-path=/etc/google-cloud-ops-agent/config.yaml
```

1

Scope: Resource Management and Monitoring

2

Mapped: A5 · K2

3

Capture stdout and exit status

4

Source: official ops-agent documentation

Ops Agent telemetry path · failure and control

FAILURE MODES



invalid YAML



agent service account lacks role



duplicate ingestion

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Agent service is healthy and expected log/metric names appear with resource labels.

Evidence-led escalation · architecture

1

Impact

Boundary component 1. State and ownership must be explicit.

2

Timeline

Boundary component 2. State and ownership must be explicit.

3

Metrics And Logs

Boundary component 3. State and ownership must be explicit.

4

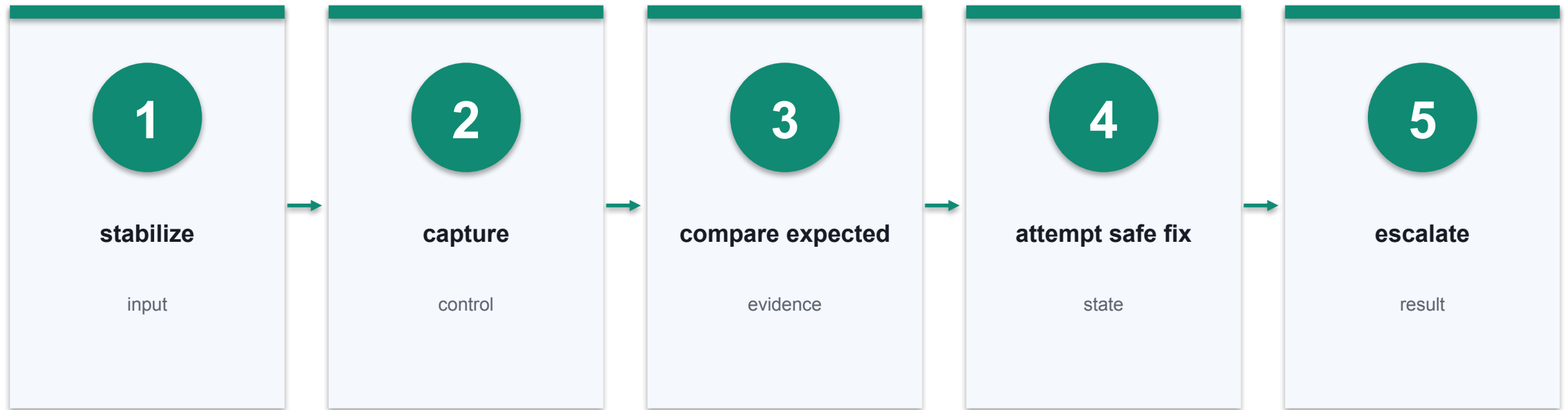
Owner And Next Action

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Escalation pack is reproducible, secret-safe and states what remains unknown.

Evidence-led escalation · runtime trace



Escalation pack is reproducible, secret-safe and states what remains unknown.

Evidence-led escalation · concrete contract

```
gcloud logging read 'timestamp>=\"2026-09-12T00:00:00Z\" AND severity>=ERROR' --limit=100 --format=json > incident-logs.json
```

01

1

Scope: Resource Management and Monitoring

2

Mapped: A5 · K2

3

Capture stdout and exit status

4

Source: official incident-response documentation

Evidence-led escalation · failure and control

FAILURE MODES



screenshots without timestamps



credentials in logs



no rollback record

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Escalation pack is reproducible, secret-safe and states what remains unknown.

Create Monitoring Evidence and an Incident Pack

SCENARIO

Create a log-based error metric and alert policy, trigger a synthetic fault and assemble escalation evidence.

YOU WILL PRODUCE

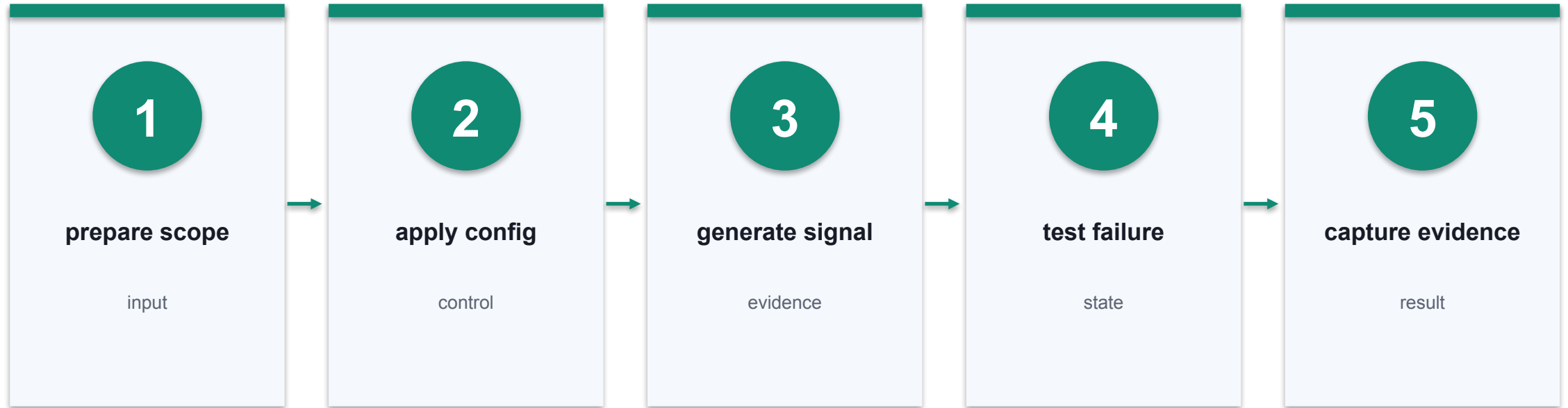
alert-policy.json, incident timeline and filtered log bundle

alert-policy.json

incident_timeline.csv

Full click-by-click instructions are in the Learner Guide and Activity PDF. This slide shows the work product, not the procedure.

Activity 9 system trace



Create a log-based error metric and alert policy, trigger a synthetic fault and assemble escalation evidence.

Activity 9 worked artifact

```
{
  "displayName": "ACE application error rate",
  "combiner": "OR",
  "conditions": [{"displayName": "Errors above
threshold", "conditionThreshold": {"filter":
"metric.type=logging.googleapis.com/user/app_errors
", "comparison": "COMPARISON_GT", "thresholdValue":
2, "duration": "60s"}}}]
```

1

Use placeholders for project-specific values

2

Run only in the isolated training project

3

Capture both success and failure output

4

Keep secrets out of submitted evidence

Activity 9 acceptance gate

FAILURE MODES



wrong project or region



insufficient IAM or disabled API



observed output does not meet target

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Accept when alert-policy.json, incident timeline and filtered log bundle are complete, timestamped and reproducible.

Lunch and resource check

Before you step away

Stop test traffic

Check active resources

Save evidence

Do not leave tokens on screen

Day 3 evidence recap

1

Topic 7

Managed Services

2

Topic 8

Kubernetes

3

Topic 9

Resource Management and
Monitoring

4

Evidence

Record expected versus
observed output.

DECISION RULE

A configuration is not complete until its observable state matches the acceptance criterion.

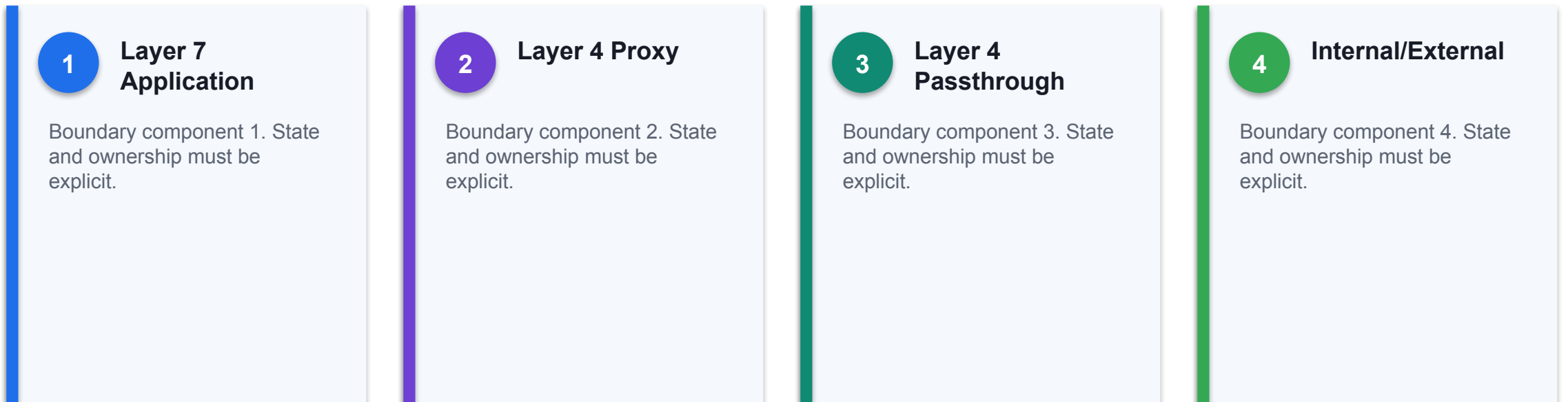
TOPIC 10 · A6 · K2

Load Balancing and Autoscaling

Operate scalable and highly available services



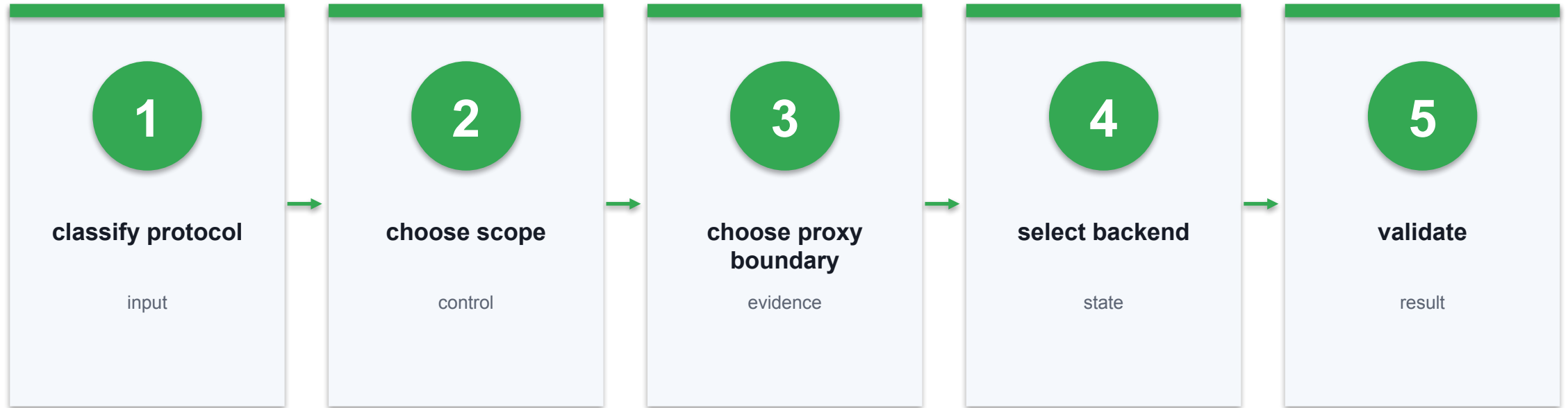
Application versus Network Load Balancers · architecture



DECISION RULE

Scope first: Choice states OSI layer, client reachability, scope and source-IP requirement.

Application versus Network Load Balancers · runtime trace



Choice states OSI layer, client reachability, scope and source-IP requirement.

Application versus Network Load Balancers · concrete contract

```
gcloud compute forwarding-rules list --  
format='table(name,loadBalancingScheme,IPProtocol,IPAd-01  
dress,backendService) '
```

1

Scope: Load Balancing and Autoscaling

2

Mapped: A6 · K2

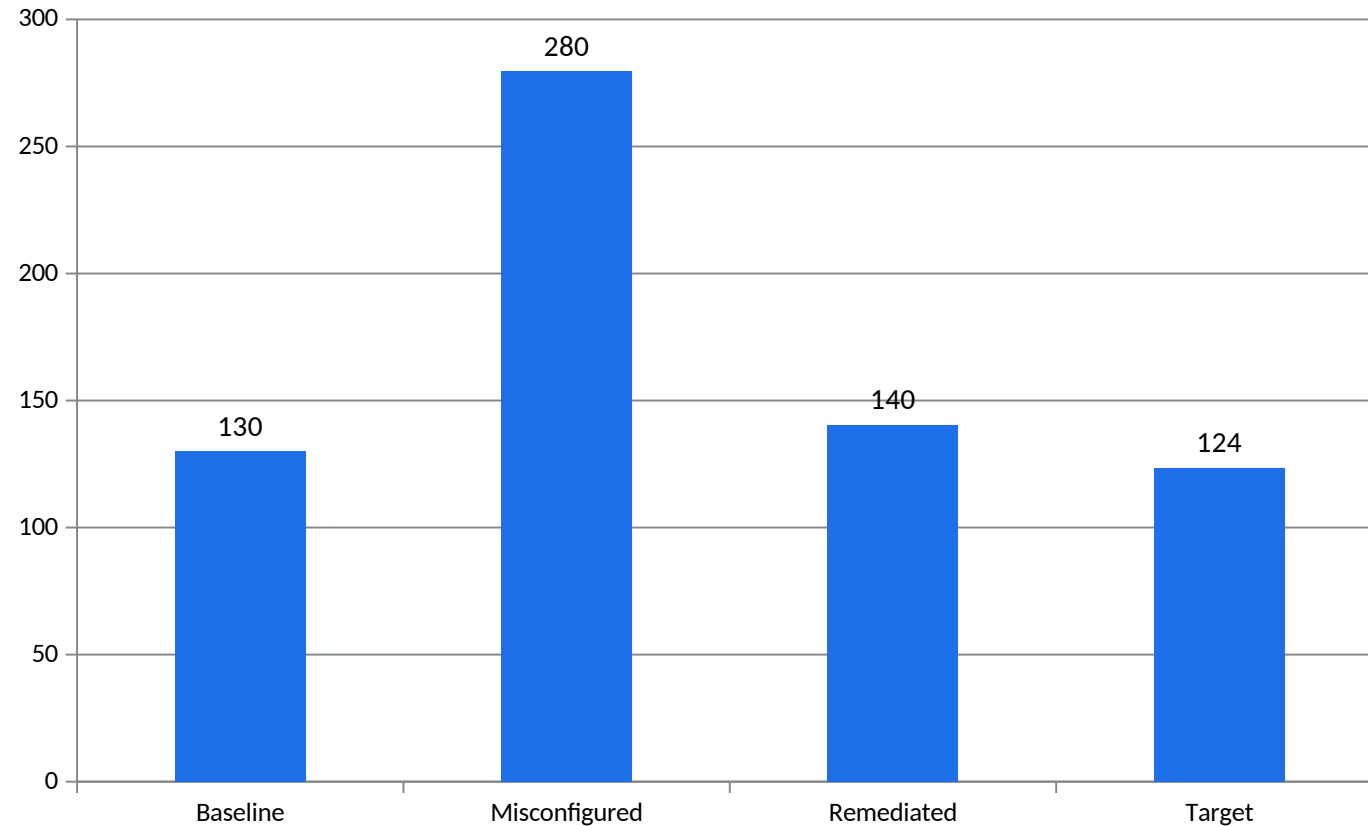
3

Capture stdout and exit status

4

Source: official load-balancing documentation

Topic 10 evidence profile

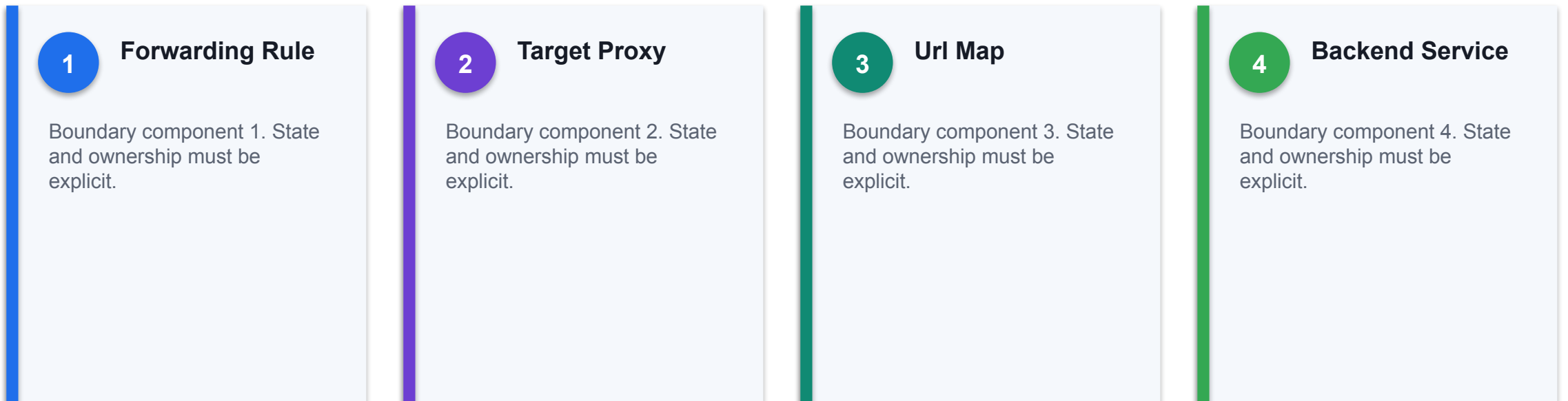


WHAT THE DATA SHOWS

A controlled fault should move the signal away from baseline. The remediation is accepted only when it returns close to target and the exact configuration change is recorded.

Indicative teaching data. Learners replace it with measurements from the aligned activity.

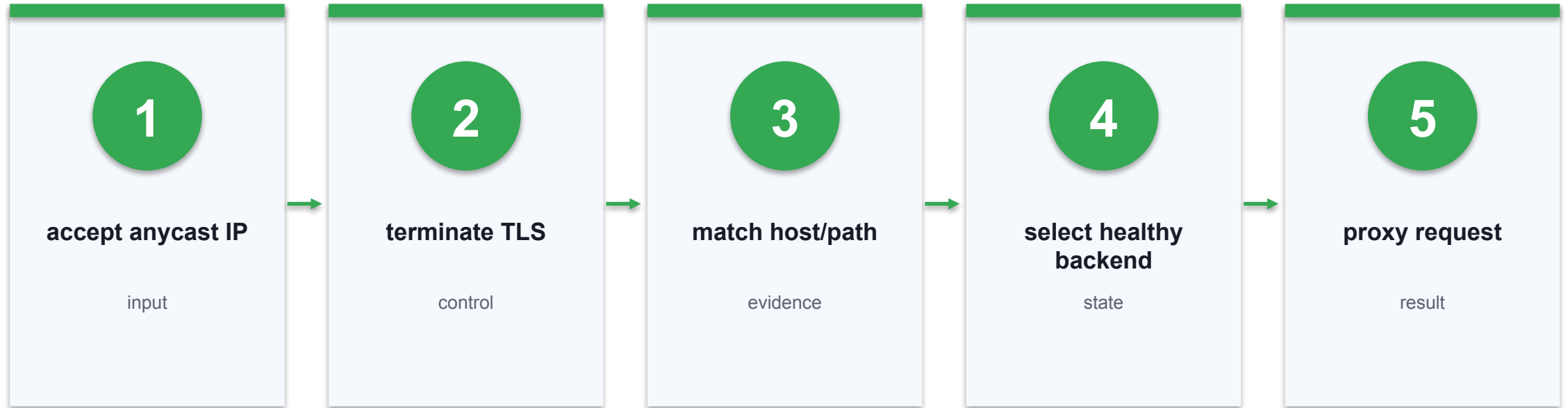
Application Load Balancer request path · architecture



DECISION RULE

Scope first: URL-map route and backend health explain where a request is sent.

Application Load Balancer request path · runtime trace



URL-map route and backend health explain where a request is sent.

Application Load Balancer request path · concrete contract

```
gcloud compute url-maps describe ace-map --global 01
gcloud compute backend-services get-health ace- 02
backend --global
```

1

Scope: Load Balancing and Autoscaling

2

Mapped: A6 · K2

3

Capture stdout and exit status

4

Source: official alb documentation

Application Load Balancer request path · failure and control

FAILURE MODES



certificate not active



URL rule mismatch



health check firewall blocked

VERIFICATION EVIDENCE



Expected state is explicit



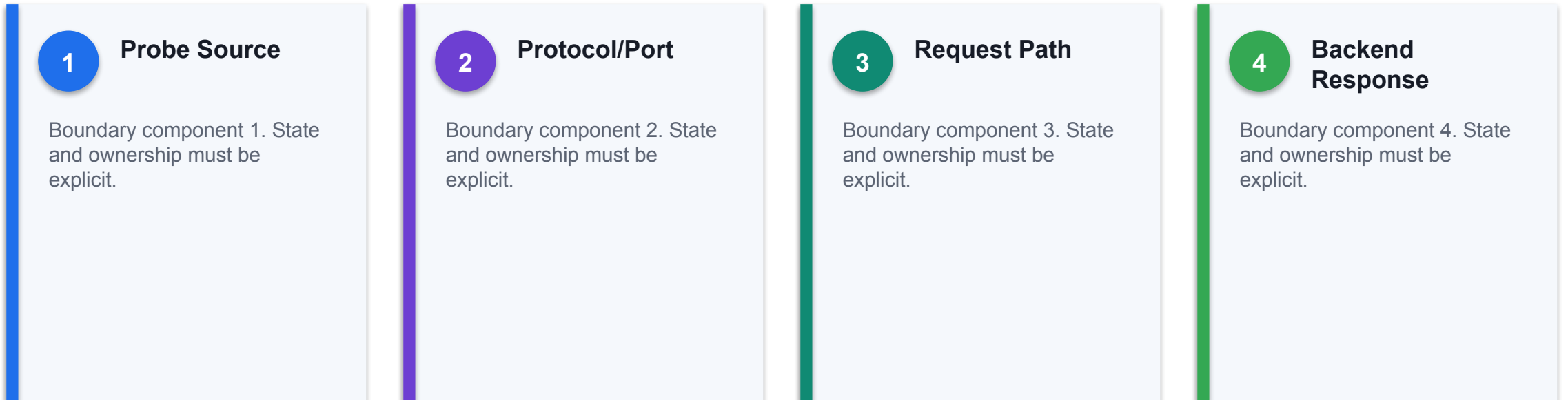
Observed output is captured



Timestamp and scope are recorded

URL-map route and backend health explain where a request is sent.

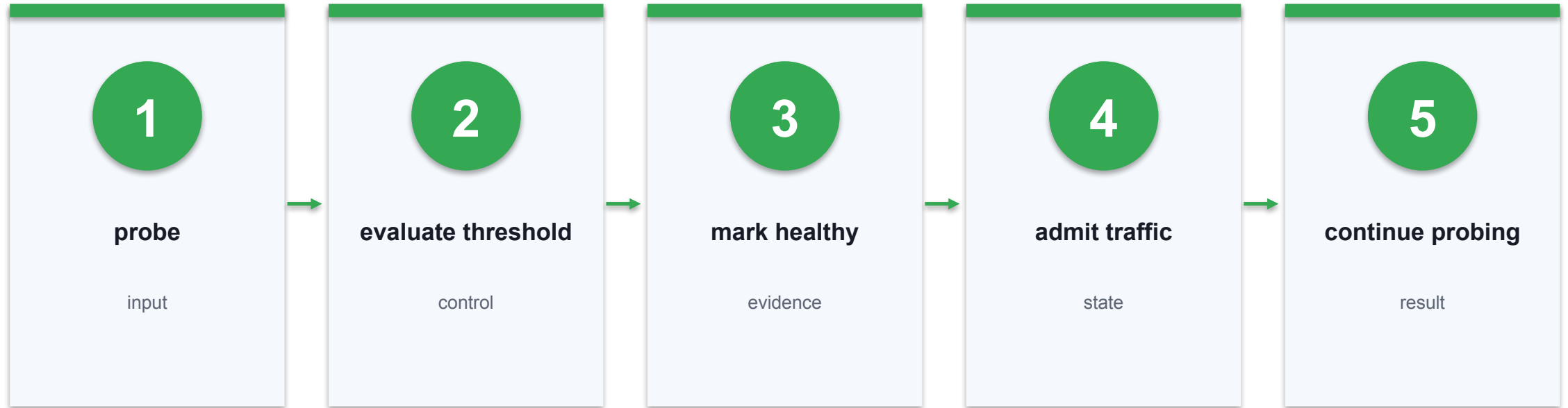
Health checks versus application readiness · architecture



DECISION RULE

Scope first: Probe path tests a dependency-light readiness endpoint and healthy thresholds match recovery needs.

Health checks versus application readiness · runtime trace



Probe path tests a dependency-light readiness endpoint and healthy thresholds match recovery needs.

Health checks versus application readiness · concrete contract

```
gcloud compute health-checks describe ace-hc --global
```

1

Scope: Load Balancing and Autoscaling

2

Mapped: A6 · K2

3

Capture stdout and exit status

4

Source: official health-checks documentation

Health checks versus application readiness · failure and control

FAILURE MODES



probe hits redirect



app listens different port



probe requires authentication

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Probe path tests a dependency-light readiness endpoint and healthy thresholds match recovery needs.

Managed instance group autoscaler · architecture

1

Metric

Boundary component 1. State and ownership must be explicit.

2

Target

Boundary component 2. State and ownership must be explicit.

3

Cooldown

Boundary component 3. State and ownership must be explicit.

4

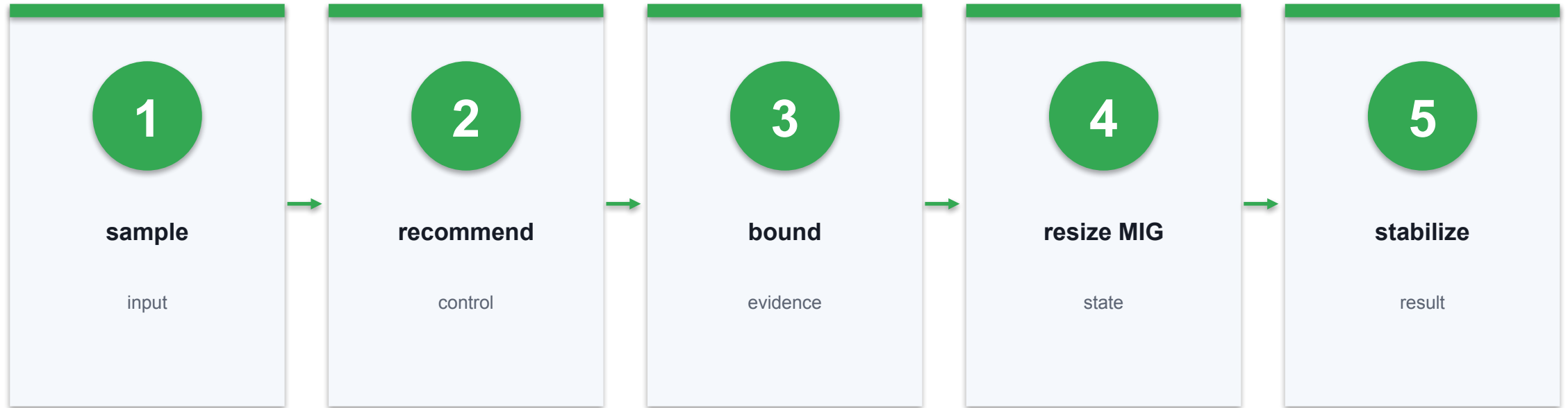
Min/Max Replicas

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Observed group size rises under sustained load and returns after the stabilization period.

Managed instance group autoscaler · runtime trace



Observed group size rises under sustained load and returns after the stabilization period.

Managed instance group autoscaler · concrete contract

```
gcloud compute instance-groups managed set-  
autoscaling web-mig --region=asia-southeast1 --min-  
num-replicas=2 --max-num-replicas=10 --target-cpu-  
utilization=0.60 --cool-down-period=90
```

01

1

Scope: Load Balancing and Autoscaling

2

Mapped: A6 · K2

3

Capture stdout and exit status

4

Source: official mig-autoscaling documentation

Managed instance group autoscaler · failure and control

FAILURE MODES



load below target



cooldown masks change



max quota reached

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Observed group size rises under sustained load and returns after the stabilization period.

Capacity math and backend saturation · architecture

1

Requests Per Second

Boundary component 1. State and ownership must be explicit.

2

Service Time

Boundary component 2. State and ownership must be explicit.

3

Concurrency

Boundary component 3. State and ownership must be explicit.

4

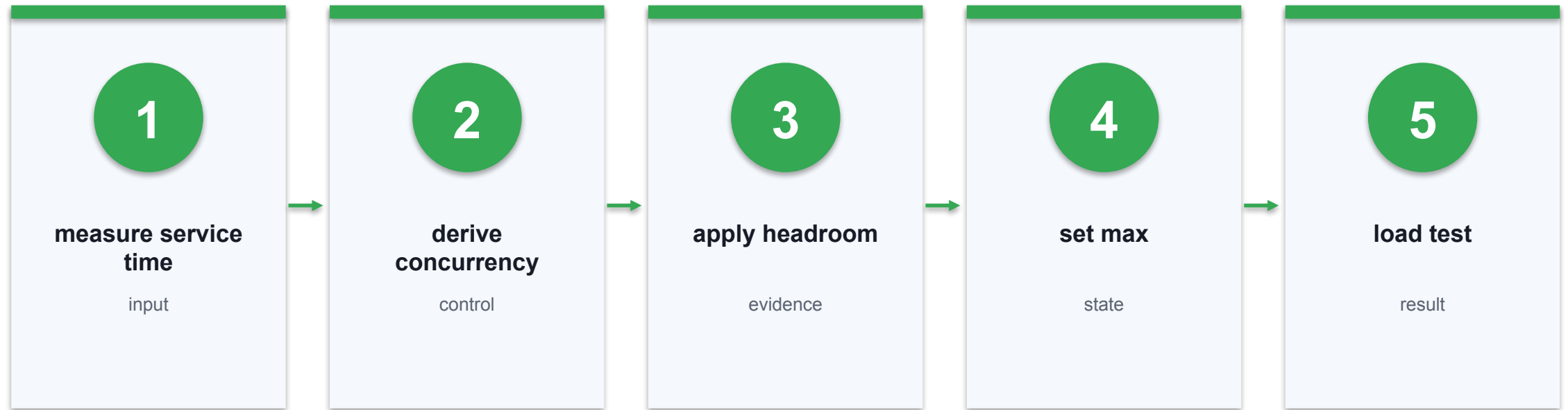
Instance Capacity

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Capacity estimate is compared with p95 latency and error rate from a controlled test.

Capacity math and backend saturation · runtime trace



Capacity estimate is compared with p95 latency and error rate from a controlled test.

Capacity math and backend saturation · concrete contract

```
python3 -c 'rps=240; seconds=.25; headroom=.7; 01  
print(round((rps*seconds)/headroom,1))'
```

1

Scope: Load Balancing and Autoscaling

2

Mapped: A6 · K2

3

Capture stdout and exit status

4

Source: official capacity documentation

Capacity math and backend saturation · failure and control

FAILURE MODES



average hides p95



downstream bottleneck



no warm-up

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Capacity estimate is compared with p95 latency and error rate from a controlled test.

Traffic distribution and session affinity · architecture

1

Client Tuple

Boundary component 1. State and ownership must be explicit.

2

Hash/Affinity

Boundary component 2. State and ownership must be explicit.

3

Healthy Backend Set

Boundary component 3. State and ownership must be explicit.

4

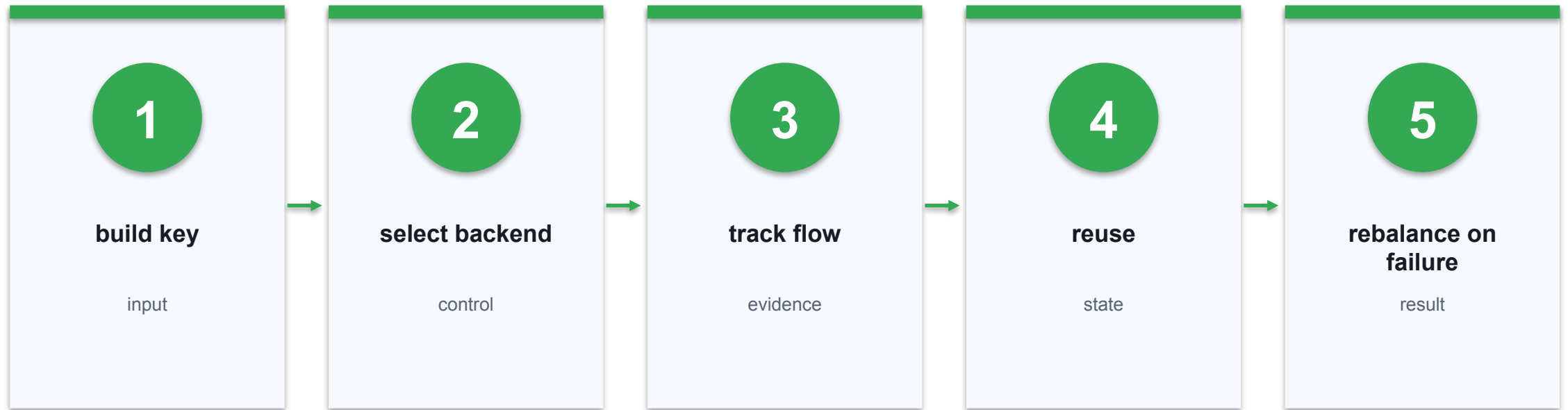
Connection Tracking

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Affinity is enabled only when application state or protocol requires it.

Traffic distribution and session affinity · runtime trace



Affinity is enabled only when application state or protocol requires it.

Traffic distribution and session affinity · concrete contract

```
gcloud compute backend-services describe ace-backend01  
--global --  
format='value(sessionAffinity,localityLbPolicy,timeou  
tSec)'
```

1

Scope: Load Balancing and Autoscaling

2

Mapped: A6 · K2

3

Capture stdout and exit status

4

Source: official traffic-distribution documentation

Traffic distribution and session affinity · failure and control

FAILURE MODES



sticky session hides imbalance



source IP changes through proxy



state not externalized

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Affinity is enabled only when application state or protocol requires it.

Autoscaling load-test evidence · architecture

1

Load Generator

Boundary component 1. State and ownership must be explicit.

2

Cloud Monitoring

Boundary component 2. State and ownership must be explicit.

3

Mig Size

Boundary component 3. State and ownership must be explicit.

4

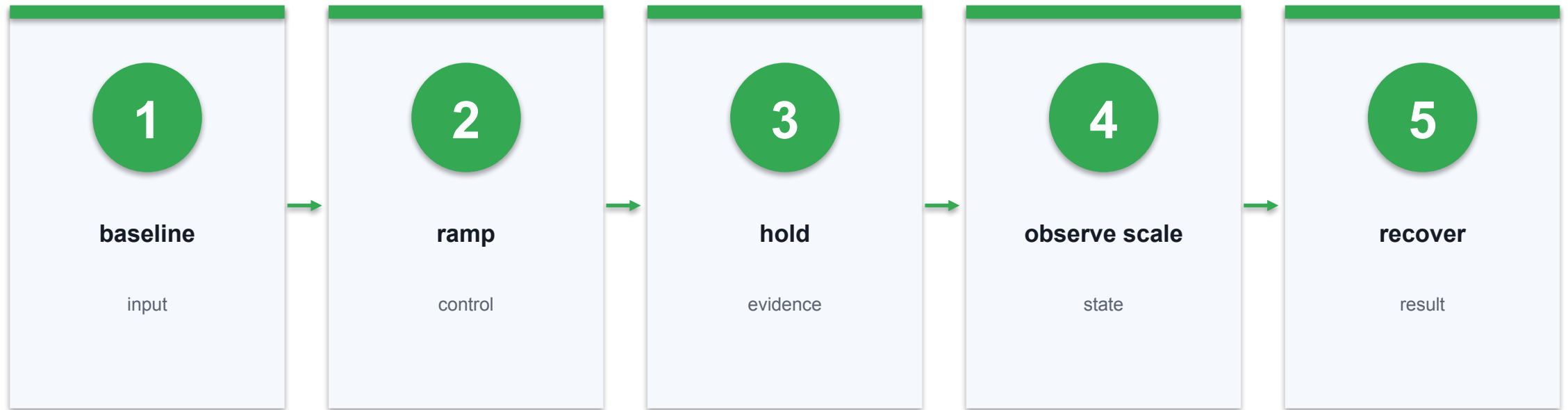
Backend Latency

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Test records target RPS, actual RPS, p95 latency, error rate and instance count over time.

Autoscaling load-test evidence · runtime trace



Test records target RPS, actual RPS, p95 latency, error rate and instance count over time.

Autoscaling load-test evidence · concrete contract

```
bash load_test.sh --url=$LB_URL --rps=240 --duration=300 > load-results.csv 01
```

1

Scope: Load Balancing and Autoscaling

2

Mapped: A6 · K2

3

Capture stdout and exit status

4

Source: official load-testing documentation

Autoscaling load-test evidence · failure and control

FAILURE MODES



client becomes bottleneck



DNS cached old IP



test too short

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Test records target RPS, actual RPS, p95 latency, error rate and instance count over time.

Escalation when scaling fails · architecture

1 Autoscaler Status

Boundary component 1. State and ownership must be explicit.

2 Quota

Boundary component 2. State and ownership must be explicit.

3 Health

Boundary component 3. State and ownership must be explicit.

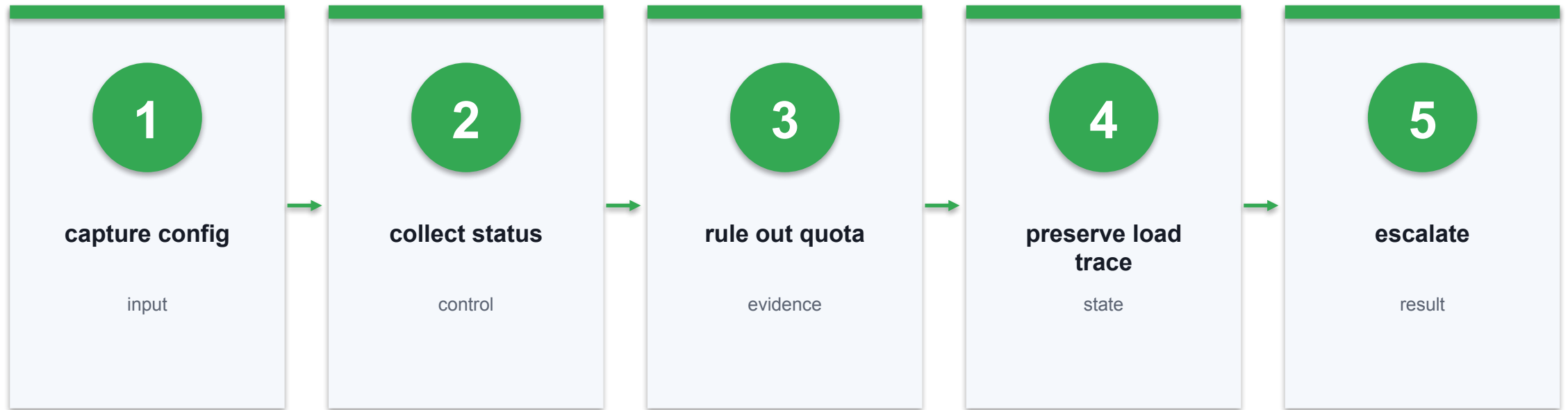
4 Audit Log

Boundary component 4. State and ownership must be explicit.

DECISION RULE

Scope first: Escalation identifies expected trigger, observed signal, blocking condition and safe attempts already made.

Escalation when scaling fails · runtime trace



Escalation identifies expected trigger, observed signal, blocking condition and safe attempts already made.

Escalation when scaling fails · concrete contract

```
gcloud compute instance-groups managed describe web-01  
mig --region=asia-southeast1 --format=json > mig.json  
gcloud compute operations list --filter='status!=DONE'
```

1

Scope: Load Balancing and Autoscaling

2

Mapped: A6 · K2

3

Capture stdout and exit status

4

Source: official troubleshooting documentation

Escalation when scaling fails · failure and control

FAILURE MODES



manual resize destroys evidence



wrong region



quota error omitted

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Escalation identifies expected trigger, observed signal, blocking condition and safe attempts already made.

Load Test a Managed Instance Group and Escalate

SCENARIO

Apply health checking and CPU autoscaling to a managed instance group, run a controlled load test and escalate a scaling failure.

YOU WILL PRODUCE

load-results.csv, MIG configuration and escalation report

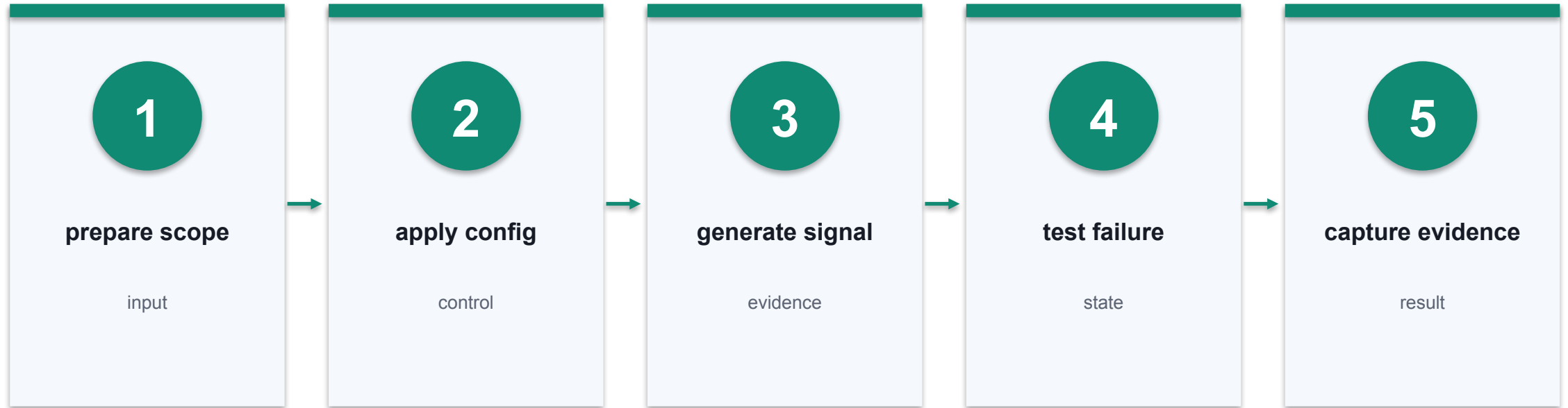
`load_test.sh`

`load-results.csv`

`escalation-template.md`

Full click-by-click instructions are in the Learner Guide and Activity PDF. This slide shows the work product, not the procedure.

Activity 10 system trace



Apply health checking and CPU autoscaling to a managed instance group, run a controlled load test and escalate a scaling failure.

Activity 10 worked artifact

```
#!/usr/bin/env bash                                01
set -euo pipefail                                  02
URL=${1:?Usage: load_test.sh URL}                   03
for i in $(seq 1 120); do                           04
    code=$(curl -sS -o /dev/null -w '%{http_code}' "$URL") 05
    printf '%s,%s,%s\n' "$(date -u +%FT%TZ)" "$i" "$code" 06
done                                                 07
```

1

Use placeholders for project-specific values

2

Run only in the isolated training project

3

Capture both success and failure output

4

Keep secrets out of submitted evidence

Activity 10 acceptance gate

FAILURE MODES



wrong project or region



insufficient IAM or disabled API



observed output does not meet target

VERIFICATION EVIDENCE



Expected state is explicit



Observed output is captured



Timestamp and scope are recorded

Accept when load-results.csv, MIG configuration and escalation report are complete, timestamped and reproducible.

Lunch and resource check

Before you step away

Stop test traffic

Check active resources

Save evidence

Do not leave tokens on screen

Day 4 evidence recap

1

Topic 10

Load Balancing and
Autoscaling

2

Evidence

Record expected versus
observed output.

3

Evidence

Record expected versus
observed output.

4

Evidence

Record expected versus
observed output.

DECISION RULE

A configuration is not complete until its observable state matches the acceptance criterion.

Course feedback and TRAQOM survey

1

Open survey

Use the trainer-provided QR or link.

2

Rate delivery

Focus on clarity, practice and support.

3

Comment

Name the topic or activity when possible.

4

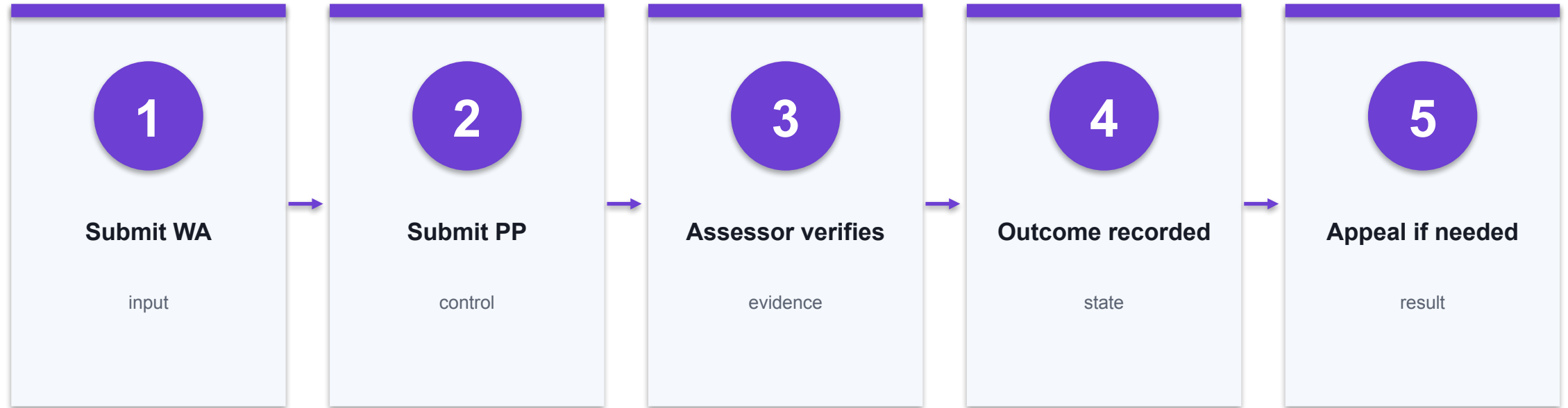
Submit

Wait for the confirmation screen.

DECISION RULE

Feedback is used to improve delivery while approved outcomes and assessment mappings remain controlled.

Assessment close-out



The LMS receives question-paper submissions only; trainer keys remain protected.

Digital attendance with TRAQOM

1

Assessment QR

Scan the correct assessment attendance record.

2

Identity

Confirm the displayed name.

3

Timestamp

Submit inside the announced window.

4

Readback

Show confirmation if the roster is missing.

DECISION RULE

Do not leave until the trainer confirms both submission and assessment attendance.

THANK YOU

Operate from evidence, not assumption.

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